Machine Translated by Google

Refrigeration (refrigerator)	2,20 * 1	2,20
Total		**5,05**

Fuente: M. Dumitru and M. Bibu, “Researches on the digesters and reactors which can be used in a farm scale biogas plant,” vol. 14, no. 3, pp. 63–68, 2014.

To meet these needs, a certain number of animals is required. The number of animals needed for a family of 5 is: 13 cattle, or 39 pigs, or 365 poultry, or proportional mixtures thereof. If the needs are limited to biogas production solely for cooking, the number of animals for 1.5 m³

. It is reduced to 4 cattle, which can be kept on a plot of two hectares of land.

10.4.2. Commercial Feasibility

Although the use of biogas for the residential sector in the country has not been promoted, its implementation potential is high in warm and temperate zones where any biodegradable organic matter or organic waste from the countryside can be used in the generation of biogas, but depending on the type of waste, its biogas yield and the control of the process can be demanding and require care.

The cooking needs of a rural family are met by one cubic meter of biogas per day. The raw material used, which contains the microorganisms necessary for biogas production, is animal excrement, especially cattle, pig, and chicken manure, as well as human excrement. The continuous use of antibiotics in humans can inhibit biogas production, so using these excrements is not advisable. Under these conditions, 30 kilograms of cattle manure or 20 kilograms of pig manure provide the raw material needed to produce one cubic meter of biogas.

Regarding technology, it is recommended to start with tubular digesters, which should be protected by fences or walls to prevent damage from animals; it is also important to start in warm areas such as the plains and the coast or the inter-Andean valleys below 1,000 meters above sea level, which have an ambient temperature close to the ideal for sustaining mesophilic microorganisms of 35°C. The process can be carried out in colder conditions, but it is necessary to heat the biodigester with the help of solar radiation and insulate it to avoid heat loss.

Many potential users will want to use the biogas to meet other needs, such as powering a lamp or a refrigerator for food preservation. In these cases, the cost of the biodigester is higher, and the raw material requirements are also greater. Therefore, it is important to assess the situation of each of the additional services requested and to what extent the program can contribute to the extra costs.

When dealing with small settlements such as rural communities or isolated populations, the feasibility of installing lagoon-type biodigesters with networked biogas distribution systems can be evaluated. As a result, individual costs can be lower. This alternative approach raises the possibility of implementing

Machine Translated by Google

pilot projects that demonstrate the existing market potential for the emergence of a commercial niche in the medium term.

10.4.3. Economic Feasibility

This energy source has not been promoted as an alternative for cooking in Colombian homes, but it has been used in industrialized processes, since it requires significant amounts of biomass to function, which would imply that households subject to substitution have animals or crops that provide the raw material necessary for its operation.

Identifying households suitable for implementing the technology is considered an economically feasible measure, taking into account factors such as climate and the availability of raw materials. From this perspective, the alternative is viable, given that the average cost of a biodigester to meet a household's cooking needs is around $1,500,000, with an average lifespan of 10 years, under a regime of periodic maintenance to ensure its durability.

Taking into account the estimated economic savings in the health and environmental sectors, the return on investment for the application of the technology would be 7.9 months.

Table 74. Monthly savings from biogas implementation per household

Technology	Monthly cost of initial stove investment	Energy cost	Healthcare costs	Environmental cost	Total Cost
Traditional stove	$0	$160.000	$ 184.976	$ 8.935	$ 353.912
Biogas stove	$12.500	$160.000	$0	$0	$172.500
Saving	**-$12.500**	**0**	**$184.976**	**$8.935**	**$181.412**

Source: Prepared by the consulting team

10.4.4. Environmental Feasibility

Providing a cost-environmental assessment of each technology, in this case the implementation of biogas stoves to replace traditional stoves, the monetary savings would amount to $8,935 per month, which is equivalent to the environmental savings in emissions of 2.

Table 75. Environmental cost savings with the implementation of biogas

Technology	Environmental cost
Traditional stove	$ 8.935

Machine Translated by Google

Biogas	$ 0
Saving	**$ 8.935**

Source: Prepared by the consulting team

10.4.5. Feasibility in Health

The health system would save $184,976 per month per household served in the case of replacement with biodigesters.

Table 76. Savings in health costs with the implementation of biogas

Technology	Healthcare costs
Traditional stove	$ 184.976
Biogas	$ 0
Saving	**$ 184.976**

Source: Prepared by the consulting team

10.4.6. Social Feasibility

In the case of biogas, households would go from collecting firewood from long distances to storing the waste and excrement produced by their animals in a closer area, one that is easily identifiable by the families. While it wouldn't necessarily reduce the time spent acquiring energy, the implementation of biodigesters would improve their quality of life by eliminating exposure to the smoke generated by traditional wood-burning stoves. In this way, they could use a modern stove powered by free gas produced by themselves.

Machine Translated by Google

11. Responsible Institutions and Stakeholder Map

Once Contract *UPME C-031-2019 is completed,* the implementation phase of the Strategy defined by the study will begin in its three stages: short term (2020-2022), medium term (2023-2030), and long term (2030-2050). To ensure success in the implementation stages, it is necessary to design an appropriate institutional framework that defines all the elements of a program of this nature, such as objectives and goals, Plan costs, subsidies involved in its implementation, participating government entities and their corresponding responsibilities, and the agent(s) that will assume the Management and Operation of the Firewood Substitution Program.

The Firewood Substitution Plan has unique characteristics that require coordinated action from different branches of the National Government to ensure the successful implementation of the strategy in the short, medium, and long term. This chapter addresses that topic.

11.1. The Progressive Replacement Plan for Firewood as a public policy

For the design of the Policy and the institutional framework, the following characteristics of the *Progressive Replacement Plan for Firewood should be taken into account:*

1. The Plan will positively impact a significant number of Colombian families. It is estimated that 1.66 million households, equivalent to more than 5 million people, will see the benefits of the Plan and experience a change in their food preparation habits, and more importantly, will see a significant improvement in their health and the quality of life of their family environment.

2. The implementation of the Fuel Replacement Plan extends beyond the current administration and will, in fact, span several more four-year terms. Furthermore, while some technologies, such as LPG, are ready for immediate deployment—aided by their past use as the basis for similar programs—others, such as biogas, require training for potential users and pilot testing to enhance their capacity as a large-scale solution.

3. The effectiveness of the Plan will depend on the actions of several government entities, which entails demanding inter-institutional coordination work so that each actor in the program carries out its task according to the established objectives.
4. In addition to the aforementioned inter-institutional coordination, it is essential that the Plan have political leadership so that it is clear to all participants that this is a priority program for the Government. We recommend that the guidance and leadership be provided by the National Government and, potentially, by an Advisor with specific responsibilities for the Firewood Substitution Program.

   As an example, analyses of the program to replace firewood with LPG in India consider that the leadership of Prime Minister Narendra Modi and the treatment of this program as a State Policy have been fundamental to the results achieved to date.

Machine Translated by Google

5. The implementation of the Program must be supported by an Implementing Agency with a clear scope and objectives. This should not entail the creation of a new state entity, but rather the contracting of an Organization or Working Group for each of the Plan's implementation periods: short, medium, and long term. This Implementing Agency will be responsible for designing the contracting processes with the various agents who will deliver the different energy sources to the beneficiary families, according to the Substitution Plan Strategy.

6. The replacement of firewood goes beyond simply changing the fuel source for cooking and heating in some areas of the country. Its impacts include improvements in health, well-being, and quality of life, as well as cultural influences. Therefore, the participation of the communities that will benefit from the program is essential. To this end, the implementation phases must be accompanied by a community outreach program. Otherwise, the solutions provided may not be integrated into household practices, and over time, residents may revert to cooking with firewood if they do not feel ownership of the change process.

7. The Progressive Firewood Substitution Program is a public interest program aimed at addressing a problem with national impact on human lives, health conditions, and economic costs. Therefore, the Firewood Substitution Program is part of a Public Policy designed to resolve a specific problem.

In conclusion, the firewood substitution policy includes decisions aimed at improving the well-being of families who consume firewood, with proposals to replace traditional energy sources with other energy resources, in order to improve their current health and pave the way for greater prosperity and quality of life. This entails establishing a process of institutional coordination among the state agencies that must be involved and contracting the implementation with an Implementing Agency, all within the framework of a Government Policy that ensures the achievement of the objectives.

11.2. Approval of the Plan – CONPES Document

Given the above, the Firewood Substitution Plan should be elevated to the status of public policy and therefore considered, evaluated, and approved at the highest levels of government. To this end, the body that should review the Firewood Substitution Plan and propose its approval as a Public Policy is CONPES.

Some of the functions of the National Council for Economic and Social Policy (CONPES), applicable to the case at hand, are331 :

1. To present, for approval, the policies, strategies, plans, programs and projects of the National Government.

331 Source: dnp.gov.com

Machine Translated by Google

2. To present, for analysis, studies on the implementation of the National Development Plan and on the policies, strategies, programs and projects of the national government.

3. Submit, for study and approval, the bases and criteria of public investment.
4. Submit the annual operating plan for investments for study and approval.
5. The members of CONPES, as indicated in Decree 2148 of 2009, are permanent and non-permanent, invited and attendees, which allows for the participation of all government entities involved in the Firewood Substitution Plan.

The National Planning Department (DNP) serves as the CONPES Secretariat and is therefore responsible for reviewing the Plan and preparing the necessary documentation for CONPES consideration. The document submitted by the DNP must establish the implementation mechanisms for the Crop Substitution Plan, assigning responsibilities and defining the resources that must be budgeted for its execution in the different stages, beginning with the first short-term phase from 2020 to 2022.

11.3. Entities participating in the execution of the Plan

The following are the entities whose participation is considered necessary by the Consulting Team due to their contribution to the objectives of the Plan.

1. Ministry of Mines and Energy: This is the governing body for the country's energy policy. Its policy decisions can impact the supply and demand of energy sources that can replace firewood, such as LPG and electricity.

2. National Planning Department: The DNP is the entity that proposes the National Development Plan to the Government and Congress of the Republic, analyzes subsidies for public services, and evaluates public investment requirements in order to determine the National Budget. It also aligns sectoral plans with the National Development Plan.

3. Mining and Energy Planning Unit (UPME): This entity is responsible for contracting the study to define the Firewood Substitution Strategy and also participates in matters such as the National Biogas Plan. As the Planning Unit, it prepares studies and proposals for policy adoption by the Ministry of Mines and Energy. Furthermore, UPME is in charge of the administration and management of supply-side funds such as FAER and PRONE, among others, which are considered an essential element in the financing schemes of the Substitution Plan.

4. The Institute for Planning and Promotion of Energy Solutions for Non-Interconnected Zones, IPSE: As an entity attached to the Ministry of Mines and Energy, and due to its functions and area of work in Non-Interconnected Zones, it is a very relevant entity for the execution of the Firewood Replacement Program as

   This will be explained later.

5. Ministry of Health: Since improving the health conditions of families affected by firewood consumption is the cornerstone and focus of the long-term firewood substitution strategy, the Ministry's participation in the subsequent stages of the program is fundamental, particularly in areas such as raising awareness among the

Machine Translated by Google

The impacts of firewood consumption on the health of families, particularly women and children, are discussed. It is also proposed that the Ministry of Health, through Health Service Providers, promote health promotion and prevention programs to reduce the impact of firewood consumption on family health.

6. Ministry of Environment and Sustainable Development: The implementation of the strategy, In addition to prioritizing health, it must be compatible with policies established through environmental and sustainable development policies. Furthermore, the Ministry will develop the National Efficient Stoves Plan, which is one of the key components of the Firewood Replacement Plan.

7. Ministry of Agriculture: Biogas is another element of the long-term Firewood Replacement Plan and, therefore, the participation of the Ministry of Agriculture as leader of the National Biogas Plan is fundamental.

8. Ministry of Housing, City and Territory: The national housing policy must incorporate the energy concept into its designs in matters such as energy efficiency and the use of clean energy by families benefiting from new housing programs and, in general, in social interest housing programs, VIS and Priority Interest Housing, VIP, especially rural housing.

9. Ministry of National Education and Ministry of Labor - SENA: The Firewood Substitution Program must be accompanied by an educational and training component covering the following topics: a) Public awareness campaigns about the effects of cooking with firewood on families, particularly women and children. b) Development of educational programs at the primary and secondary levels on energy, energy efficiency, health impacts, and other related topics. And c) Structuring programs within SENA for the design and installation of energy solutions such as efficient stoves, biogas, and renewable energy sources, aimed at meeting the cooking and heating needs of homes.

10. Ministry of the Interior: The support of this Ministry in the relationship with the communities and ethnic groups that are part of the Substitution Plan is considered important.

11. Ministry of Culture: The Ministry of Culture in relation to cultural practices and traditions of fire and cooking, for which the use of firewood must be respected.

12. Presidential High Council for Women: Its participation is considered important given that any firewood substitution program involves considering gender issues, as a strategy to achieve substitution focused on improving the conditions that currently affect mainly the well-being of women.

11.4. The Implementing Agency

Once the National Government makes the decision to implement the Firewood Replacement Plan, the appropriate organizational model for the execution of the Plan must be established.
Before proposing specific options, it is necessary to know what activities the entity responsible for implementing the Plan will have to carry out. In this regard, the following is proposed:

Machine Translated by Google

1. Implement the goals set out in the Replacement Plan for the period 2020-2022 in with regard to:
    a. Define in detail the energy portfolio that will be part of the Substitution Program during that period and the subsidy schemes that will be applied to the process.

    b. Define the coverage in terms of geographic areas, number of families to be replaced, and the criteria for identifying the final beneficiaries. It is expected that by 2020, information from the new, updated version of SISBEN will be available, from which the required information could be obtained.
    c. Define the Budget for the Replacement Program.
    d. Propose the area allocation scheme, which could be: concession or exclusive service areas, taking advantage of the experience of previous programs such as the Gas Plan and other models that are evaluated for the implementation stage.

    e. Once the area allocation scheme has been defined, the Implementing Agency must open the call processes to select the agents or entities that will bring the energy to the beneficiary families as provided in the replacement plan.

    f. Develop the promotion of the Program to cooperation entities technical support and international and/or national funding.
    g. Design outreach programs with leaders, women, communities and ethnic groups that will benefit from the Replacement Program.
    h. Define the process timelines.

Because this is a program whose implementation is planned over a long period, it is proposed that each phase have a different Implementing Agency. However, to facilitate the program's continuity through its various stages, at the end of the 2020-2022 period, an evaluation will be conducted of the proposed area allocation schemes, lessons learned, and recommendations to define the scheme applicable to the 2023-2030 period.

In summary, the Implementing Agency in charge of the execution of the program in the first phase, in addition to carrying out the tasks indicated in the previous section, must leave the way ready so that an agent who will be selected around 2022, can take the information and project the execution of the program for the period 2023-2030.

Now, the question arises: What are the options for choosing the agent, or what will be called the Implementing Agency? The following options are proposed:

1. Select one of the ministries or entities identified in section 11.3.
2. To propose a contracting process using the PPP (Public-Private Partnership) scheme in order to link private capital for the execution of public infrastructure and associated services as indicated by the DNP.
3. Develop a state procurement process to select the Implementing Agency that will assume this responsibility for the first short-term period (2020-2022).

Assigning a program of this nature to a public sector entity carries the risk of it not being given high priority compared to other short-term issues and

Machine Translated by Google

of greater urgency that dominate the agenda of the public entities involved. On the other hand, having an Administrative Department such as the DNP or the UPME, as the Planning Unit attached to the Ministry of Mines and Energy, assume operational and execution responsibilities could divert resources and energy from their mission, which is planning in their respective areas of activity.

The proposal is to hire an organization dedicated exclusively to each phase of the Firewood Replacement Plan, for a fixed term that allows for the execution of structuring and contracting tasks, beginning with the first short-term phase (2020-2022). The terms of the contract would be as follows:

1. During the first half of 2020, it is proposed that the Ministry of Energy submit the application to the National Planning Department (DNP) for approval of the Substitution Plan by the National Council for Economic and Social Policy (CONPES); 2. CONPES will designate the entity responsible for contracting the Implementing Agency. Given its experience and knowledge of the subject, the Mining and Energy Planning Unit (UPME) would be the entity responsible for structuring the terms of reference, conducting the bidding processes, and proceeding with the contracting.
3. An estimated contract period of 30 months is anticipated, starting in July 2020. and would last until December 2022.
4. As indicated above, starting in 2023, a new Implementing Agency would take responsibility for the execution of the Program for the period 2023-2030, for which the Agency contracted for the first period would formally hand over the Substitution Program with the results of the First Stage and the lessons learned, in order to guarantee continuity in the activities.

As has been repeatedly stated in this chapter, the success of the Implementing Agency will be related to the effectiveness of its interaction with the different State entities with which it must interact in fulfillment of its objectives.

In this regard, coordination mechanisms such as Governing Councils or Executive Committees must be designed. Likewise, Inter-administrative Agreements may be signed with State entities whose institutional mandate contributes to the success of the Program. IPSE, for example, can be leveraged to take advantage of its strengths and experience in similar programs in Non-Interconnected Zones, which can be applicable to other regions of the country. Furthermore, IPSE can serve as a link with international entities to channel technical cooperation and funding resources for the implementation of the Firewood Substitution Plan.

Machine Translated by Google

12. Strategy of the Progressive Replacement Plan for Firewood

12.1. Elements of the strategy

The Firewood Replacement Plan is based on the target population that it aims to cover in the different stages:

- Short Term (2020-2022)
- Medium Term (2023-2030)
- Long Term (2031-2050)

The target population results from the scenario analysis carried out in Deliverable II, in the section that responds to objective 7 of this project, with respect to the analysis of the information and figures existing regarding the consumption of firewood, to define references and unification of information, whose results are found in chapter 6.

In this section, two scenarios of figures on firewood consumption were identified and proposed: a first scenario "Scenario A" in which the total number of households that state that their main fuel for cooking is firewood is taken, according to the National Survey of Quality of Life -ECV- of 2018, and the total firewood consumption of the residential sector presented in the BECO 2018, and the average firewood consumption per household and/or per person is determined, according to the number of people per household of the ECV 2018.

As shown in Table 77, according to the 2018 ECV (Living Conditions Survey), 1,664,229 households responded to the question *"What energy or fuel do you primarily use for cooking?"* with *"firewood, wood."* On the other hand, the Colombian Energy Balance (BECO) for 2018 shows a firewood consumption for the residential sector of 6,123 kilotons, which is equivalent to 10.1 kg of firewood per day per household, or 3.1 kg of firewood per person per day, considering an average household size of 3.2 people.

Table 77. Summary of figures for households cooking with firewood "Scenario A"

Homes	15,493,441 Number of Households
Households that cook with firewood	1,664,229 Number of Households
Percentage of households that cook with firewood	11%
Firewood consumed in the residential sector per year	6.123 kTons
People per household	3.2 Number of People
Firewood consumption per household	3.7 Ton/year
Firewood consumption per household	10.1 kg/day
Firewood consumption per person	3.1 kg/day

Source: Own calculations based on DANE (2019)332 and UPME (2019)333 .

A second scenario, "Scenario B", was proposed based on the figures recorded in the 2018 ECV, specifically considering the response to the question *"additionally, do you use*

332 DANE (2019). National Quality of Life Survey 2018.
333 UPME (2019). Colombian Energy Balance -BECO- 2018.

Machine Translated by Google

"Do you use any other fuel for cooking?" Of the 1,664,000 households that primarily cook with firewood, **30% answered affirmatively to the question about using any other fuel for cooking.** Based on this information and the findings from regional workshops on cooking with firewood, which indicated that the average firewood consumption per meal is 1.1 kg, households that cook with firewood are segmented into two groups.

Table 78. Summary figures for "Scenario B" of households that use firewood for cooking

Total number of households: 15,493,441		
Households that cook mainly with firewood* 1,664,229 Number of Households		
Households that cook only with firewood : 70%		
Households that cook only with firewood 1,164,421 Number of Households		
Households that cook with firewood and diversify** 30% Percentage		
Households that cook with firewood and diversify** 499,809 Number of Households		
People per household 3.2 Number of People		
Firewood consumption per menu: 1.1 kg/menu/day		

*The ECV asks, *"What energy or fuel do you primarily use for cooking?"* These are the data that answer this question with: *"firewood, wood".*

**In response to the ECV question , *"Additionally, do you use another fuel for cooking?"* This is the number of households that answered affirmatively. Of these households, 93% use LPG as a secondary fuel.

Source: Own calculations based on DANE (2019)334 and data from the consulting team.

On the other hand, according to information from the DANE Quality of Life Survey itself, there are approximately 252,765 families who cook only with firewood and do not have electricity service because they are in Non-Interconnected Zones (ZNI) or because they are located in isolated areas in the National Interconnected System.

Detailed analysis of the information from Scenario B allows us to identify 3 groups of users of firewood for cooking that must be considered in the strategy planning:

- Group 1: Households that cook only with firewood and have access to the service of electrical energy (911,656).
- Group 2: Households that cook only with firewood and do not have access to electricity service (252.765).335
- Group 3: Households that cook with firewood and diversify with another energy source, mainly LPG (499,809).

This analysis, based on Scenario B, provides us with the elements for defining the strategy that will be presented in section 12.2. Some of these elements are the following:

334 DANE (2019). National Quality of Life Survey 2018.
335 The sum of families in Groups 11 and 2 is 1,164,229, which corresponds to 70% of the total who cook only with firewood.

Machine Translated by Google

Group 1 consists of families who cook exclusively with firewood and have access to electricity. Located within the area served by the National Interconnected System, they are expected to have access to transportation that allows them to receive LPG by land. They will also have access to alternative fuel options such as improved cookstoves, biogas, and electricity, thus providing them with all available energy sources. In summary, this group, due to its location, has access to all the energy options outlined in the firewood replacement strategy.

Group 2 comprises families who rely solely on firewood and currently lack access to electricity. These families reside in non-interconnected zones or areas isolated from the National Interconnected System (SIN). In these areas, access to alternative fuels is limited due to transportation costs and logistical challenges. For example, LPG requires long-distance transport via various transportation systems, increasing the price for end users. Given the current lack of electricity, fixed fuel options are proposed, such as improved cookstoves or biogas. It is worth noting that, in accordance with the Sustainable Development Goals, 100% of the Colombian population is projected to have access to electricity by 2030. However, even if they gain access to this service in the future, these families will continue to face the same difficulties in obtaining technologies that allow the replacement of firewood because they are located in isolated areas, which is why fixed solutions such as Improved Stoves and Biogas would continue to be the preferred options.

Group 3 exhibits an energy consumption pattern in which LPG is used for one or two meals a day, with the remaining meals cooked with firewood. Testimonies gathered at the Regional Workshops indicate a preference for LPG, particularly for the first meal of the day when food needs to be cooked quickly. In general, families try to moderate their LPG use so that the 15 kg cylinder lasts as long as possible considering its cost. Given that the families in this group already have LPG-based cooking equipment, the preferred option is to maintain and increase their consumption of this energy source to encourage a greater substitution of firewood with LPG.

12.2. The Replacement Plan Strategy

The Strategy must outline the actions to be taken in each of the three periods mentioned at the beginning of the chapter (short, medium, and long term), for the three household groups defined in the previous section, and considering the four energy sources or technological alternatives identified as options in the Substitution Plan. In other words, the strategy comprises three main areas:

- Implementation timeframe
- Population group(s) to which the Plan is directed
- Energy substitute(s)

Machine Translated by Google

Regarding these key points, which will determine the strategy to follow, there are some comments:

• Regarding the timeframe, the Plan contemplates an implementation period between 2020 and 2050.
The strategy will outline replacement proposals and estimated costs throughout the period, as indicative, considering that as progress is made through the different periods, there may be replacement options that have not been previously identified.
identified at present that may prove viable later, or the opposite, that is,
that some of the energy sources proposed to date may not be viable later on.

• In the short term horizon (2020-2022), which coincides with the period of the current National Development Plan -PND-, the strategy must propose replacement results for at least 100,000 families as postulated by the PND in the program *Consolidation of the public household service of combustible gas* of the Pact for the quality and efficiency of public services.

The Ministry of Environment, in coordination with environmental authorities, is responsible for establishing the national program for replacing wood-burning stoves with efficient or improved stoves, according to the 2018-2022 National Development Plan. This means that, at least for the short term (2020-2022), there will not be a formal plan to scale up the improved stove solution to an industrial level that would allow for its mass commercialization. However, even without knowing the results of the program managed by the Ministry of Environment and Sustainable Development (MADS), this Strategy for Wood Replacement anticipates a significant contribution from improved stoves in the short term and considers that in the medium and long term, they will be a significant contribution to the Wood Replacement Plan.

In particular, it is considered that Improved Cookstoves have a natural market niche among Group 2 families, where the current lack of electricity and difficulties in transporting LPG suggest fixed solutions in the area. In fact, there are entities like Cornare that will continue their efforts to promote the use of Improved Cookstoves, and, on the other hand, at the regional level, as in the case of Boyacá, there is an initiative for the construction of Improved Cookstoves using coal, which partially mitigates indoor air pollution through flues for evacuating wood combustion fumes. As indicated in section 12.1, although the national goal is to have 100% of the Colombian population with access to electricity by 2030, families in the so-called Group 2 will continue to experience similar isolation conditions to those they currently face, which is why fixed solutions such as Improved Cookstoves and Biogas would remain the preferred options.

• Regarding Biogas, to date there is no National Biogas Plan developed by the
Ministry of Agriculture or another entity that allows for the scaling up of industrial production of the elements and materials for the development of Biogas so that this energy source becomes a massive solution for replacing

Machine Translated by Google

Firewood. In fact, section 12.3 stipulates that the first phase of the Biogas project will emphasize training and the development of Pilot Projects in the short term (2020-2022). It is expected that, in subsequent periods, once the results of the Pilot projects and the training of end users are known, greater coverage can be achieved. For the purposes of designing the strategy, the following biogas solution targets are considered: 1,500 pilot projects in the short term; 15,000 in the medium term; and 100,000 in the long term.

Regarding electricity, section 9 presented a calculation of the costs associated with using this type of energy for cooking. However, despite the cost advantages and the proven efficiency of electricity and new induction cooktops, the power demand of an induction cooktop places demands on the grid from users who use electricity for cooking, typically located at the end of a radial distribution system. In these areas, where electricity is primarily used for lighting and cooling, the new demands for cooking can overload the system, causing outages and service interruptions. Therefore, while not ruling out the use of electricity, a program to replace firewood with electricity requires a careful evaluation of the distribution network to ensure that the radial system can handle the new loads.

While all firewood users experience poverty and significant difficulties in acquiring alternative energy sources, the degree of poverty differs among the groups discussed in this chapter. Group 2 comprises families living farther from urban centers with less access to energy services and high LPG supply costs, whereas families in Group 1 have electricity, and those in Group 3 already have access to LPG. Therefore, families more dependent on firewood for cooking experience higher levels of poverty than those located in less isolated areas with greater access to energy resources to replace firewood.

• Cost analyses of technological replacement options, when health, environmental and social costs are also internalized, showed advantages of electricity and LPG over solutions such as Improved Stoves and Biogas.
However, the strategic approach of the Replacement Plan is to combine all the proposed options in order to achieve an inclusive portfolio that also encourages technological developments in improved stoves and biogas and allows consideration of the cultural characteristics of households, presented in section 5.

Based on the elements of the previous analyses, the substitution strategy is proposed as follows:

Machine Translated by Google

12.2.1. Replacement targets for each phase

Based on data from the 2018 National Quality of Life Survey, three groups of households have been defined, in which different strategies should be used to achieve the replacement of firewood for cooking in Colombian households by 2050.

Table 79. Number of households by group

Group	Number of households	Features
G1	911.656	Firewood, they don't diversify and they have electricity
G2	252.765	They use firewood and currently have NO electricity or LPG.
G3	499,809	Firewood and diversifying with LPG
	1.664.230	

Source: Prepared by the author based on DANE 2019.

Based on these groups and the deadlines defined above for the plan, targets for households with firewood substitution have been defined for each phase, which are presented in Table 80.

The **short-term period of 2020-2022** presents the **greatest challenges,** as it is the **program's initial** phase , during which it is necessary to define the field implementation methods and inter-institutional coordination mechanisms. This first short-term phase aims to achieve **coverage of approximately 10% (160,000)** of the total number of households to be replaced, exceeding the target set by the National Development Plan.

Subsequently, in the **medium-term stage (2023-2030)** which comprises a period of 8 years, the replacement of **650,000 households is proposed,** which corresponds to **39%** of the total.

Finally, for the period **2031-2050** there would be **854,230 households** to be replaced that currently consume firewood, which is equivalent to **51%** of the number of households contemplated in the Plan.

Table 80. Firewood replacement targets by phase and by household group

Group	Phase			Total
	2020-2022	2023-2030	12031-2050	
G1	100.000	300.000	511.656	
G2	30.000	150.000	72.765	
G3	30.000	200.000	269.809	
Total Phase	160.000	650.000	854.230	***1.664.230***

Machine Translated by Google

Source: Own elaboration.

In the period 2020-2022, the reasons that support the replacement strategy with a Coverage Plan for 160,000 families are the following:

a. Priority is given to meeting the minimum goals of the National Development Plan with a coverage of 100,000 households in the short term in Group 1. b. Firewood substitution begins in Groups 2 and 3, with 30,000 families in Group 2 (12% of the total households in the group) and 30,000 families in Group 3 (6% of the total households in the group).

c. After the first stage, with the proposed coverage of 160,000 households, 1,504,230 families remain to be replaced in Phases II and III.

For the period 2023-2030, the plan is to replace 650,000 households, distributed as follows:
manner:

a. In Group 1, which includes households that consume firewood and have access to electricity, the goal is to replace cooking energy in 300,000 households, which represents approximately 37% of the number of households still to be replaced in this group.

b. In Group 2, that is, those households that consume firewood and do not currently have electricity service, 150,000 households will be replaced, approximately 67% of the households to be replaced in this group, which maintains the policy guideline of accelerating their exit from a situation of only firewood to diversify with another energy source.

c. In Group 3, which includes families that consume firewood and diversify with another energy source, usually LPG, a total of 200,000 households are proposed, approximately 43% of the number of households to be replaced in this Group.

As a result of the medium-term stage, 650,000 households are replaced, and 854,230 remain for complete replacement in Phase III or the long-term stage, which covers the period 12031-2050.

12.2.2. Technological alternative by phase and by group

This section proposes the contribution of each technological option by phase, for which the needs of each of the 3 Groups are determined. Table 81 presents the total for each technology and/or energy source to be implemented in each phase of the Plan, where **LPG would be the proposed replacement alternative for approximately 53% of households, improved stoves for 22%, electricity for 18%, and biogas for 7%.**

Table 81. Distribution of proposed replacement technologies or energy sources by phase and household group.

Technology	2020-2022	2023-2030	12031-2050 Total	
GLP	100.000	350.000	436.465	886.465
Improved stoves	29.250	150.000	182.765	362.015

Machine Translated by Google

Electrical energy	29.250	135.000	135.000	299.250
Biogas	1.500	15.000	100.000	116.500
Total Phase	**160.000**	**650.000**	**854.230**	**1.664.230**

Source: Own elaboration.

The allocation criteria for these technologies and/or energy sources for each household group described in Table 79 are as follows:

1. Improved Stoves will be used preferentially in Group 2 when families do not currently have access to electricity and are located in isolated areas, either in ZNI or in the National Interconnected System, for which fixed solutions are required considering that there is no electricity service and that LPG has high transportation costs.

2. For biogas, a gradual evolution over time is considered, starting with user training and the development of pilot projects (presented in section 12.3). This report outlines its contribution as follows:

   a) 1,500 units in the short term, in the period 2020-2022 (pilots).
   b) 15,000 units in the medium term, in the period 2023-2030.
   c) 100,000 units in the long term, in the period 2031-2050.

   It is proposed that biogas solutions be used preferentially in Groups 1 and 2, where there are currently no substitutes for the firewood used for
   cooking.

3. LPG will be used preferentially in Groups 1 and 3. The latter group includes households that consume firewood and diversify with LPG; thus, considering that families have already made the effort to acquire the LPG cooking kit, an additional effort will be made to achieve greater penetration and permanence.
   until the complete and sustained replacement of firewood in this segment.

4. Electricity may be used in population groups and areas where it is determined that the grids can meet the demand resulting from the use of electric stoves. For this purpose, its use is proposed primarily in Group 1. If electricity solutions cannot be developed in some areas, other available energy sources that offer the lowest cost, such as biogas or improved stoves, will be used where conditions warrant. Improved stoves could be a solution when the electrical grids cannot support the additional demand.

Machine Translated by Google

5. As biogas projects, which are used in Group 2, become more established, the Improved Stoves may also be used in Group 1.

Table 82 presents the technological alternatives by Phase, whether short, medium or long term, for each of the 3 household groups in which the analysis has been proposed.

Table 82. Distribution of alternatives by household groups and phases

Group	Technology	2020-2022	2023-2030	12031-2050
G1	Biogas	750	7.500	50.000
	Electrical energy	29.250	135.000	135.000
	Improved Stoves	-	7.500	160.000
	GLP	70.000	150.000	166.656
	Total Group	**100.000**	**300.000**	**511.656**
G2	Pilot biogas	750	7.500	50.000
	Electrical energy	-	-	-
	Improved Stoves	29.250	142.500	22.765
	GLP	-		
	Total Group	**30.000**	**150.000**	**72.765**
G3	Pilot biogas	-	-	-
	Electrical energy	-	-	-
	Improved Stoves	-	-	-
	GLP	30.000	200.000	269.809
	Total Group	**30.000**	**200.000**	**269.809**
Total Phase		***160.000***	***650.000***	***854.230***

Source: Own elaboration.

12.2.3. Goals of the Replacement Plan by Regions

To provide more detail on these targets for the number of households replacing firewood for cooking with other energy sources or technologies, an approximation has been made of the targets set in section 12 at the regional level. The following table presents the grouping of departments considered for each of the proposed regions.

Table 83. Departments of Colombia grouped by Region

Region	Departments
Caribbean	Atlántico, Bolívar, Cesar, Córdoba, La Guajira, Magdalena, Sucre and San Andrés.
Central	Antioquia, Caldas, Caquetá, Huila, Quindío, Risaralda and Tolima.

Machine Translated by Google

Oriental	Boyacá, Cundinamarca, Meta, Norte de Santander, Santander and Bogotá.
Orinoquia–Amazon	Amazonas, Arauca, Casanare, Guainía, Guaviare, Putumayo, Vaupés and Vichada.
Pacific	Cauca, Chocó, Nariño and Valle del Cauca.

Source: Own elaboration based on DANE (2019b)336 .

For the three phases, the households proposed for the replacement of firewood in the previous section were distributed, taking into account the departments in which their replacement could occur in each phase of time and were grouped by regions according to Table 83.

This distribution was based on the technologies identified according to their characteristics. In the first phase, presented in Table 84, the solutions are distributed for the proposed 160,000 households.

- Biogas: In this phase, which corresponds to pilot projects, the solutions will be located in warm, low-lying areas where there is a greater probability of success and where easy access is expected to maintain ongoing monitoring of the pilot projects' progress. In Group 1 (households with access to electricity), the pilot projects will focus on the central region and the eastern plains. In Group 2 (households without access to electricity), the solutions will focus on the departments in the Caribbean region with the highest number of households that use firewood for cooking.

- Electricity: This is allocated to departments in Group 1 with the highest number of households using firewood for cooking, where robust electrical systems are in place. Electricity solutions are not allocated to the Electricaribe group of departments due to the current challenges facing this region.

- Improved stoves: These are allocated in group 2 (households without access to electricity), in the departments with the highest number of households in this category.
- LPG: It is allocated to households in groups 1 and 3; considering that the LPG commercial chain has broad national coverage, the proportional distribution was made between the departments of both groups, without the participation of the ZNI,
  where the costs could be very high.

Table 84. Number of households replacing firewood with technology, group and region Phase I (2020-2022)

	Pilot biogas			Improved stoves			Electrical energy			GLP		
Region	**G1**	**G2**	**G3**	**G1**	**G2**	**G3**	**G1**	**G2**	**G3**	**G1**	**G2**	**G3**
Caribbean	- 750			- 24.030					- 25.748			3.584
Central	188					- 12.796			- 15.592			9.660

336 DANE (2019b). Press release National Quality of Life Survey (ECV) 2018.
Retrieved from: https://www.dane.gov.co/files/investigaciones/condiciones_vida/calidad_vida/CP_ECV_2018.pdf

Machine Translated by Google

Oriental	374						7.829		- 13.179			8.230
Orinoquia–Amazonia	188									1.198		548
Pacific					5.220		8.625		- 14.283			7.978
	750 750		-	**- 29.250**		**- 29.250**		-	**- 70.000**		-	**30.000**
												160.000

Source: Own elaboration.

The second phase assumes that the biogas pilot tests have concluded and a satisfactory solution has been found for Electricaribe's problems. Table 85 presents the distribution of households.

- Group 1: Biogas efforts continue to be distributed among departments in low-lying areas, including those from the 2020-2022 period, to maintain continuity with the initial effort. The number of improved cookstove solutions (7,500) is distributed among the departments with the greatest need for replacement, and electricity solutions are allocated to departments with the most robust electrical systems, including those served by Electricaribe, for which it is assumed that the system has overcome its difficulties and regained robustness during this period. LPG solutions are distributed proportionally to the remaining number of households to be served.

  considering the criterion expressed above on the efficiency of the LPG commercial chain in all departments, except those of the ZNI.
- Group 2: The sum of the improved stove and biogas solutions was distributed proportionally among the number of households in Group 2 and then, according to the contribution of each energy source to the total, the distribution was made between improved stoves and biogas.

- LPG: LPG in group 3 is distributed proportionally to the balance of households to be served, considering the criterion expressed above on the efficiency of the LPG commercial chain in all departments, except those of the ZNI.

Table 85. Number of households replacing firewood with technology, group and region Phase I (2023-2030)

	Biogas			Improved stoves			Electrical energy			GLP		
Region	**G1**	**G2**	**G3**	**G1**	**G2**	**G3**	**G1**	**G2 G3**		**G1**	**G2**	**G3**
Caribbean	3.500	3.826		3.000	72.680		54.087			53.918		23.623
Central	1.000	861		3.000	16.377		30.681			32.839		63.682
Oriental	1.500	611			11.638		27.692			27.700		54.273
Orinoquia–Amazonia	1.000	1.082			20.540					2.931		5.823
Pacific	500 1.120			1.500	21.265		22.540			32.612		52.599
	7.500	7.500		7.500 142.500		- 135.000			- 150.000		- 200.000	
												650.000

Source: Own elaboration.

Machine Translated by Google

In the third phase

- Biogas: Biogas solutions are distributed proportionally between the two groups G1 and G2. In the long term, it is considered that the solutions of Biogas is already in the commercial phase, with distribution channels throughout the country.
- LPG: A distribution mechanism is applied proportionally to the number of households in the group that have not had firewood replacement from group 1 and group 3.
- Improved cookstoves and electricity: To allocate electricity and improved cookstove solutions in Group 1, the sum of both solutions is distributed proportionally among the number of households still to be served. Then, based on each energy source's contribution to the total, the distribution is made between improved cookstoves and electricity. Improved cookstove solutions cover the remaining households in Group 2.

Table 86. Number of households replacing firewood with technology, group and region Phase I (12031-2050)

	Biogas			Improved stoves			Electrical energy			GLP		
Region	**G1**	**G2**	**G3**	**G1**	**G2**	**G3**	**G1**	**G2**	**G3**	**G1**	**G2**	**G3**
Caribbean	17.387	25.503	-	52.537	11.625	-	48.107	-	-	59.904	-	31.867
Central	10.591	5.742	-	32.000	2.604	-	29.298	-	-	36.486	-	85.911
Oriental	8.932	4.085	-	26.991	1.858	-	24.715	-	-	30.774	-	73.216
Orinoquia–Amazonia	2.572	7.208	-	16.694	3.281	-	3.783	-	-	3.257	-	7.853
Pacific	10.518	7.462	-	31.778	3.397	-	29.097	-	-	36.235	-	70.961
	50.000	50.000	-	160.000	22.765	-	135.000	-	-	166.656	-	269.809
												854.229

Source: Own elaboration.

12.3. Proposal for a National Biogas Program in Colombia337

In order to formulate and implement a firewood replacement program, it is very important to understand current cooking processes and the characteristics of their use, which

337 This section was authored by José María Rincón Martínez and Diana Marcela Durán Hernández, from the Industrial Development Center-Tecsol. Bogotá- Colombia

Machine Translated by Google

These were mentioned in the first report of this consultancy, which needs to be highlighted again in order to assess the appropriate replacement.

Firewood and forest residues consist of branches, bark, shavings, sawdust, leaves, stumps and roots, which have been for centuries the most important energy source for humanity for cooking food, heating homes, and applications in small industries, especially in the food industry338 .

For good combustion, it is necessary to standardize both the stove and the fuel; therefore, it is useless to have the best stove if the fuel is not suitable: for example, you can have damp wood or wood with variable ash or moisture content.

Biogas is a homogeneous fuel that, at present, internationally, is the most suitable fuel for isolated rural areas and marginal urban areas, which is why several developing countries in Latin America, Africa and Asia, in collaboration with other countries, especially European ones or through South-South cooperation, have established National Biogas Programs (NBPs).

The following are some guidelines for establishing a National Program of Biogas in Colombia (PNBC)

12.3.1. General Objective

The overall objective of the PNBC is to establish a biogas production program that serves as a sustainable alternative to firewood, helps fertilize the soil and mitigate greenhouse gases.

12.3.2. Specific objectives

1. Organize an inter-institutional and interdisciplinary group that promotes the use of biogas as a substitute for firewood, which does not cause environmental problems and improves the quality of life of the inhabitants of rural areas.
2. Design and build pilot plants to carry out demonstration tests of biogas production and use as a substitute for firewood in its different uses at the rural level and evaluate the results according to the thermal floor: hot, temperate and cold.
3. Install demonstration plants and establish biogas networks in marginal rural settlement areas.

The program will be developed in stages, which will be carried out consecutively and independently.

338 DCC Concha, María Cecilia; Pabón, Giovanni; Cerón Viviana. -- Bogotá and 2015. Ministry of Environment and Sustainable Development, “Guidelines for a National Program of Efficient Wood-Cooking Stoves,” 2015.

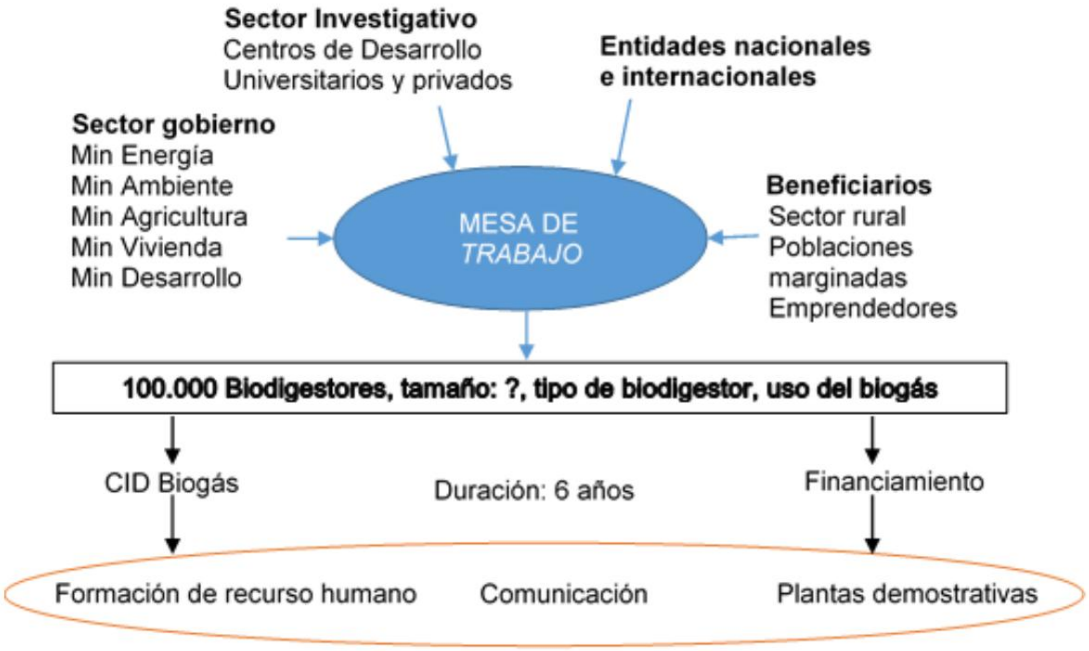
Machine Translated by Google

- Stage 1 or human resource training and evaluation of the potential use of biogas. Design, construction, evaluation of raw materials and selection of technology through pilot tests in different thermal zones and evaluation of the potential of families for the installation of anaerobic digestion plants.

  Duration: 2 years
- Stage 2 or transfer stage. Dissemination of the technology regarding its socioeconomic benefits and demonstration trials in different zones and thermal zones of the country. Time: Permanently for the duration of the program.
- Stage 3 or commercial scaling. Design and construction of modular plants for different users, including small towns, with biogas distribution networks.

Figure 15 illustrates the participating entities from various sectors, including government, research and development organizations, technology financing and transfer providers, and beneficiaries, including entrepreneurs. These participants will form an executive committee; the research and development sector, in conjunction with the financing entities, will be responsible for carrying out the activities, from bench tests to commercial plants in marginalized urban areas.

Figure 15. Organizational scheme for the implementation of the National Biogas Plan in Colombia, PNBC



Source: Prepared by the authors of this section

Machine Translated by Google

Finally, the activities by objective, which are linked to each stage, are described below.

- For objective 1: To organize an inter-institutional and interdisciplinary group to promote the use of biogas as a substitute for firewood that does not cause environmental problems and improves the quality of life of rural residents. This is a cross-cutting objective involving various interested ministries such as the Ministry of Energy, the Ministry of Environment and Sustainable Development (MADS), the Ministry of Agriculture and Rural Development (MADR), the Ministry of Housing and Urban Development (MINVIVIENDA), the public and private research sector, as well as potential investors as special guests. The specific structure may vary depending on the project's development stage. The government sector will obviously be responsible for the initial organization, while the industrial sector will have greater influence during the scaling-up phase. The program is estimated to take six years to develop.

- For objective 2: Design and build pilot plants for the production of biogas as a substitute for firewood in its different uses at the rural level and evaluate the results according to the thermal floor: hot, temperate and cold: the following activities must be carried out, which can be completed in a reasonable time of 2 years.
  For this 2-year period, it is considered that it should start in a warm climate, since under these high temperature conditions the biogas production rate is practically double that of the cold climate, so the residence time is shorter and the plant size is also smaller.

  or Select a reliable technology that has a minimum lifespan of 15 years.

  or Build demonstration pilots that serve to show the environmental benefits of cooking food in different thermal zones.
  or Establish dissemination systems in the area of influence.
  o Socioeconomic evaluation of pilot results and technology selection.

  or Conduct technology outreach courses and staff training of the areas for the construction of biogas plants.
  or Establish maintenance and learning processes for families that have installed biogas plants

- For objective 3: Design and construction of modular plants for different users, including small villages, with biogas distribution networks. It is estimated that this objective can be achieved in 4 years, for a minimum coverage of 100,000 rural families, including 2 marginalized communities where biogas networks will be built. In reality, the main limitation for this objective, which largely corresponds to stage 3 or scaling-up, is the financial aspect and the financial organization for the projects in the villages. For this objective, the business model is fundamental to the project's progress. The business model could consider the sale of low-carbon emissions generated by the program's savings, for example, to Ecopetrol, oil production companies, or coal-fired power plants.

Machine Translated by Google

Some specific activities are listed below:

- or Selection of technology according to the thermal floor, raw materials and population size.
- o Design the biogas distribution network.
- or Install a minimum of 2 demonstration plants on each of the thermal floors selected.
- or Conduct an evaluation of the tests performed and establish a plant construction program, in accordance with the evaluation performed.
- or Develop an outreach program

The estimated duration of the program is 10 years.

Machine Translated by Google

13. Estimated Costs for Implementing the Plan

Once the strategy for technologies and target households for each of the Progressive Firewood Substitution Plans has been established, the costs of implementing this plan are identified, not only in terms of the technologies implemented, but also considering the subsidies that would be required for its operation, and the other components that will allow the Plan to achieve its objectives, reach all the households that have been proposed and remain over time.

13.1. Costs of technologies

Section 9.1.1 presents the investment costs required for each of the four technologies proposed to implement the Progressive Wood Replacement Plan.

In order to know the total costs of the implementation of the program, these values are taken by technology and applied to the total number of beneficiaries that the program would have per technology, according to section 12, specifically with what is presented in Table 81 with respect to the technologies that are recommended to be implemented in each phase (short term, medium term and long term) of the development of the Plan.

In the first phase, corresponding to the **period 2020-2022,** according to the strategy presented above, **cooking with firewood will be replaced in 160,000 homes,**
using the four available energy alternatives. Section 9.1.1 shows the unit costs of cooking equipment for the different technologies. This section aggregates the costs according to the coverage of each technology in the different phases of the Replacement Project. Thus, the cost for the technologies that the Plan would provide to households for this first phase is approximately $154,625 million pesos.

Table 87. Cost of replacement technologies in the 2020-2022 phase (pesos)

Technology	Cost per kit	Number of	Total Cost
Biogas	1.500.000	**households:**	1,5002.250.000.000
Improved stoves	2.000.000	29.250	58.500.000.000
Electrical energy	1.500.000	29.250	43.875.000.000
GLP	500.000	100.000	50.000.000.000
			154.625.000.000

Source: Own elaboration.

Machine Translated by Google

In the second phase, corresponding to the **period 2023-2030,** according to the strategy presented in section 12, cooking with firewood will be replaced in **650,000 households,** using the four technologies presented in the following table. The cost of the technologies that the Plan would provide to households for this second phase is approximately **$700 billion pesos, based on the unit values in section 9.1.1.**

Table 88. Cost of replacement technologies in the 2023-2030 phase (pesos)

Technology	Cost per kit	Number of	Total Cost
Biogas	1.500.000	**households:** 15,000	22.500.000.000
Improved stoves	2.000.000	150.000	300.000.000.000
Electrical energy	1.500.000	135.000	202.500.000.000
GLP	500.000	350.000	175.000.000.000
			700.000.000.000

Source: Own elaboration.

For the third phase, which will take place between 2031 and 2050, the plan is to replace firewood for cooking in **854,230 households,** specifying the technologies to be used according to what is presented in Table 89. The cost for the technologies that the Plan would deliver to households, for this phase, is approximately **$936,262.5 million pesos.**

Table 89. Cost of replacement technologies in phase 12031-2050 (pesos)

Technology	Cost per kit	Number of	Total Cost
Biogas	1.500.000	**households** 100,000	150,000,000,000
Improved stoves	2.000.000	182.765	365.530.000.000
Electrical energy	1.500.000	135.000	202.500.000.000
GLP	500.000	436.465	218.232.500.000
			936.262.500.000

Source: Own elaboration.

In total, in terms of the technologies that will replace traditional cooking with firewood, **the strategy will have an approximate cost for its three phases of 1,790,887.5 million pesos or 1,791 trillion pesos** in its thirty years of development.

Machine Translated by Google

Table 90. Cost of replacement technologies in the Progressive Firewood Replacement Plan by phase (millions of pesos)

Technology	2020 - 2022	2023 - 2030	12031 - 2050	Total Cost
Biogas	2.250,0	22.500,0	150.000,0	174.750,0 365.530,0
Improved stoves	58.500,0	300.000,0	724.030,0	202.500,0 448.875,0
Electrical energy	43.875,0	202.500,0	218.232,5	443.232,5 936.262,5
GLP	50.000,0	175.000,0	**1.790.887,5**	
	154.625,0	700.000,0		

Source: Own elaboration.

13.2. Costs of energy subsidies

In addition to the costs of the technologies to be implemented to progressively replace firewood, the Plan must have resources for subsidies provided to households for the monthly costs of some of the energy sources proposed for the replacement of firewood.
Specifically, it has been proposed to include a subsidy for LPG and electricity in homes where the traditional wood-burning stove is replaced by LPG or electric stoves, according to the characteristics of the households' access to energy.

Of the four technological options to progressively replace cooking with firewood, biogas and improved stoves do not have established economic costs in the consumption of energy for cooking (firewood or biomass), although in section 9.1.2 some cost components are presented that households generally do not value economically.

However, alternatives to traditional wood-burning stoves, such as LPG or electric stoves, do involve energy costs. As presented in that section, the monthly cost of LPG for cooking in an average household is approximately $60,000 pesos, and the cost of electricity is $66,000 pesos for conventional stoves. Therefore, it is proposed to incentivize their use with a 50% subsidy of the energy cost required for their operation each month, which would be $33,000 pesos for electricity and $30,000 pesos for LPG, as shown in Table 91.

Table 91. Suggested monthly subsidies for the operation of the proposed energy and/or technological alternatives (pesos)

Technology	Monthly Subsidy
Biogas	
Improved stoves	
Electrical energy	33.000
GLP	30.000

Source: Own elaboration.

Machine Translated by Google

Taking into account the number of households where technologies using these two energy sources would be implemented to replace cooking with firewood in the three phases of the Plan's development, *Table 92, "Subsidies Required per Phase, Option A (Millions of Pesos),"* presents the **value of the subsidies for electricity and LPG consumption proposed to be provided to households benefiting from the Plan with these solutions to incentivize their transition to cleaner energy sources.** It is important to note that **the remaining 50% of energy consumption will be borne by the users, for a total cost equal to the cost of the subsidies.**

In total, the energy subsidy costs for this thirty-year Plan amount to $10,711,701 million pesos ($10.71 trillion pesos). This figure is derived from considering the annual subsidy for these technologies until 2050, taking into account the number of units replaced in each phase.

Table 92. Subsidies required per phase option A (millions of pesos)

Technology	2020 - 2022	2023 - 2030	12031 - 2050	Total Cost
Electrical energy	34.749,0	520.344,0	2.370.060,0	2.925.153,0
GLP	108.000,0	1.296.000,0	6.382.548,0	7.786.548,0
	142.749,0	**1.816.344,0**	**8.752.608,0**	**10.711.701,0**

Source: Own elaboration.

However, in order to propose an alternative that reduces these costs, an "Option B" is presented, in which this subsidy is applied for only 10 years to all households that replace their cooking methods with these technologies. To ensure the effective implementation of the subsidy, it is necessary to design mechanisms to target the subsidies to the intended population of the Replacement Program. This timeframe is considered sufficient to change household cooking habits and ensure the sustainability of the firewood replacement, even after the subsidy ends.

Under this option, the total energy subsidy costs for this thirty-year Plan would be $4,376,304 million pesos ($4.37 trillion pesos), with a reduction
of approximately 59% with respect to option A.

Table 93. Subsidies required per phase option B (millions of pesos)

Technology	2020 - 2022	2023 - 2030	12031 - 2050	Total Cost
Electrical energy	34.749,0	508.761,0	641.520,0	1.185.030,0
GLP	108.000,0	1.260.000,0	1.823.274,0	3.191.274,0
	142.749,0	**1.768.761,0**	**2.464.794,0**	**4.376.304,0**

Source: Own elaboration.

Machine Translated by Google

13.3. Other costs of implementing the Plan

In addition to acquiring the technologies that will be provided to households where firewood is being replaced, and subsidizing the energy sources that require it, implementing a plan of this magnitude requires assuming costs for other items necessary for its effective implementation and long-term sustainability. Some of the additional costs that must be considered are:

- Dissemination of the Plan.
- Logistics for missions and events.
- Coordination of the implementation of the Plan.
- Advisors for the implementation of the Plan.
- Program for monitoring, reporting and verification of goals.

In order to carry out these activities and others presented in section 11.4, it is recommended that **an Implementing Agency for the Plan be contracted,** with a suitable team and the capacity to coordinate and implement this Plan. Some of the recommended aspects regarding financing costs are:

• Human Resources: The main variable to consider in the program's operation is the human resources needed to carry out the process described in the previous chapter. These resources depend on the program's defined scope and the expected timeframe for achieving its objectives. This proposal considers the following resources: a program manager, two consultants (an economist and an engineer), and a secretary for administrative and support tasks.

• Administrative expenses: Administrative expenses include aspects such as the rental of an office or workspace for the required staff, the necessary public services and expenses for office furniture and equipment.

• Operating expenses: Operating expenses include, on the one hand, the communication component related to the program's website and other outreach activities. On the other hand, they include the transportation needs of the consultants to the regions where the firewood substitution will take place in each phase of the Plan.

Taking these factors into account, the approximate cost of these activities is on average $100 million pesos per month, which would be equivalent to a total of $37.2 billion pesos in the development of the three phases of the plan until 2050, as shown in Table 94.

Table 94. Other Plan Implementation Costs (millions of pesos)

Technology 2020 - 2022	2023 - 2030	12031 - 2050	Cost
Months	36	96	240 **Total**
Monthly Cost	100	100	100
	3.600	**9.600**	**24.000 37.200**

Source: Own elaboration.

Machine Translated by Google

13.4. Total cost of implementing the Progressive Firewood Replacement Plan

Option A

Taking into account the three types of costs previously analyzed for the implementation of the Progressive Firewood Replacement Plan, the costs for each phase of the Plan's implementation are obtained. For the first phase, which refers to the short-term goals for the period 2020–2022, the Plan has estimated costs of $300,974 million pesos, taking into account energy subsidies for all the years of the Plan's duration.

Table 95. Total cost in pesos of implementation of phase I (2020-2022)

Technology and/or energy	Number of Households	Technology Cost	Subsidy Energetic	Implementation
Pilot biogas	1,500	2.250.000.000		
Improved stoves	29.250	58.500.000.000		
Electrical energy	29.250	43.875.000.000	34.749.000.000	
GLP	100.000	50.000.000.000	108.000.000.000	
Total Phase	**160.000**	154.625.000.000	142.749.000.000	3.600.000.000
				300.974.000.000

Source: Own elaboration.

The total costs of the second phase, referring to the medium-term goals corresponding to the period 2023-2030 are approximately $2,525,944.0 million pesos.

Table 96. Total cost in pesos of implementation of phase II (2023-2030)

Technology and/or energy	Number of Households	Technology Cost	Subsidy Energetic	Implementation
Pilot biogas	15.000	22.500.000.000		
Improved stoves	150.000	300.000.000.000		
Electrical energy	135.000	202.500.000.000	520.344.000.000	
GLP	350.000	175.000.000.000	1.296.000.000.000	
Total Phase	**650.000**	700.000.000.000	1.816.344.000.000	9.600.000.000
				2.525.944.000.000

Source: Own elaboration.

Finally, the third phase has estimated total costs of $9,712,870.5 million pesos in the period corresponding to 2031-2050, which corresponds to the long term.

Table 97. Total cost in pesos of implementation of phase III (2030-2050)

Technology and/or energy	Number of Households	Technology Cost	Subsidy Energetic	Implementation
Pilot biogas	100.000	150.000.000.000		

Machine Translated by Google

Improved stoves	182.765	365.530.000.000		
Electrical energy	135.000	202.500.000.000	2.370.060.000.000	
GLP	436.465	218.232.500.000	6.382.548.000.000	
Total Phase	**854.230**	936.262.500.000	8.752.608.000.000	24.000.000.000
				9.712.870.500.000

Source: Own elaboration.

Therefore, the total cost of the Progressive Wood Replacement Plan for the thirty-year period is estimated at $12,539,788.5 million pesos ($12.54 trillion pesos).

Of this total cost, the costs for **acquiring the technological alternatives**
These correspond to **14.3%,** the **costs for subsidies throughout the term of the Plan to households in which solutions based on LPG or electricity are implemented, to 85.4%, and the implementation of the program to 0.3%.**

Table 98. Estimated total cost for the implementation of the plan (millions of pesos)

Phase	I (2020 - 2022)	II (2023 - 2030)	III (2031 - 2050)	**Total Plan**
Total Cost	300.974,0	2.525.944,0	9.712.870,5	**12.539.788,5**

Source: Own elaboration.

Option B

On the other hand, in "option B" in which subsidies would only be given for 10 years, the estimated costs of $300,974 million pesos are maintained for the first phase of the Plan, taking into account that energy subsidies are delivered during the short term period.

Table 99. Total implementation cost of phase I (2020-2022) Option B

Technology and/ or energy	**Number of**	**Technology Cost**	**Subsidy Energetic**	**Implementation**
Pilot biogas	**Households**	1,5002.250.000.000		
Improved stoves	29.250	58.500.000.000		
Electrical energy	29.250	43.875.000.000	34.749.000.000	
GLP	100.000	50.000.000.000	108.000.000.000	
Total Phase	**160.000**	154.625.000.000	142.749.000.000	3.600.000.000
				300.974.000.000

Source: Own elaboration.

The total costs of the second phase, referring to the medium-term goals corresponding to the period 2023-2030, are approximately $2,478,361 million pesos.

Machine Translated by Google

Table 100. Total implementation cost of phase II (2023-2030) Option B

Technology and/or energy	Number of Households	Technology Cost	Subsidy Energetic	Implementation
Pilot biogas	15.000	22.500.000.000		
Improved stoves	150.000	300.000.000.000		
Electrical energy	135.000	202.500.000.000	508.761.000.000	
GLP	350.000	175.000.000.000	1.260.000.000.000	
Total Phase	**650.000**	700.000.000.000	1.768.761.000.000	9.600.000.000
				2.478.361.000.000

Source: Own elaboration.

Finally, the third phase has estimated total costs of $3,425,057.5 million pesos in the period corresponding to 2031-2050, which corresponds to the long term.

Table 101. Total implementation cost of phase III (2031-2050) Option B

Technology and/or energy	Number of	Technology Cost	Subsidy Energetic	Implementation
Pilot biogas	**Households:** 100.000	150.000.000.000		
Improved stoves	182.765	365.530.000.000		
Electrical energy	135.000	202.500.000.000	641.520.000.000	
GLP	436.465	218.232.500.000	1.823.274.000.000	
Total Phase	**854.230**	936.262.500.000	2.464.794.000.000	24.000.000.000
				3.425.056.500.000

Source: Own elaboration.

Therefore, considering the subsidy costs in **Option B , the total cost of the Progressive Firewood Replacement Plan for the thirty-year period is estimated at $6,204,391.5 million pesos ($6.2 trillion pesos).** Of this total cost, the costs for **acquiring the technological alternatives correspond to 28.9%,** the costs for **subsidies over 10 years for households implementing LPG or electricity-based solutions to 70.5%,** and **the implementation of the program to 0.6%.**

Machine Translated by Google

Table 102. Estimated total cost for the implementation of the plan (millions of pesos)

Option B

Phase	I (2020 - 2022)	II (2023 - 2030)	III (2030 - 2050)	**Total Plan**	
Total Cost	300.974,0	2.478.361,0	3.425.056,5	**6.204.391,5**	

Source: Own elaboration.

Machine Translated by Google

14. Available Sources of Funding

The development of the *Progressive Firewood Substitution Plan in Colombia* requires resources to support its implementation throughout the country, both in Non-Interconnected Zones (ZNI) and in rural areas within the national grid. The following describes the general requirements for establishing the Program in Colombia and the financial mechanisms currently available for its funding.

14.1. General description of the Plan costs

The variables considered in the cost proposal for the operation of the wood replacement program in Colombia are presented in section 13. According to this operation, a cash flow is proposed in which the Wood Replacement Program will have a focus in ZNI with efficient stoves and BIOGAS, another in SIN with LPG and other technologies already mentioned.

At the end of the first three years, when approximately 160,000 solutions would be implemented as part of the program, a larger Phase 2 would begin, with 650,000 solutions, and approximately 10 years after the start of the Plan's implementation, Phase 3 would begin, lasting 20 years, during which approximately 854,230 solutions would be implemented.

The results of the Plan's cost analysis indicate an implementation cost of 100,324.7 million Colombian pesos per year in the first phase, from 2020 to 2022, considering the costs of technologies, subsidies, and Plan operation. However, in the second phase, from 2023 to 2030, the annual implementation cost would be approximately 309,795.1 million pesos, considering subsidies per household for up to 10 years from the implementation of the technology for firewood replacement (in the cases of LPG and electricity). Finally, in the third phase of the plan, from 2030 to 2050, the annual cost would be approximately 171,252.8 million pesos, taking into account that the household

14.2. Description of available mechanisms

The proposed funding approach stems from the numerous initiatives aimed at improving cooking processes, which currently result in isolated efforts lacking the benefits of economies of scale, lessons learned, and ease of replication. The following alternative funding mechanisms are initially being considered for the program's operation.

- Support fund for non-interconnected areas FAZNI, administered by the Ministry of Mines and Energy.

- Non-Conventional Energy and Efficient Energy Management Fund -FENOGE-, created by Article 10 of Law 1715 of 2014.

- Programs for the Benefit of Communities, PBC - Agreement 02 of 2017 of the Agency National Hydrocarbons Agency – ANH.

Machine Translated by Google

- International cooperation.

- General Royalties System –SGR.

- National carbon tax and the request for non-taxation due to carbon neutrality, Decree 926 of 2017 of the Ministry of Environment.

- Colombia in Peace Fund - **Decree 691 of April 27, 2017** and **CONPES Document 3850** **Colombia in Peace Fund**

- Other private initiatives.

14.2.1. Fund for Non-Conventional Energies and Efficient Energy Management - FENOGEE-,

Among the functions of FENOGE are:

- To finance plans, projects and/or *programs that promote, stimulate or encourage the development and use of non-conventional renewable energy sources (NCRE), mainly those of a renewable nature,* for the diversification of the full and timely energy supply, the competitiveness of the Colombian economy, the protection of the environment, the efficient use of energy and the preservation and conservation of renewable natural resources.

- To finance *Efficient Energy Management, improving energy efficiency* and developing demand response mechanisms.

- To promote and maximize energy efficiency, through good practices and technological conversion or *fuel substitution in any activity of production, transformation, transport, distribution and consumption* of the different forms of energy, respecting current regulations on the environment and renewable natural resources.

- To encourage greater penetration of non-conventional renewable energy sources, as well as *efficient energy management processes in non-international zones.*

- To achieve *interaction among various actors, including the public and private sectors, academia, trade associations, local banks, international banks, multilateral banks, and international organizations, in terms of financing, cooperation, technical assistance, and resource contributions, among others.*

- To finance other purposes such as energy audits, studies, administration, oversight, information systems, monitoring systems, research projects, technology transfer, training, energy efficiency centers, business schemes for production processes, heat utilization in the mining and energy industry, and final disposal of replaced equipment.

- Monitor the plans, programs and projects implemented.

Machine Translated by Google

14.2.2. Programs for the Benefit of Communities, PBC - Agreement 02 of 2017 of the National Hydrocarbons Agency - ANH

Agreement 02 of 2017 of the National Hydrocarbons Agency - ANH in its articles 78, 79 and 80 contemplates:

Article 78. Corporate Social Responsibility: In accordance with national and international provisions on the adoption and implementation of corporate policies in this area, Interested Parties, Proponents and Contractors must develop the activities and Operations of Hydrocarbon Exploration and Production under their responsibility, based on ethical values; respect for the State, its workers and contractors; society, *communities and the environment,* and the protection of the diversity and cultural identity of Ethnic Groups, as well as undertake *actions that contribute to the economic and social progress of both, and to sustainable and inclusive development.*

Article 79. Social Investment: In the execution of Contracts for the development of Hydrocarbon Exploration, Evaluation, Development, and Production activities, *works, projects, and/ or activities must be undertaken for the benefit of the communities within the Area of Influence* of the Operations, in Onshore Areas, or within the Area of Interest resulting from social baseline studies, in the case of Offshore Contracts, where such activities have an impact or effect. The Contractor must *allocate resources for the development of these projects to contribute to improving the quality of life and living conditions of the inhabitants.* In compliance with this commitment, Contractors must design and develop *Community Benefit Programs (CBPs)* established or settled within the Area of Influence of the Operations under their responsibility, or within the Area of Interest, in the case of Offshore Contracts. These programs are intended to promote sustainable development, strengthen the social, cultural, and economic environment, and improve living conditions, in accordance with the following Article. It is the responsibility of the ANH to assist the Contractors in the procedures for presenting and agreeing on such Programs with the authorities that represent said communities or with representatives of the same.

Article 80. Community Benefit Programs (CBPs): These programs must consist of the development of infrastructure, health and basic sanitation, and/or education initiatives, as well as the development of productive projects. *The investment in such programs must correspond to an amount equivalent to one percent (1%) of the total value of the Exploratory Program,* including the Minimum and Additional Programs; of any subsequent Exploratory and Evaluation Programs; and to one percent (1%) of the amount of the Annual Operations Program for all Commercial Fields in the Area or Areas in Production, during the corresponding period. These projects and activities must be different from those that the Contractor is obligated to undertake in compliance with Licenses and Environmental Management Plans, or in the execution of Management Measures agreed upon in Prior Consultation Procedures to prevent, correct, mitigate, and/or compensate for impacts arising from the execution of the Contract on communities and ethnic groups, all subject to higher-level regulations.

The benefits outlined in Article 80 have already led to the implementation of gas expansion programs. The Entretejiendo Foundation and its gas program will be carried out in partnership with Canacol Energy and Surtigas. The goal is to benefit more than 7,500 people.

Machine Translated by Google

Natural gas service, through distribution networks, connections, and internal natural gas installations, will be provided to seven rural communities in the municipality of Sahagún. This project, in addition to improving the health and quality of life of the beneficiaries, will contribute positively to the ecosystems by seeking to eradicate deforestation caused by the frequent use of firewood for cooking. It is therefore a proven source of funding with concrete results.

14.2.3. Multilateral or bilateral cooperation

International cooperation resources are another mechanism for financing the Program's operation. In this case, economic cooperation entities from different countries should be considered, along with their interest in funding programs that contribute to environmental and productivity objectives, such as energy management in these types of initiatives.

Funding from international cooperation could be supported based on the Program's link to actions against climate change due to the reduction of emissions produced by renewable energy and energy efficiency projects.
In this case, consideration should be given to the impact that the cooperation entity would have on the actions carried out by the Program coordination, and the time horizon of the funding that said entity would be willing to provide to the Program.

Among the financial mechanisms for carrying out projects related to renewable energy and energy efficiency, various sources can be found from multilateral or bilateral cooperation entities. One example is the Swiss Embassy's Economic Cooperation and Development (SECO) office, which has made a fundamental financial contribution to several environmental initiatives in the country.

The following are some agencies that have contributed to renewable energy, energy efficiency and/or planning projects in the country.

- IDB: The Inter-American Development Bank is a financial organization whose objective is to achieve development in a sustainable and climate-friendly manner.
  Their priority interests include three development challenges – social inclusion and equality, productivity and innovation, and economic integration – and three cross-cutting themes – Gender equality and diversity, climate change, and environmental sustainability. Among the activities that have received financing for the public sector, the IDB has three loan categories: investment loans, loans to support policy reforms, and special development financing.
- IFC: The International Finance Corporation (IFC), a member of the World Bank Group, is the leading international development institution focused exclusively on the private sector in developing countries. The IFC offers a variety of loans to the private sector, tailored to the specific sector and country. This could be a source of financing for private sector-led projects.

Machine Translated by Google

• CAF: The Development Bank of Latin America promotes a sustainable development model through credit operations, non-reimbursable resources, and support in the technical and financial structuring of public and private sector projects in Latin America. Loans are CAF's main operating modality and can be short-term (1 year), medium-term (1 to 5 years), and long-term (more than 5 years). These loans can include different types: trade loans (pre-shipment and post-shipment) and working capital loans, project loans, and limited-guarantee loans.

USAID: The United States Agency for International Development supports the Colombian government's efforts to reduce threats to biodiversity and ecosystems and strengthen environmental governance to conserve protected areas and ethnic territories. USAID programs strengthen the capacity of the government, community organizations, and civil society to manage, protect, and ensure the sustainability of livelihoods based on natural resources; mitigate, adapt to, and/or reduce vulnerability to the long-term impacts of climate change; and promote the use of clean and renewable energy, as well as low-carbon economic growth.

• Other agencies to which projects under the Replacement Plan could be submitted
Key elements for obtaining bilateral cooperation resources are:

• IKI *(International Climate Initiative of the Federal Ministry for the Environment, Nature Conservation and Nuclear Safety):* This German initiative has funded climate and biodiversity-related projects in developing, newly industrialized, and transition countries. Funding focuses on projects related to: mitigating greenhouse gas emissions, adapting to the impacts of climate change, conserving natural carbon sinks, with a focus on reducing emissions from deforestation (REDD+), and conserving biodiversity.

• DANIDA (Danish International Development Agency): Operated by the Danish government, it supports initiatives to limit greenhouse gas emissions, including the promotion of sustainable energy and, more generally, the organization of green economic growth. As part of this, Denmark supports, among other things, climate-adapted activities in agriculture, limiting deforestation, reforestation, and disaster prevention.

• NORAD (Norwegian Agency for Development Cooperation): The thematic areas of cooperation of this agency that are related to the themes of the Firewood Replacement Plan are the energy area, which considers that access to affordable and reliable energy services is crucial for economic and social development, and the climate, environment and natural resources area, which starts from the premise that global climate change and the loss of biodiversity are the greatest environmental threats facing the world.

Machine Translated by Google

• GIZ – Germany: Commissioned by the German Federal Ministry for the Environment, Nature Conservation and Nuclear Safety (BMZ), GIZ implements a number of projects and programs in Colombia funded by the International Climate Initiative (IKI). Since October 2016, GIZ has supported the Colombian government's peacebuilding efforts, also commissioned by the European Union, which contributes funds to the peacebuilding program financed by the BMZ.

However, the number of international cooperation agencies is quite large, and other alternatives could emerge in this field. Depending on the specific entity and/or program in which the project participates, the resources allocated to projects may be for financing or co-financing. Therefore, these financing mechanisms can be considered for projects that may or may not generate financial returns, as long as they are aligned with the goals of the funding agency.

14.3. Funding proposal

Based on these sources of funding, some are established that would form the basis of the program and others that would strengthen the actions undertaken within it.

Table 103. Funding Proposal

<table>
<tr><td>Non-Conventional Energy Fund and Management Energy Efficient -FENOGE-</td><td rowspan="2">Bases of the Replacement Program
Progressive Firewood In Colombia</td><td rowspan="8">Office of Implementation of the Program Substitution Progressive of Firewood</td></tr>
<tr><td>Support Fund for Non-Interconnected Areas (FAZNI)</td></tr>
<tr><td>Programs that Benefit Communities,
PBC - Agreement 02 of 2017 of the Agency National Hydrocarbons Agency – ANH</td><td rowspan="6">Other Solutions Implemented within the framework of Program</td></tr>
<tr><td>International cooperation</td></tr>
<tr><td>General Royalties System - SGR</td></tr>
<tr><td>National carbon tax and the request for non-taxation due to carbon neutrality, Decree 926 of 2017 of the Ministry of Environment</td></tr>
<tr><td>Colombia in Peace Fund - Decree 691 of the 27th April 2017 and CONPES Document 3850 Colombia in Peace Fund</td></tr>
<tr><td>Other private initiatives</td></tr>
</table>

Machine Translated by Google

1.1.1. Base program

According to the previous analysis, FENOGE is proposed as the financial basis of the program, in conjunction with resources from the *Programs for the Benefit of Communities, PBC,*
This is contemplated in agreement 02 of 2017 by the ANH. Thus, the Mining and Energy Planning Unit (UPME), together with the National Hydrocarbons Agency (ANH), would present a financing proposal to FENOGE for the funding of the Implementing Agency of the Firewood Substitution Plan in Colombia.

- PHOENIX (Associated with the Sun, a phoenix gains new life by rising from the ashes of its predecessor).

This implementing agency would lead a one-stop shop, which, as envisioned by FENOGE, will achieve *the interaction of various actors, including the public and private sectors, academia, trade associations, local banks, international banks, multilateral banks, and international organizations, in terms of financing, cooperation, technical assistance, and resource contributions, among others.*

Under this guideline, the implementing agency, through the FENIX program, would be responsible for engaging with exploration and production area operators, who are responsible for managing the resources allocated under Article 80 of ANH Agreement 02. Similarly, it would present the program to various international cooperation partners to obtain support resources, technical assistance, access to lessons learned, and other resources. This work would take place during the first year of operation and would guarantee resources for implementing the solutions in this phase, through a combination of these three funding sources (FENOGE, ANH, and International Cooperation).

1.1.2. Implementation of other phases and expansion of program coverage

Once the implementing agency is operational (with the FENIX339 Program), it will have among its obligations the linking of other sources of funding
mentioned above:

General Royalties System - SGR

Legislative Act Project Number 40 of 2019 Senate, approved on October 15, 2019, defines that 1% will be allocated for the protection of environmental areas, strategic ecosystems and *the national fight against deforestation;* these resources were not contemplated in article 361 of the constitution.

National Carbon Tax and the request for non-taxation due to carbon neutrality, Decree 926 of 2017 of the Ministry of Environment

All taxpayers who are required to pay the carbon tax and who are interested in demonstrating reductions in emissions or removals of GHGs can access the total or partial non-accrual of the carbon tax, following the guidelines.

339 This name is proposed, as an allusion to the Phoenix, which revives from its ashes, invoking in this case the ashes produced by the combustion of wood.

Machine Translated by Google

procedural requirements established in Decree 926 of 2017 (See section 4.4). In this sense, all actors in the fossil fuel value chain, including consumers or end users, may opt for non-accrual within the framework of the aforementioned regulations.

Colombia in Peace Fund - Decree 691 of April 27, 2017 and CONPES Document 3850 Colombia in Peace Fund

The Colombia in Peace Fund (FCP) is an independent trust fund whose purpose is to serve as an entity for the execution and coordination of resources for the implementation of the Final Peace Agreement. Created through Decree Law 691 of 2017, it is attached to the Administrative Department of the Presidency of the Republic and has a duration of 10 years.

The Fund under Conpes 3850 includes projects aimed at adapting and increasing the resilience of communities to climate change (infrastructure, production systems), recovering and maintaining the resilience of ecosystems (conservation, restoration, land-use planning), and adapting ecosystems to climate change. It also includes forest carbon projects, low-carbon technologies and production projects, etc.

14.4. Activities to be carried out to access financing

- UPME-ANH cooperation agreement.
- FENOGE proposed structure.
- Presentation and Feasibility of the proposal.
- Call for Implementing Agency – FENIX Program.
- Hiring the implementing organization.
- Survey and acquisition of resources for International Cooperation.
- Dissemination and call to Operators (ANH Agreement 02) to be part of the program.
- Linking donors to International Cooperation.
- Design and implementation of early wins.
- Structuring new projects for financing in SGR and resources from the carbon tax and the Colombia in Peace Fund
- Structuring projects to present to other initiatives or funds.

Machine Translated by Google

14.5. Generation of Carbon Bonds

A potential source of funding for the Progressive Firewood Substitution Plan is the generation of carbon credits, which could be sold to those who need to mitigate emissions, either to fulfill required payments for not being subject to the carbon tax or to meet international commitments. Commercially, it has been estimated that reducing one ton of CO2 could cost $16,000 pesos, a value that will be used to estimate the potential funding for the Plan's implementation that could be achieved through the sale of carbon credits. However, transaction costs, which could be high due to the program's dispersed nature, must be taken into account. Once the strategy guiding the Progressive Firewood Substitution Plan is defined, an estimate of the emissions that would be saved through its implementation has been made. The types of activities that could be eligible for this funding will be coordinated with the Ministry of the Environment to avoid double counting.

Furthermore, an emissions quantification exercise must be carried out, and in the case of the first part of the program, where the operators contemplated in ANH Agreement 02 are proposed as funders, they would become the “owners” of these avoided emissions. This program would then have a complementary character to the programs and efforts of the Ministry of the Environment.

14.5.1. CO2 emissions

The factors that determine CO2 emissions are the following:

- Fuel. The fuel is characterized by its LHV (Lower Heating Value), expressed as the energy (MJ) per unit of fuel, whether kg or m³ . For electricity, the LHV is expressed in kWh. The following table shows these heating values expressed in MJ/kg, MJ/m³, and MJ/kWh.

Table 104. Lower heating values (LHV) of fuels

Combustible	LHV	LHV	HW
	MJ/kg	MJ/m3	MJ/kWh
Generic biogas		24,433	
Generic charcoal	28,76		
Electricity			3,6
Generic LPG	45,415		
Generic firewood	16,993		

Source: SIAME.340

340 http://www.siame.gov.co/Inicio/C%C3%A1lculofactordeemisi%C3%B3n/tabid/77/Default.aspx

Machine Translated by Google

- Fuel emissions. Fuel emissions have been considered per unit of energy, in production as described in the LCA, plus the emissions corresponding to the transport of the fuel to the consumer, called transport/distribution emissions. These emissions are expressed in tCO2/GJ.

Machine Translated by Google

Table 105. Fuel Emissions

Combustible	Production (t COÿ/ GJ)	Transport/ Distribution (t COÿ/ GJ)	Total (t COÿ/ GJ)
Generic biogas	0.0844	negligible	0.0844
Generic charcoal	0,0881		0,0881
Electricity	0,1056	0,0116	0,1172
Generic LPG	0,0672	0,00028	0,50675
Generic firewood	0,0895		0,0895

Source: Transportation and distribution estimates. For Production[341] .

• Fuel consumption. For cooking activities at the household level, information is available on fuel consumption when using only one fuel and also in combination (for example, firewood and LPG). This fuel consumption is associated with the cooking technologies used and the efficiency of the appliances.

The efficiency of the devices corresponds to current stoves (not the latest technology), and in the case of firewood, it includes both the traditional "3-stone firewood" open-fire stove and the "Improved Firewood" stove, which is assumed to consume 70% of the firewood used by the traditional "3-stone firewood." Efficiency is used to verify that all fuels deliver the same amount of useful energy.

Table 106. Monthly fuel consumption and stove efficiencies

Combustible	Consumption			Energy		Useful Energy Efficiency	
	m3/month	kg/month	kWh/month	MJ/day	MJ/month	MJ outdoor/MJ	MJ useful/month
Generic biogas	30			24,43	732,98	37,2%	272,67
Generic charcoal		100,00		95,87	2876,00	9,5%	273,22
Electricity			120,00	14,40	432,00	63,0%	272,16
Electricity Stove Induction			90,00	10,80	324,00	22,71 84,0%	272,16
Generic LPG		15,00		681,22	207,99	40,0%	272,49
Generic firewood - 3 stones		367,20		6239,83	145,60	4,4%	274,55
Generic firewood - Improved		257,04		4367,88		6,3%	272,99

Source: Own elaboration.

[341] phttp://www.siame.gov.co/Inicio/C%C3%A1lculofactordeemisi%C3%B3n/tabid/77/Default.aspx

Machine Translated by Google

The calculation of emissions per household is then given by the following expression:

Emissions (tCO2/a) = Consumption (unit/a) *Calorific value (MJ/unit) *Emissions (tCO2/MJ)

14.5.2. Emissions from the wood replacement strategy

The strategy for replacing traditional wood-burning stoves, called “Leña 3 piedras”, is structured to be implemented over three time periods starting in 2020, 2023 and 2031, and until the years 2022, 2030 and 2050, respectively.

The first period ends with the validity period of the National Development Plan, the second ends in 2030, the year for which commitments to reduce CO2 emissions have been signed at COP 21 in Paris; and the third ends in 2050.

There are three user groups: G1, G2 and G3, who will have their use of firewood replaced with cooking technology options such as *improved stoves, electricity, biogas* and *LPG.*
The total number of "3-stone firewood" users to be replaced for each group and technology is given in the following table. It should be noted that G3 currently consists of users who employ a mix of "3-stone firewood" and LPG stoves in equal energy proportions.

Table 107. Replacement groups and number of stoves to be replaced per period

Group	Technology	2020-2022	2023-2030	2030-2050
G1	Pilot biogas	750	7.500	50.000
	Electrical energy	29.250	135.000	135.000
	Improved Stoves	-	7.500	160.000
	GLP	70.000	150.000	166.656
	Total Group	**100.000**	**300.000**	**511.656**
G2	Pilot biogas	750	7.500	50.000
	Electrical energy	-	-	-
	Improved Stoves	29.250	142.500	22.765
	GLP	-	-	-
	Total Group	**30.000**	**150.000**	**72.765**
G3	Pilot biogas	-	-	-
	Electrical energy	-	-	-
	Improved Stoves	-	-	-
	GLP	30.000	200.000	269.809
	Total Group	**30.000**	**200.000**	**269.809**
Total Phase		***160.000***	***650.000***	***854.230***

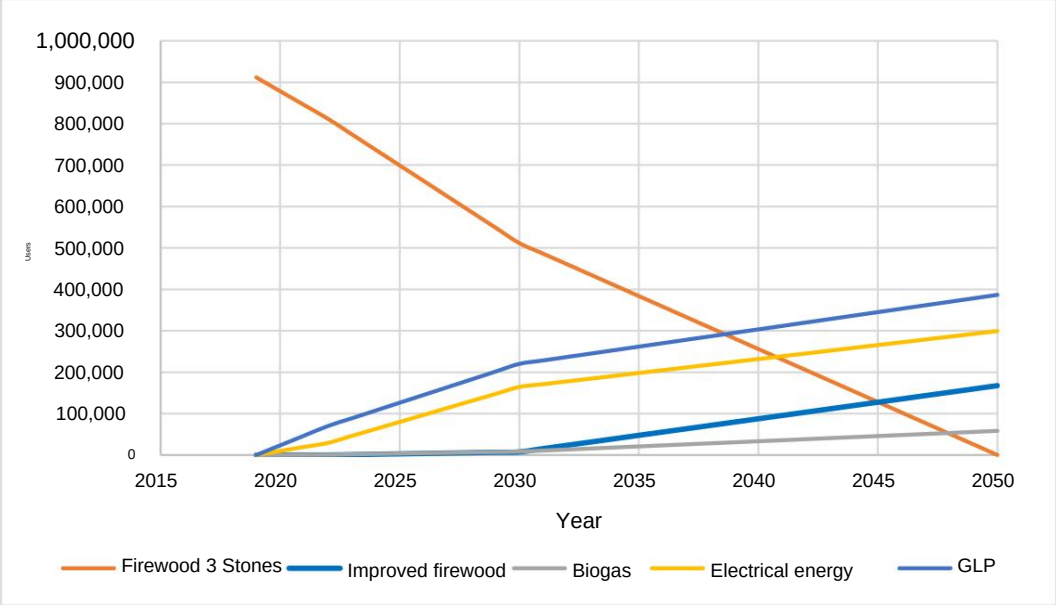
Source: Own elaboration.

Machine Translated by Google

G1 CO2 emissions

In this user group, the aim is to replace 911,656 "3-stone firewood" stoves with biogas stoves, electric stoves, LPG stoves, and improved wood-burning stoves. Graph 26 shows the proposed evolution of the number of users for G1. For each period, the number of technology replacements has been assumed to be linear throughout the period.

Chart 26. Evolution of the number of users by Group G1

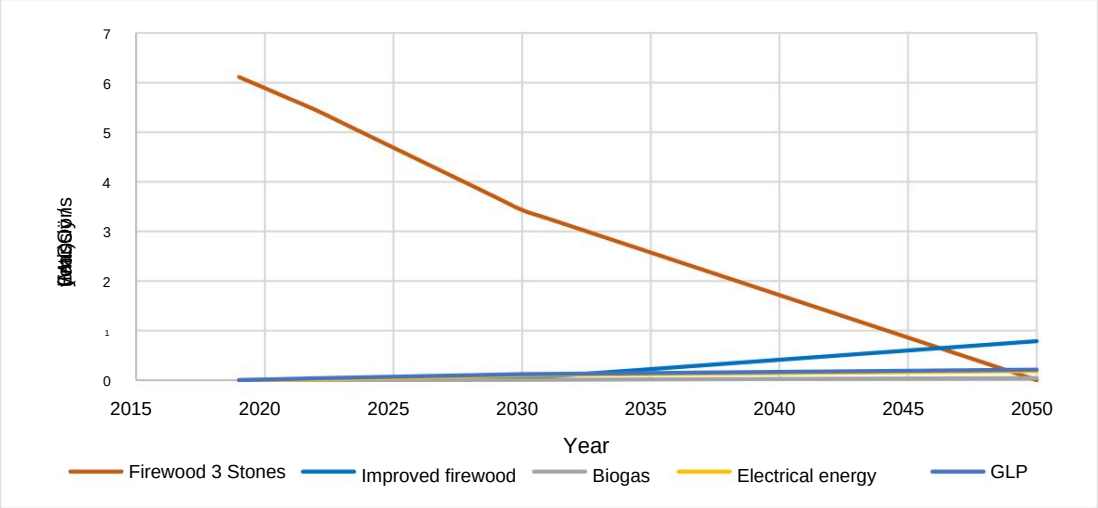


Source: Own elaboration.

Current emissions from the 911,656 users of "3-stone" wood stoves amount to 6.11 MtCO2/ year. Graph 27 shows the evolution of CO2 emissions due to the penetration of different technologies until emissions from "3-stone" wood stoves are reduced to 0 in 2050.

Machine Translated by Google

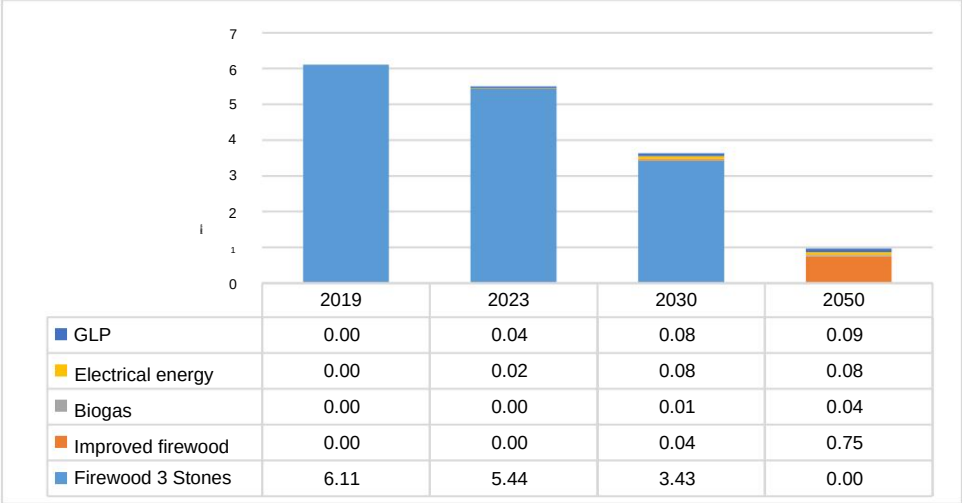
Figure 27. CO2 emissions from stoves - G1



Source: Own elaboration.

The following graph shows the annual emissions of this G1, at the end of the years 2022, 2030 and 2050.

Chart 28. CO2 Emissions – Group G1



	2019	2023	2030	2050
GLP	0.00	0.04	0.08	0.09
Electrical energy	0.00	0.02	0.08	0.08
Biogas	0.00	0.00	0.01	0.04
Improved firewood	0.00	0.00	0.04	0.75
Firewood 3 Stones	6.11	5.44	3.43	0.00

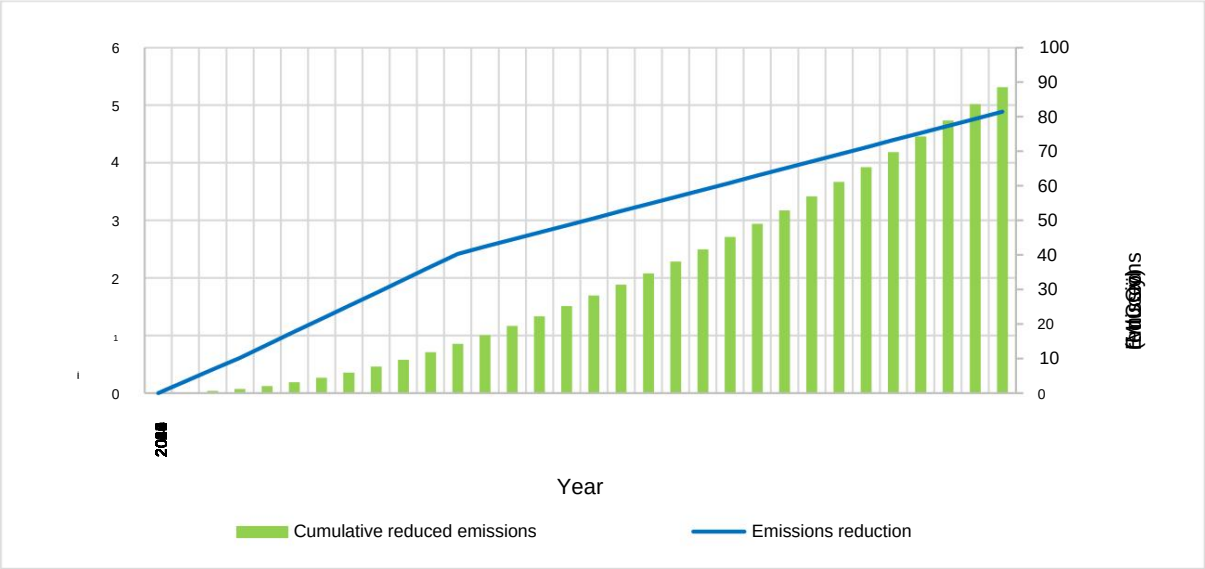
Source: Own elaboration.

The current CO2 emissions from these 911,656 stoves are 6.11 MtCO2/year. For Group G1, emissions reductions begin at 0 in 2019, increase annually as the various technologies are introduced, and reach a maximum of 4.89 MtCO2/year in 2050, when this strategy is fully implemented. Throughout the period 2020-

By 2050, cumulative emissions in the baseline case are 195.56 MtCO2, while emissions in G1 are 106.99 MtCO2. Therefore, emissions will have been reduced by 88.57 MtCO2.

Machine Translated by Google

Chart 29. Reduction of CO2 emissions – G1



Source: Own elaboration.

The evolution of the target number of users replacing traditional wood-burning stoves and the reduction of emissions is shown in the following graph.

Figure 30. G1 User Goals



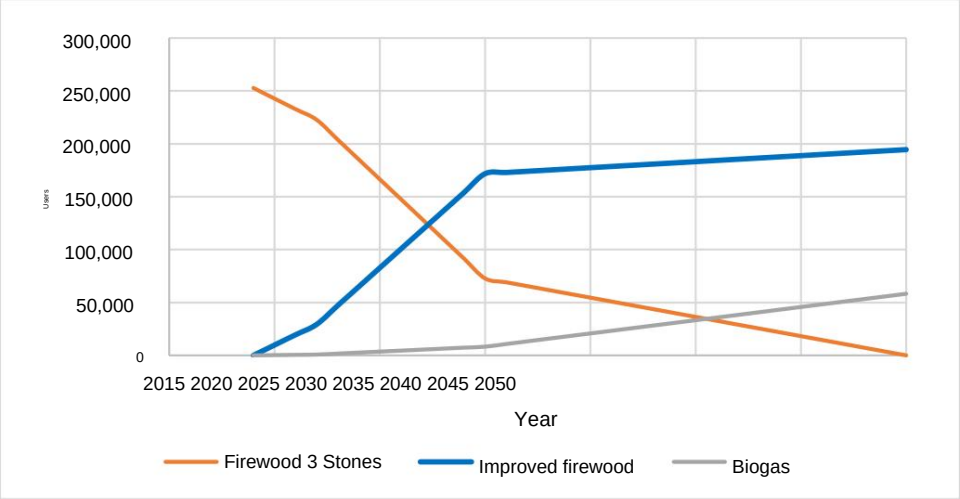
Source: Own elaboration.

Machine Translated by Google

G2 CO2 emissions

In this G2 group, the aim is to replace 252,765 "three-stone firewood" stoves with biogas stoves and improved wood-burning stoves. Graph 31 shows the evolution of the number of users for G2.

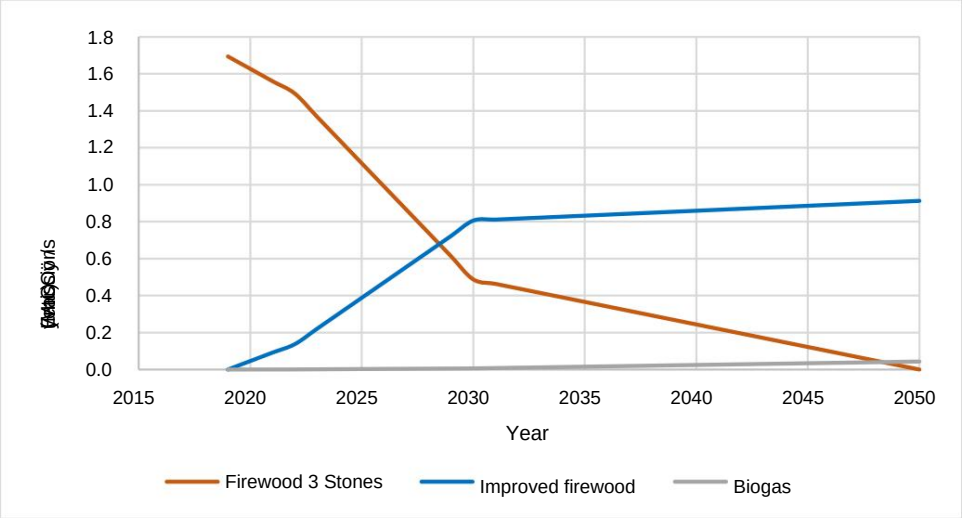
Chart 31. Evolution of the number of users by Group G2



Source: Own elaboration.

Current CO2 emissions from these 252,765 "Three-stone" wood stoves in the G2 amount to 1.69 MtCO2/a. Graph 32 shows the evolution of CO2 emissions due to the penetration of different technologies until emissions from traditional "Three-stone" wood stoves are reduced to 0 by 2050.

Figure 32. CO2 emissions from stoves – G2

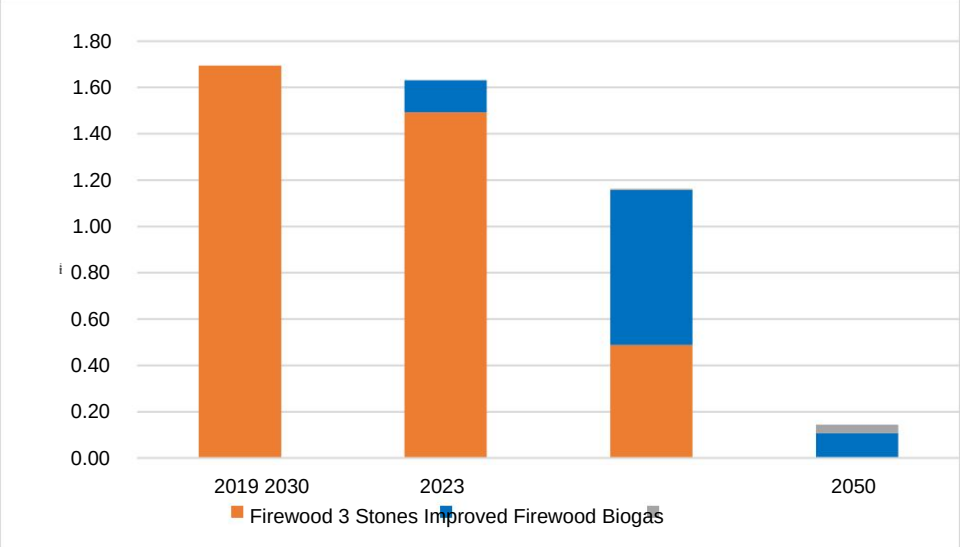


Machine Translated by Google

Source: Own elaboration.

The following graph shows the annual emissions of this G2 group at the end of the years 2022, 2030 and 2050.

Chart 33. CO2 Emissions – Group G2

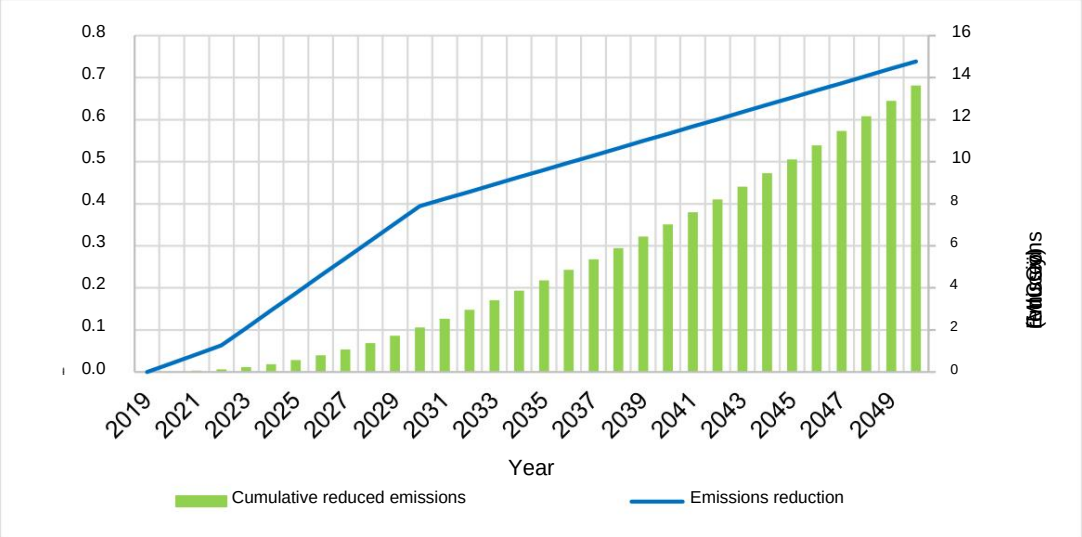


Source: Own elaboration.

The current CO2 emissions from these 252,765 “3 stone firewood” stoves are 1.69 MtCO2/year Currently, if these stoves continue operating as they do now, they will have emitted 54.22 MtCO2 by 2050. Over the entire period 2020-2050, the baseline emissions are 54.22 MtCO2, while the G2 emissions are 40.60 MtCO2. Therefore, emissions will be reduced by 13.62 MtCO2.

Machine Translated by Google

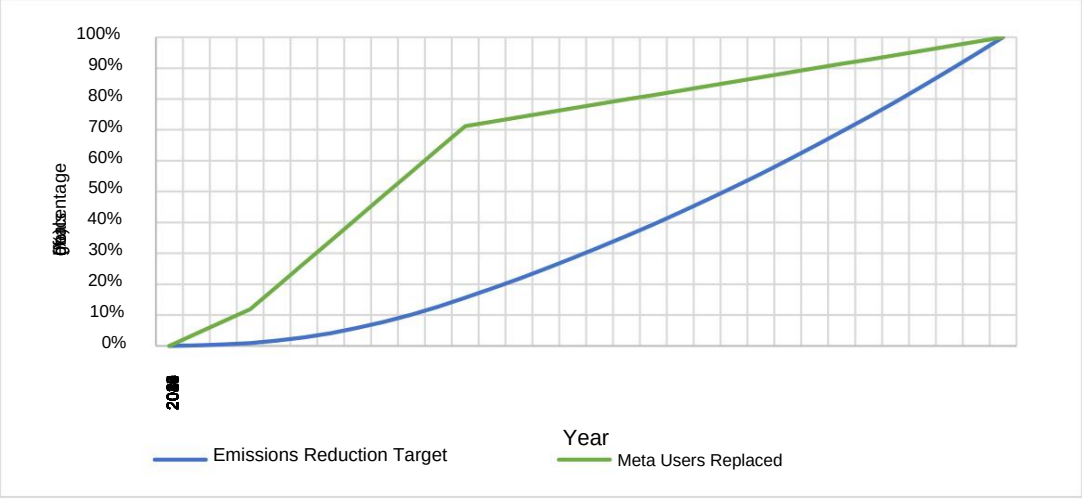
Chart 34. Reduction of CO2 emissions – G2



Source: Own elaboration.

The evolution of the target number of users replacing "3 stone firewood" stoves and the reduction of emissions are shown in the following graph.

Chart 35. G2 Goals



Source: Own elaboration.

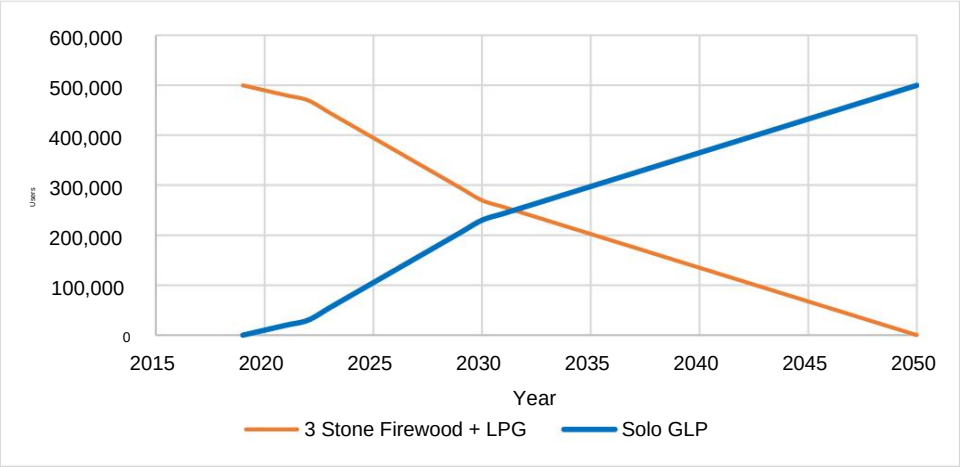
G3 CO2 emissions

This group aims to increase LPG use among 499,809 users. These users currently employ both traditional wood-burning stoves (3 stones) and LPG stoves (equally in terms of

Machine Translated by Google

energy) and the goal is for them to use only LPG. Graph 36 shows the proposed evolution of the number of users for G3.

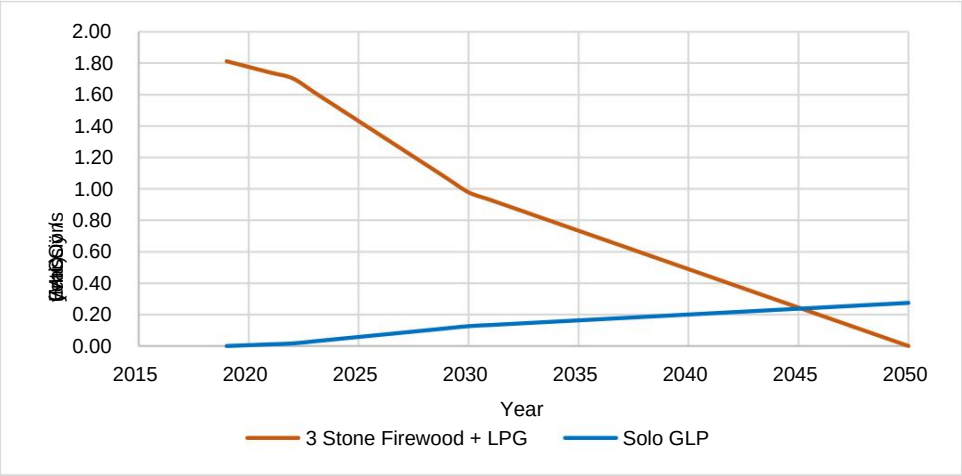
Chart 36. Evolution of the number of users by Group G3



Source: Own elaboration.

The current CO2 emissions from these 499,809 users amount to 1.81 MtCO2/a. Graph 37 shows the evolution of CO2 emissions due to the penetration of LPG use until the emissions from these users “3-stone firewood” + LPG are reduced to 0 in 2050.

Figure 37. CO2 emissions from stoves – G3

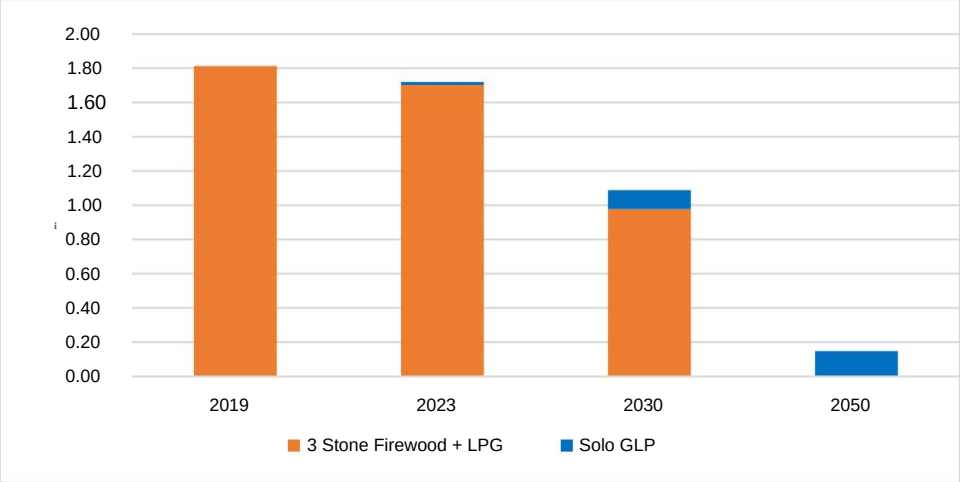


Source: Own elaboration.

The following graph shows the annual emissions of this G3, at the end of the years 2022, 2030 and 2050.

Machine Translated by Google

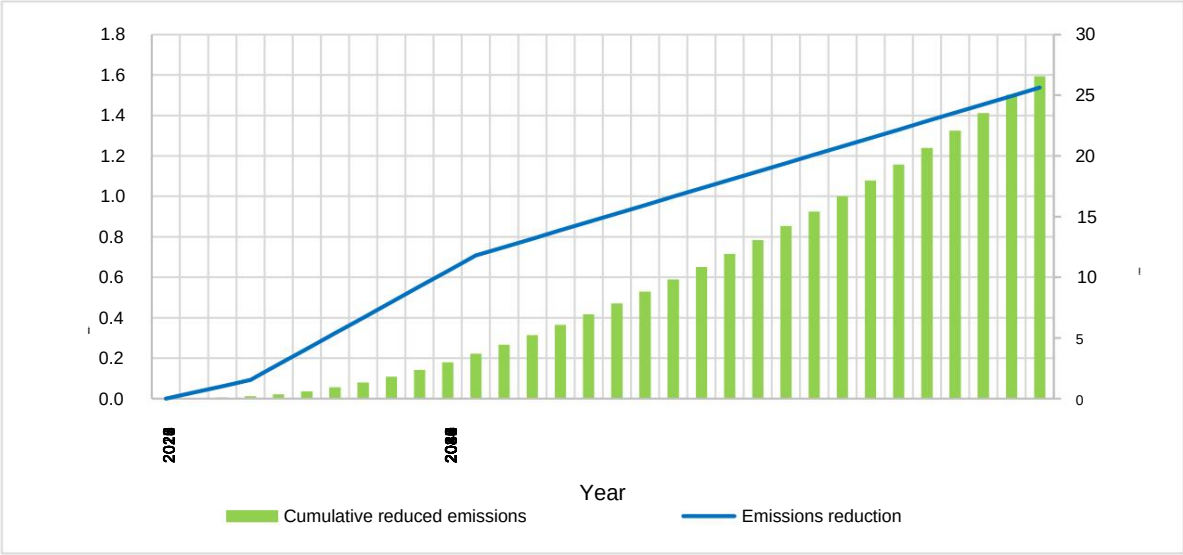
Chart 38. CO2 Emissions – Group G3



Source: Own elaboration.

The CO2 emissions from these 499,809 users of "3-stone firewood" + LPG are currently 1.81 MtCO2/year. Over the entire period 2020-2050, the cumulative emissions for the baseline scenario are 58.0 MtCO2, while the G3 emissions are 31.43 MtCO2. Therefore, cumulative emissions will be reduced by 26.57 MtCO2.

Chart 39. Reduction of CO2 emissions – G3

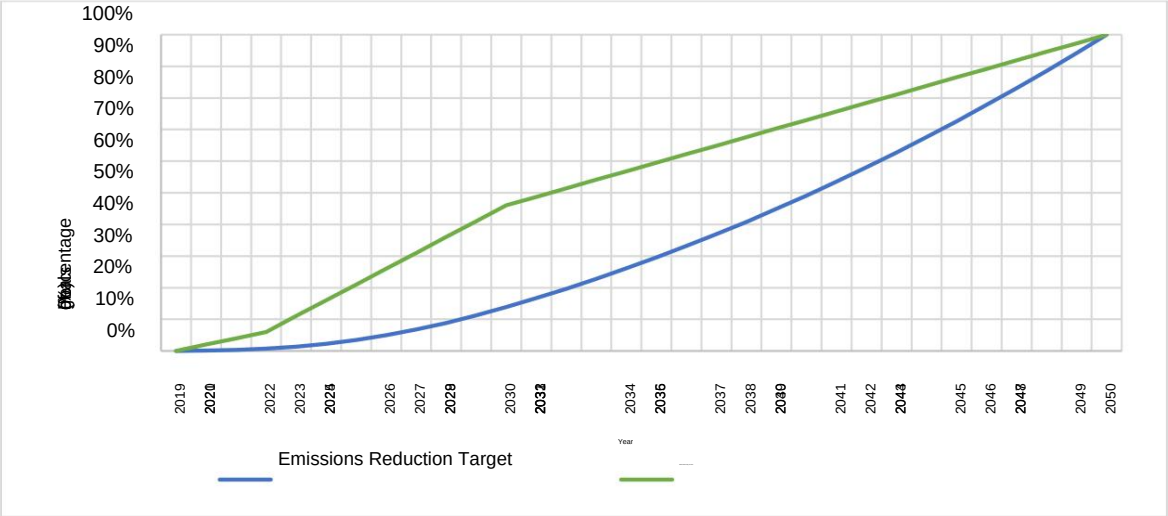


Source: Own elaboration.

The evolution of the target for users who only use LPG and the target for emissions reduction are shown in the following graph.

Machine Translated by Google

Chart 40. G3 Goals

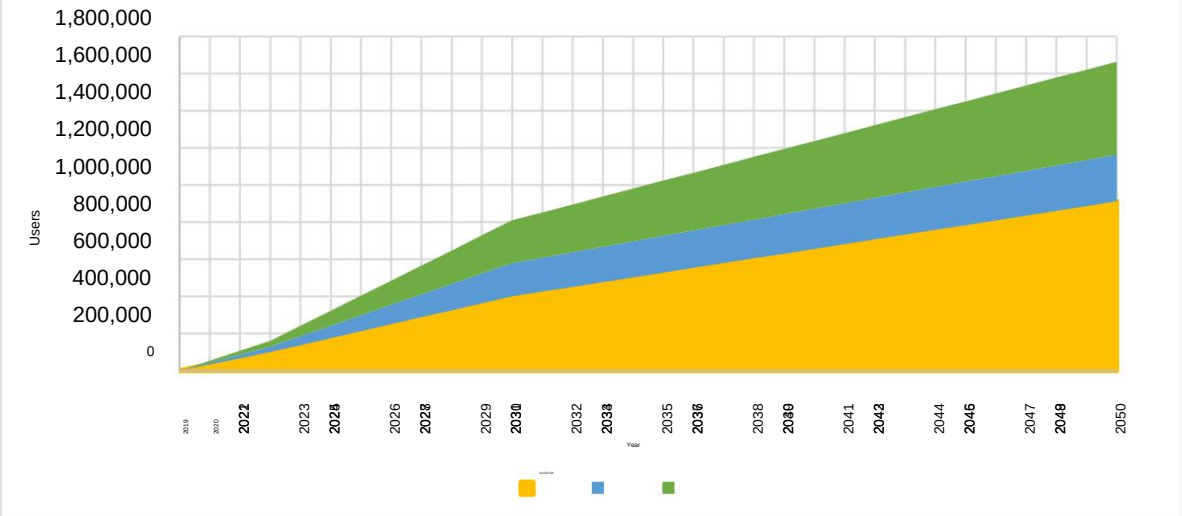


Source: Own elaboration.

Summary of CO2 emissions from groups G1, G2 and G3

In total, the strategy aims to reduce firewood consumption for 1,664,230 users, using the technologies and fuels specified for each of the three groups. Chart 41 shows the evolution of the total number of users.

Chart 41. Evolution of the number of users of all Groups

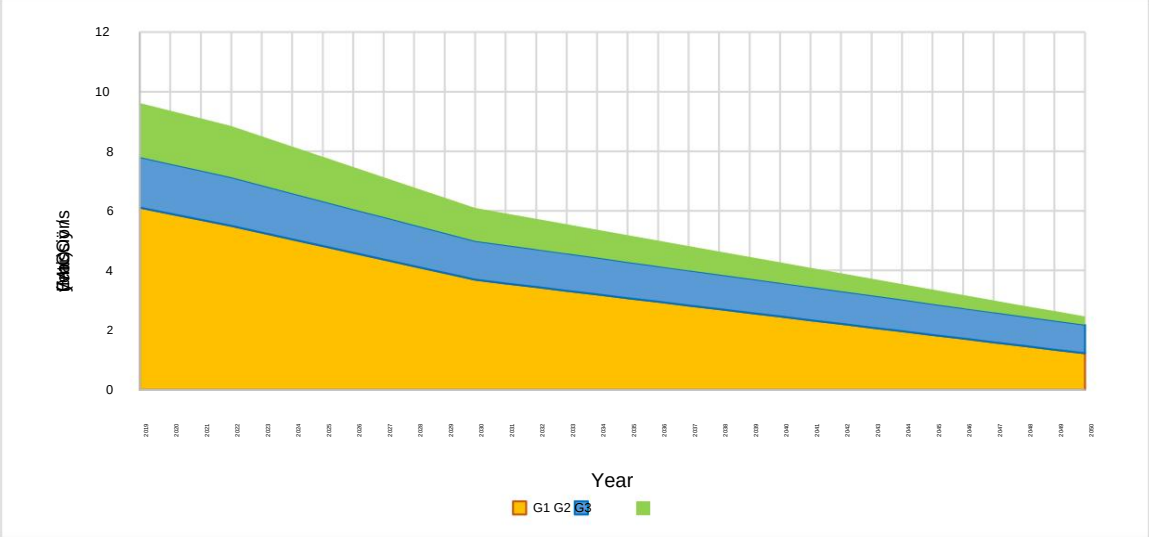


Source: Own elaboration.

Machine Translated by Google

The current CO2 emissions of these 1,664,230 users amount to 9.62 MtCO2/year.
Figure 42 shows the evolution of CO2 emissions for each Group. Ultimately, in 2050, if this strategy is implemented, emissions would be 2.45 MtCO2/year.

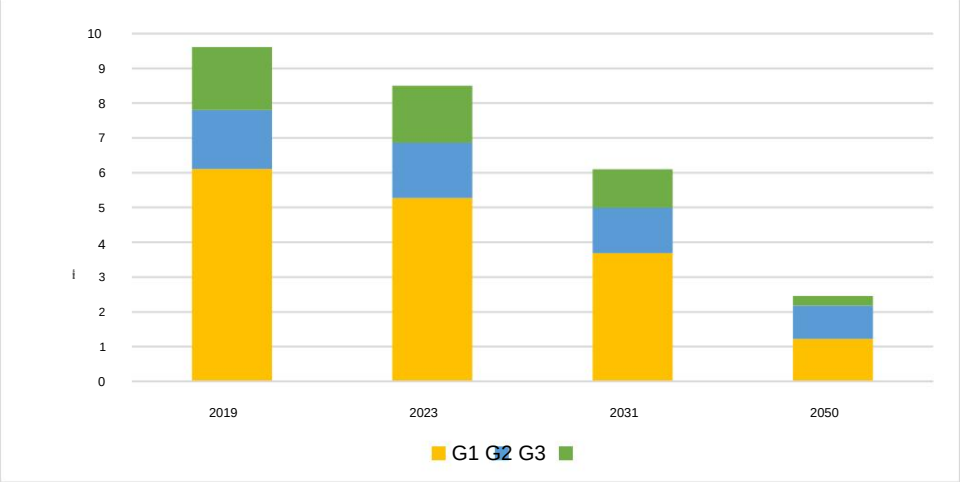
Chart 42. Evolution of annual CO2 emissions from all Groups



Source: Own elaboration.

The following graph shows the annual emissions of all Groups at the end of the periods considered, in the years 2022, 2031 and 2050.

Chart 43. CO2 emissions of all Groups in the years 2031, 2031 and 2050

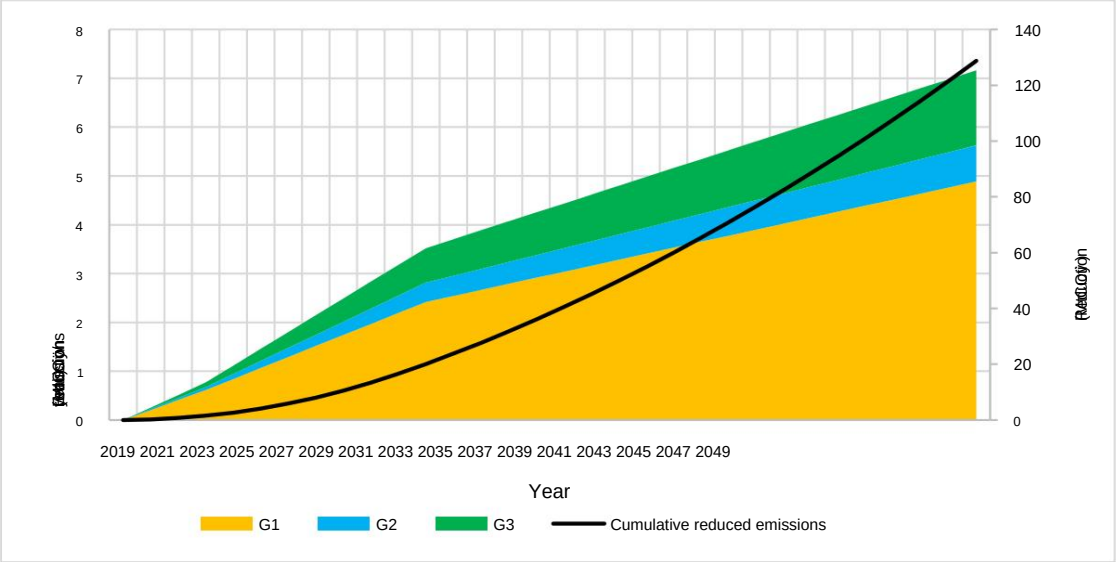


Source: Own elaboration.

Machine Translated by Google

Current CO2 emissions from all Groups are 9.62 MtCO2/year. Over the entire period 2020-2050, cumulative emissions from the baseline scenario are 307.78 MtCO2, while emissions avoided by the replacement plan for all Groups over the same period are 128.7676 MtCO2. Therefore, emissions will be reduced to 179,179.06 MtCO2.

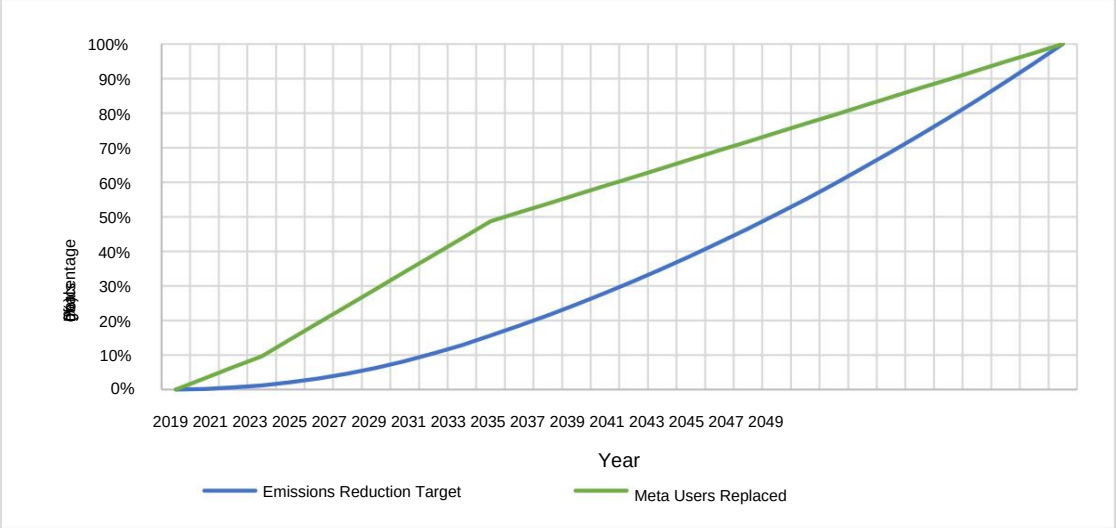
Chart 44. CO2 emissions reduction of all Groups



Source: Own elaboration.

The evolution of the user target for all Groups and the emissions reduction target are shown in the following graph.

Chart 45. Goals for all Groups



Source: Own elaboration.

Machine Translated by Google

14.5.3. Estimation of carbon bond issuance

With the full implementation of the Progressive Firewood Replacement Plan strategy presented in section 12, it is expected that a total of 128.76 megatons of CO2 emissions will be reduced by replacing firewood for cooking in 1,664,230 households. Table 108 shows the expected CO2 reduction in each phase of the Plan.

Table 108. Total emissions reduced by phase of the Progressive Replacement Plan for Wood

	Tons of reduced CO2	Tons of CO2 accumulated
Fase I (2020-2022)	1.538.370	1.538.370
Phase II (2023-2030)	18.540.197	20.078.567
Phase III (2030-2050)	108.682.341	128.760.908

Source: Own elaboration.

Based on the emissions reductions expected to be achieved in each phase of the Plan, taking into account a commercial value of $16,000 pesos per ton of CO2 reduced, it is obtained that a total of bonds with a value of 2,060,175 million pesos could be generated.
Table 109 presents the value of the carbon credits that could be generated in each phase of the Plan.

Table 109. Carbon credit revenues that could be generated by the emission reduction of the Progressive Replacement Plan for Cooking Wood

	Carbon Credits per phase (millions of pesos)
Fase I (2020-2022)	24.614
Phase II (2023-2030)	296.643
Phase III (2030-2050)	1.738.917

Source: Own elaboration.

Taking into account the total costs of the strategy of the Progressive Replacement Plan for cooking firewood in the residential sector, the income that could be obtained from the sale of carbon credits corresponds to approximately 16.4% of the total investment cost of the Plan, considering option A, and 33% considering option B.

Machine Translated by Google

15. Bibliography

Aristizábal Hernández, JD (2014). Validation and comparative evaluation of the efficiency of an improved wood stove under controlled conditions and field test. Technical Reporter (Colombia) Volume 78, 12-24.

Aristizábal, J. (2010). Improved cook stoves and fuelwood lots: an alternative of fuel self-supply for small farmers dependent of oak forests in the Colombia eastern cordillera. Colombia Forestal, 245–256.

Aristizábal, J. “Adoption levels and impact of an improved wood stove in rural communities of the department of Santander, Colombia”, 2018

B. Del and C. Clim, Means of implementing the UNFCCC in Colombia. 2017.

Banco Mundial, Environmental Health Costs in Colombia: Changes from 2002 to 2010.

Barrera Luna, Raul. The concept of Culture: definitions, debates and social uses. Autonomous University of Barcelona. February 15, 2013.

Biobolsa, “Catalog.”.

Biogas Nicaragua, “Nicaragua Biogas Program,” 2018. [Online]. Available: http://programabiogasnicaragua.org.

BLUNCK, M. et al. Carbon markets for improved cooking stoves. 4th Edition. Eschborn (Germany): GIZ – HERA, 2011. p. 5.

C. Serrano Camacho, “Alternatives for the Use of Biogas in Sanitary Landfills in Colombia,” Rds.Org. Co, p. 93, 2006.

Population Census 2000. National Institute of Statistics (2002) Guatemala City

ECLAC. Contribution of energy services to the achievement of the Millennium Development Goals and the mitigation of poverty in Latin America and the Caribbean, 2009.

Cid Jurado, Alfredo Tenoch. Culinary semiotics and cultural heritage: cuisine Colombian. 2018. Chilean Journal of Semiotics.

Concha, María Cecilia; Pabón, Giovanni; Cerón Viviana. -- Bogotá and 2015. Ministry of Environment and Sustainable Development, “Guidelines for a National Program of Efficient Wood-Burning Stoves,” 2015.

Congress of Colombia, “LAW 1715 OF 2014,” law, vol. 151, no. 4. pp. 1–46, 2014.

Cooking for Life, “Elements for the formulation of a Cooking Energy Policy in Rural Areas of Colombia”. Luis Augusto Yepes G. International LPG Congress – GASNOVA, August 2018.

Regional Autonomous Corporation of the Negro and Nare River Basins (Cornare). Management report submitted to the Comptroller General of the Republic, January 1, 2001 to June 20, 2010.

Machine Translated by Google

Curiel. Ochi. Gender, race, sexuality: contemporary debates. University Conference of the Rosary.

DCC Concha, María Cecilia; Pabón, Giovanni; Cerón Viviana. Ministry of Environment and Sustainable Development, Guidelines for a National Program of Efficient Wood-Burning Stoves. Bogotá, 2015.

D. P. Garcia, J. C. Caraschi, G. Ventorim, and F. H. A. Vieira, "Trends and challenges of Brazilian pellets industry originated from agroforestry," Cerne, vol. 22, no. 3, pp. 233–240, 2016.

DANE (2007). Colombia A Multicultural Nation. Its Ethnic Diversity. May 2007.

DANE (2015). Non-fetal deaths 2014 – Preliminary. Bogotá: DANE. Table 5. Archive Excel

DANE (2014). National Quality of Life Survey 2013. Op. Cit.

DANE (2018). Indigenous Population of Colombia. Results of the 2018 National Housing Census.

DANE (2018). Black, Afro-Colombian, Raizal and Palenquera Population. 2018 Population and Housing Census.

DANE (2019). National Quality of Life Survey 2018.

DANE (2019b). Press release National Survey of Quality of Life (ECV) 2018. Retrieved from: https://www.dane.gov.co/files/investigaciones/condiciones_vida/calidad_vida/CP_ECV_2018.pdf

Emergency Decree No. 069-2009, Peru

Delgado Salazar, Ramiro and Delgado Giraldo, Miguel. Of Gustemas and Tecnemas. Culinary Identity of Knowledge and Flavors.

Díaz, R., Berrueta, V., Masera, O. 2011. Thematic Notebooks No. 3 on Bioenergy: Wood Stoves. Mexico City

Díaz-Jiménez. 2000. Firewood Consumption in the Residential Sector of Mexico. Evolution History and CO2 Emissions. Mag Sc Thesis. Mexico City, MX, UNAM. P 113.

DNP (2012). Guide for the incorporation of the ethnic variable and the differential approach in the formulation and implementation of plans and policies at the national and territorial level.

DNP (2017). "Installation of efficient stoves for rural housing", Bogotá.

DNP (2018). "Economic valuation of environmental degradation in Colombia 215- Urban air pollution, indoor air pollution, and deficiencies in access to water and sewage systems."

E. Ponce (2016). "Simple methods for obtaining rural biogas and its conversion into electricity," Idesia, vol. 34, no. 5, pp. 75–79.

Machine Translated by Google

Electrificadora del Meta (2018). “Construction of eco-efficient stoves and establishment of dendroenergetic banks and protective insulation in the area of influence of the 115 kv line Suria substation – Puerto Gaitán substation, Meta department.”

Energy Information Admnistration. www.eia.doe.gov

Supreme Court (2018). Supreme Court ruling on peasant protection. STP2028-2018 Case No. 96414. February 13, 2018.

FAO (2011). “Biogas Handbook”.

FAO (2008). Forests and energy: key issues. Rome: FAO. p. 12.

FAS-Staff (2016). “Russian Federation Biofuels Annual Biofuels Sector Update.” p. 21.

Rural Mission Foundation. Analysis of different theoretical conceptions of the peasant and their forms of organization. 2013. Study funded by the Ministry of Agriculture.

Natura Foundation. Efficient wood-burning stoves as a contribution to quality of life, efficient energy use and the reduction of GHG emissions in rural areas of Antioquia and Santander, Colombia: Systematization document and technical supplement / Gómez, Roberto León; Aristizábal, Javier Darío and Cárdenas, Luis Mario.

Bogotá, DC Colombia, Fundación Natura. 2015. 130 p

GM Karanja and EM Kiruiro, “Biogas Production 107,” Statistics (Ber). pp. 100-1 107–208, 2003.

GACC (Global Alliance for Clean Cookstoves, US). 2012. Peru Market Assessment: Sector mapping (online). Accessed September 2, 2019.

GASNOVA. “BASIS FOR THE PROJECT: REPLACING THE CONSUMPTION OF FIREWOOD, COAL AND WASTE WITH LPG”. COLOMBIAN LPG ASSOCIATION, 2018

GASNOVA (2019). Statistical Report October 2019.

GEF – ADESCOLAB – UNDP. Adescolab's experience with the project: Protection and recovery of the La Barra forest and establishment of an Agroforestry System.
(2004) Colonia La Barra, Metapán, El Salvador.

GMI, “Global Methane Initiative,” 2014. [Online]. Available: https://www.globalmethane.org/.

Gómez Boas, Ana Dalila. Romani People of Colombia. National Planning Department. 2010.

Government of Canada, “Biomass Program Continues to Support Transition to Renewable - Canada,” 2016. Energy in Manitoba https:// [Online]. Available: www.canada.ca/en/agriculture-agri-food/news/2016/07/biomass-program-continues-to-support-transition-to-renewable-energy-in-manitoba.html.

Research Group on Identity Studies. Who are the peasants today in Colombia? School of Human Sciences. University of Rosario

Machine Translated by Google

HCHR, taken from: https://www.hchr.org.co/index.php/76-boletin/recursos/2470-ique-es-el-enfoque-diferencial.

ICBF (2017). Differential Approach Model.

ICE, "Group ICE," 2017. http://www.grupoice.com/wps/wcm/connect/669c79e9-84c9-4682-b63b-136238ebc7e2/tarifas+ice+gaceta+alcance+106+junio+27+2016.pdf?MOD=AJPERIS. [Online]. Available: Program Biogas.

IICA. Project Monitoring and Evaluation Guide. s. l: IICA – AEA – Ministry of Foreign Affairs Foreign Affairs of Finland, 2012. p. 11.

Final Report University of the Andes Agreement No. 88 of 2005 Ecopetrol – Ministry of Environment, Housing and Territorial Development – University of the Andes, May 2006

JB CARRASCO LEAL, "Mechanism for the Voluntary Mitigation of Greenhouse Gas Emissions for Colombia," Corporación Ambient. Empres., pp. 1–18, 2015.

JM Rincón and E. Silva, Editors; Bioenergy. Bogotá, 2014.

JM Rincón Martínez, DM Durán Hernández, O. Quintero Montoya, CS Duarte González, PO Guevara Patiño, and ME Velásquez Lozano, "Availability of residual biomass and its potential for biogas production in Colombia," pp. 16–25, 2018.

J. Stern, "The Future of Gas in Decarbonising European Energy Markets: the need for a new approach, no. January". 2017.

Johnson, M., Edwards, R., Frenk, C. A. & Masera, O. (2008). In-field greenhouse gas emissions from cookstoves in rural Mexican households. Atmospheric Environment, 42(6). 1206-22. DOI: 10.1016/j.atmosenv.2007.10.034.

Juan Fernando Ramírez Quirama and Adolfo León Taborda Vergara, "Consumo de leña en fogones tradicionals en familias campesinas del oriente antioqueño", Prod. + limpia, vol. 9, pp. 99–114, 2014.

K.Thoday, P. Benjamin, M.Gan, E. Puzzolo. "The Mega Conversion Program from Kerosene to LPG in Indonesia: Lessons learned and recommendations for future clean cooking energy expansion". Energy for Sustainable Development 46 (2018) 71–81

Laurent, A, "Standardism as government: voluntary carbon certification and the Peruvian cookstove sector. Tesis doctoral, Sciences-Po Paris", 2014

Lozano, M et al., "Estimation of the potential for conversion to biogas of biomass in Colombia and its use final report interdisciplinary team made up of the Colciencias research groups", Bogotá, 2017.

Ministry of Environment, "MAIN QUESTIONS REGARDING THE NATIONAL CARBON TAX AND THE REQUEST FOR NON-CAUSATION DUE TO CARBON NEUTRALITY.".

Machine Translated by Google

M. Dumitru and M. Bibu, “RESEARCHES ON THE DIGESTERS AND REACTORS WHICH CAN BE USED IN A FARM SCALE BIOGAS PLANT,” vol. 14, no. 3, pp. 63–68, 2014.

Mario Enrique Velásquez Lozano et al., “Estimation of the potential for conversion to biogas of biomass in Colombia and its use final report interdisciplinary team made up of the Colciencias research groups”, Bogotá, 2017.

Masera, CO, & Fuentes, GA (2006). Current state of bioenergy in Mexico. Bioenergy in Mexico. A catalyst for sustainable development. Conafor, Mexican Bioenergy Network, 1-10.

Mejía Barragán, Fabiola. 2011. Environmental implications of the use of firewood as domestic fuel in the rural area of Usme. Master's thesis in environment and development.

Memorandum GAS-PES-079, from the Special Programs Area to the Vice Presidency of International Trade and Gas. December 6, 1997

MEMyM, “Supply of Electricity from Renewable Sources through Cammesa on behalf of the Distribution Agents and Large Users of the Wholesale Electricity Market.” 2016.

Microsol, “Report on the relationship between short-lived climate pollutants and rural cooking in the residential sector. Prospective study applied to Colombia, Mexico and Peru”, 2018.

Ministry of Environment and Sustainable Development (2017). Resolution 2254 of 2017 By which the ambient air quality standard is adopted and other provisions are issued.
Retrieved from: http://www.minambiente.gov.co/images/normativa/app/resoluciones/96-

res%202254%20de%202017.pdf

Ministry of Industries and Productivity. Call for expressions of interest and qualification of induction cooker suppliers for the efficient cooking program, 2014

Ministry of Environment and Sustainable Development. National Stove Survey, 2014

Ministry of Mines and Energy, Comprehensive Audit of the Subsidy Program for Users of Strata 1 and 2 for the Consumption of Liquefied Petroleum Gas, LPG, distributed through cylinders in the Departments of Caquetá, Nariño, Putumayo and Archipelago of San Andrés, Providencia and Santa Catalina Contract 245 of 2016.

Ministry of Health. 2017. The use of efficient stoves and their impact on health promotion in the Colombian context.

Ministry of the Interior of Colombia. 2014. Origin myths of indigenous peoples of Colombia.

Montealegre, Diana and Urrego, Jaime. 2011. Differential approaches to Gender and Ethnicity. National University of Colombia.

Machine Translated by Google

Moreno, A. “Webinar: “Advances in clean cooking. Latin American Network and of Caribbean of Clean Kitchens”, 2018.

Murcia, GUG; Huertas, MC; Rodríguez, JM; Castellanos, HO 2010. Multitemporal changes in forests and other land cover in the Colombian Amazon, at a scale of 1:100,000, from 2002 to 2007. SINCHI Amazonian Institute for Scientific Research. Bogotá, DC, available online: http://siatac.co/c/document_library/get_file?uuid=a0ec40be-0617-4ee8-8f27-1b66890bfcdc&groupId=762

Muro, HA, Paredes, JN, & Bravo, CA (2010). Environmental impact produced by the use of firewood in the Vilacota-Maure regional conservation area of the Tacna region. IV Latin American Conference on Solar Energy (IV ISES_CLA) and XVII Peruvian Symposium on Solar Energy (XVII-SPES) (pp. 1-5). Cusco: Jorge Basadre Grohmann National University.

NEB, “NEB - Canada’s Adoption of Renewable Power Sources - Energy Market Analysis - Biomass,” 2019. [Online]. Available: https://www.neb-one.gc.ca/nrg/sttstc/lctrct/rprt/2017cnddptnrnwblpwr/bmss-eng.html.

Ocaña Figueroa, RE, & Linares Castillo, É. L. (2005). Dendroenergetic plant species used by the inhabitants of Encino, Santander, Colombia. Acta Biológica Colombiana Vol. 10, no. 1, 97.

OLADE, “Technical assistance on lessons learned and recommendations for the development of efficient stove projects in Guatemala, El Salvador, Honduras, Nicaragua and Panama”, 2010.

OLADE, “Available Technologies to Implement Rural Biogas Programs In Latin America,” no. 11

Ordoñez Caicedo, Carlos. The Great Book of Colombian Cuisine. Volume 9. Ministry of Colombian culture.

Oxford, “Biogas Contribution decarbonising gas markets,” no. June 2017, 2017.

P. Schaube, W. Ortiz, and M. Recalde, “Status and future dynamics of decentralised renewable energy niche building processes in Argentina,” Energy Res. Soc. Sci., vol. 35, no. November 2017, pp. 57–67, 2018.

Presidency of the Republic of Colombia, “President Duque presented the 'Colombia E2_ Entrepreneurship x Energy' program, to strengthen the renewable energy industry,” 2019.

PROBIOMASA, “Project for the promotion of energy derived from biomass,” 2018. [Online]. Available: http://www.probiomasa.gob.ar.

Rural LPG Program-ECOPETROL. IG-Itansuca-Geoengineering Consortium. May 2007

Gas for the Countryside Program - Vice Presidency of International Trade and Gas. Final Report. May 1999.

Machine Translated by Google

R. J. Patinvoh, J. V Arulappan, F. Johansson, and M. J. Taherzadeh, “Biogas digestersÿfrom plastics and bricks to textile bioreactor — A review,” J. Energy Environ. Sustain., vol. 4, pp. 31–35, 2017.

Ramírez Quirama, JF, & León Tabor, A. (2014). Firewood consumption in traditional stoves in rural families of eastern Antioquia. Producción + Limpia, 99•114.

Restrepo, J., Reyes, P., and De Ochoa, P. 1983. Pneumoconiosis due to inhalation of smoke from firewood.

Rodríguez Puentes, Ana Luz. 2009. The Do No Harm Approach. National University from Colombia.

Rodriguez, H. Productive Uses of Renewable Energy in Guatemala – PURE. Fundación Solar, Guatemala City, 2013

Rosas Pachon, Javier. The kitchen: a stage for the reconstruction of memory. Pontifical Javeriana University.

Sarika Jain (2019). *Global potential of biogas.*

SEMARNAT, “Energy recovery from urban waste (EnRes),” 2018. [Online]. Available: https://www.semarnat.gob.mx/.

Sierra, F. Mejia F and Guerrero C. 2011. Firewood as domestic fuel in rural areas From Usme. Bogotá. University of La Rioja.

Silva Herran, D., & Nakata, T. (2008). Renewable technologies for rural electrification in. International Journal of Energy Sector Management, 139 - 154.

Smith, K; Sagar, A. Making the clean available: Escaping India’s Chulha Trap, 2014

Sun Liying, “Status and prospect of biomass energy development in China´s rural areas,” 2016.

T. Bond and M. R. Templeton, “History and future of domestic biogas plants in the developing world,” Energy Sustain. Dev., vol. 15, no. 4, pp. 347–354, 2011.

TR Preston, “Build your own low-cost biodigester,” 2005.

Torres Duque C, Maldonado D, Pérez Padilla R, Ezzati M and Viegi G. Biomass Fuels and Respiratory Diseases, on behalf of the Forum of International Respiratory Societies Report, 2008.

Tripathi A, Sagar A, 2019, Ujjwala, V2.0: What should be done next?, Policy Brief, CCAPC/2019/02, Collaborative Clean Air Policy Centre, New Delhi

TRONCOSO, Karin. Factors associated with better adoption of clean stoves: a gender perspective. Bucaramanga, 2014. p. 29.

UNESCO, taken from: http://www.unesco.org/new/es/culture/themes/dialogue/intercultural-dialogue/).

Machine Translated by Google

UNESCO. Culture: Key to Sustainable Development. May 17, 2013.

UPME (2017). Liquefied petroleum gas (LPG) supply chain.

UPME (2018). ALLEY 2017. Retrieved from of: http://www1.upme.gov.co/InformacionCifras/Paginas/BalanceEnergetico.aspx

UPME (2019). Colombian Energy Balance -BECO- 2018.

UPME. Mining and Energy Balance 1975 – 2012. Bogotá UPME – SIMEC, 2012. Archive Excel.

Valderrama, E., & Linares, É. L. (2008). Use and management of firewood by the peasant community of San José de Suaita (Santander). Revista Colombia Forestal Vol. 11, 19-34.

Varios Autores. Forum of International Respiratory Societies Report Biomass Fuels and Respiratory Diseases. 2008.

WLPGA website, 2019, https://www.wlpga.org/cfl-mediaroom/the-story-of-the-national-lpg-program-of-india/, accessedon 27 August 2019.

WLPGA. (2018). Substituting LPG for Wood: Carbon and Deforestation Impacts. Neuilly-sur-Seine, France: WLPGA.

Wood Pellet Association of Canada, “Government organizations contribute to our success: WPAC,” 2017. [Online]. Available: https://www.pellet.org/wpac-news/wpac-government-organizations-contribute-to-our-success.

World Biogas Association, “What does the future hold for biogas in China?” 2019. [Online]. http:// [Accessed: www.worldbiogasassociation.org/what-does-the-future-hold-for-Available: biogas-in-china/. 14-Aug-2019].

Machine Translated by Google

16. Annexes

Annex 1 – Regional Workshops

16.1.1. Caribbean Regional Workshop

Reason: Montería Regional Workshop

Date and time: September 26, 2019, 9:00 am – 1:00 pm.

Location: Montería, Córdoba.

Assistants

No. Name 1 María Elena Pineda	Entity	Email
Castillo 2 Miguel Angel Ortiz Padilla 3 Carlos	CVS	ele.016@hotmail.com
A. Silgado Mendoza 4 Diego Miguel	UPB	miguel.ortiz@ubp.edu.co
Piñeros 5 José Rafael Negrete G.	Sonnen Sol	carlosasilgado@gmail.com
	MME	dmpineros@minminas.gov.co
	N&S Constructions	junegreteg@nysconstrucciones.com
6 Carmelo García 7 Diego	N&S Constructions	
A. Camargo T.	UPB	diego.camargo@upb.edu.co
8 Fabián Villafañe	University of Cordoba	fabianjvilla@gmail.com
9 Janer Rodriguez	Corpoema	Janer_012095@hotmail.com
10 Mabel Lopez	Firewood User	
11 Carlos Márquez	ESO Engineering SAS	proyectesosas@gmail.com
12 Luis Augusto Yepes	Corpoema	layg15@gmail.com
13 Fabio González	Corpoema	fgonzalezb@unal.edu.co
14 Laura Hortua	Corpoema	laura.hortua@corpoema.net

Goals

- Socialize the project with stakeholders from regions with the highest consumption of firewood for cooking in the residential sector, in order to identify lessons learned from previous programs, proposals from the communities and information that will enrich the program proposal, as well as to identify and prioritize actions planned for the progressive firewood substitution program.

- Discuss the cultural, gender, and ethnic aspects related to the use of firewood as an energy source for cooking in Colombia, listening to the perceptions of the attendees regarding the customs and/or traditions that should be taken into account in the identification of energy and technological alternatives.
- Generate a dialogue on the energy, technological and institutional alternatives that would allow the implementation of a new program for the substitution and/or more efficient and cleaner use of firewood in the residential sector of the country.

Machine Translated by Google

Workshop order

Presentation

The workshop began with a brief presentation of the project and its objectives to contextualize the attendees on the state of the study and the objective of the workshop, which was not to present results, but on the contrary, to discuss and listen to their perception on the topic of current firewood consumption and the alternatives for its replacement identified so far, as well as the aspects to take into account in a future program of progressive replacement of firewood as an energy source for cooking.

During the presentation, a graph was shown relating cooking energy sources to prosperity, development, and quality of life, on the one hand, and to the reduction of indoor air pollution, on the other. Regarding this graph, a professor and researcher specializing in renewable energy commented on the importance of including a third axis related to renewable energy sources.

This presentation addressed the perspective on biomass and a discussion about sustainability based on preconceived notions about emissions generated by burning firewood. Some people consider these emissions negligible because they believe these plants capture carbon before being used as firewood. This is a point that must be addressed in depth in the proposed plan to prevent it from becoming an argument against replacing firewood as a cooking fuel.

• Related to the topic of sustainability in firewood consumption, the discussion with attendees led to the assertion that firewood could be sustainable as long as the wood consumed for cooking is replanted to continue capturing carbon, and this brought up the topic of woodlots. However, when discussing the need to link improved cookstoves, where firewood consumption cannot be replaced with other energy sources, with strategies such as woodlots, the discussion by the Ministry of Environment and the Ministry of Energy was revisited regarding the difficulty the National Planning Department (DNP) has had in providing financial viability to projects that link improved cookstoves to woodlots.

The culinary customs of the region

With the presence of a woman who uses firewood from the village of Las Palomas, located near the city of Montería, we asked her some questions and discussed regional cooking customs. Initially, Ms. Mabel López explained that the houses are traditionally built of palm thatch and lack a chimney; the stove is located inside the house, and the entire house fills with smoke from the hearth. Her household consists of three adults and three children, and she is the one who cooks the food.

One of the problems he finds with the wood-burning stove is that the house and its contents remain dirty, and the soot is very difficult to clean. However, at 45 years old, he mentions that he has experienced health problems, especially vision problems, which according to

Machine Translated by Google

The doctors who treated her could be related to her being in the smoke.

Illustration 16. Traditional palm dwelling in the Colombian Caribbean



Source: Credencial Magazine (2019) Traditional Housing in the Colombian Caribbean. By Rodrigo Arteaga. Retrieved from: http://www.revistacredencial.com/credencial/historia/temas/la-vivienda-tradicional-en-el-caribe-colombiano

Regarding her habits, Mrs. Mabel commented that she prefers to keep the stove burning all day, starting when she begins making coffee in the morning, and that she collects firewood herself approximately every three days, when she notices her supply is running low. As for her opinion on cooking with firewood, she stated that she cooks with it because it is the only option available to her and that she can afford, and that in the area where she lives, everyone cooks with firewood.

The workshop participants explained the specifications of the region's traditional stove or hearth. She described it as a "termite nest" placed on top of the stove or "chicken." Therefore, the region's traditional kitchen or "stove" consists of a wall or counter that provides height, and one or more termite nests placed on top of it that serve as a hearth.

Machine Translated by Google

Illustration 17. Termite infestation and traditional cooking in Córdoba, Colombia



Source: Luis Naranjo (2012). Blog: The Lord of the Jungle. Retrieved from: http://elsenordelajungla.blogspot.com/2012/

When asked about the possibility of moving her stove outside her house or changing her cooking energy source, this woman replied that building a separate structure just for the kitchen is too expensive and beyond her means. Regarding the energy change, she said she would be willing to use alternative technologies, but none have been offered to her yet. When asked if she would be willing to pay a certain amount to cook with something other than firewood, she said yes, though she did not specify the amount.

Additionally, she was asked about the free time she would have if she used another type of cooking, to which she replied that she would dedicate the time to other household chores and helping her son with his homework.

Alternatives

Regarding technological and energy alternatives to replace firewood as an energy source in the residential sector, participants were asked about their experience with other sources. Professor Diego Camargo from UPB, who has worked on cooking solutions in both Medellín and Montería since 2004, mentioned that solar stove prototypes were developed in Medellín with the National University of Colombia. These stoves cost approximately 80,000 pesos and could have a lifespan of 5 years; however, their operation depended on weather conditions.

At UPB, some preliminary work has been done to improve the efficiency of cooking with firewood, such as using biomass burners. So far, no commercial initiatives have emerged, but efforts have been made to formulate these projects to make them economically viable. This professor believes that firewood is preferable to LPG because he considers that firewood does not generate emissions, while LPG does. However, it is clarified that the consulting team will prioritize health as the main focus when defining medium- and long-term strategies for replacing firewood.

Another professor, Fabián Villafañe from the University of Córdoba, mentions that the problem lies not with LPG itself, but with its value chain. Its [unclear - possibly "its supply chain" or "its quality control"] should be guaranteed.

Machine Translated by Google

Uninterrupted availability to users so they can see the benefits of using LPG compared to firewood. It's also important to consider the cost of this energy source for households with fewer economic resources. Attendees mentioned that most rural residents work as day laborers earning approximately 15,000 pesos daily, making it very difficult for them to purchase LPG, along with the other necessary equipment, at market prices.

Regarding biogas, some attendees mentioned that commercial solutions exist on the market and have been implemented; however, these haven't reached the most vulnerable populations due to a lack of communication and the promotional costs that producers and/or vendors would have to bear. One business initiative discussed by engineer Carlos Silgado involves improved wood-burning stoves that also generate enough energy to power three or four light bulbs with the steam produced by the combustion chamber. A biogas program being implemented in Antioquia was also mentioned, but it appears to have stalled.

In addition to energy and technological alternatives, workshop attendees mentioned the importance of implementing programs that last over time and that not only provide users with an alternative, but also involve them in the solution and train them in its use, maintenance, and benefits.

Conclusions

Finally, it was concluded that there are four factors that must be taken into account when determining alternatives to replace cooking with firewood in the households in the country, and taking these factors into account, the quality of life of the users could be maximized, considering that one of the main axes of the approach of this project is health.

- The energy resource and its availability in the area and for the home.
- Cultural roots and education, so that the community has an active role in the solution.
- The energy distribution chain and its value chain.
- The (environmental) sustainability of the alternative.

The proposed solutions must guarantee the availability, the value chain of the energy substitute for firewood, and sustainability over time.

Machine Translated by Google

Photographic Images

PHOTOS REGIONAL WORKSHOP - CARIBBEAN
 
Photo: Presentation of the Regional Workshop

Photo: Workshop attendees

Machine Translated by Google

16.1.2. Nariño Regional Workshop

Reason: Regional Pastry Workshop

Date and time: October 3, 2019, 2:00 pm – 6:00 pm.

Location: Pasto, Nariño.

Assistants

N°	Name	Entity	Email
1	Andrés de La Vega	G.C.C. Montagas	andres.delavega@montagas.com.co
2	Rosa Myriacey Jojoa	The Owl Nature Reserve	coobuho@gmail.com
3	Concepción Matabanchoy Reserve Network		cmatabanchoy@gmail.com
4	Javier Revelo	University of Nariño	javierrevelof@udenar.edu.co
5	Erika Tumal	Montagas Marketing	erika.tumal@montagas.com.co
6	Karol Torres	SENA environmental support	katorresv@sena.edu.co
7	Sophia Caicedo	C&C	sofia@caicedoycaicedo.com
8	Adriana Caicedo	C&C	adriana@caicedoycaicedo.com
9	Mario Fernando Martínez	University of Nariño	fercho4562@hotmail.com
10	Felipe Belalcázar	IDSN	entornossaludables.idsn@gmail.com
11	Amanda Martinez	Secretariat of Agriculture	amanda.martinez207@hotmail.com
12	Julián Gómez	SEYPREC SAS	juliangomez@greenpomona.com
13	Diana Bolaños	S.G.A.	dianabcabrera@hotmail.com
14	Darío Fajardo	University of Nariño University	dario@udenar.edu.co
15	Malory Nazarene	of Nariño University of Nariño	mncarchana4@gmail.com
16	Karen Delgado	University of Nariño	karendps@gmail.com
17	Jesus Daniel Latorre	Corpoema Corpoema	jlatorre.9009@gmail.com
18	Luis Augusto Yepes	Corpoema	layg15@gmail.com
19	Fabio González		fgonzalezb@unal.edu.co
20	Laura Hortua		laura.hortua@corpoema.net

Goals

- Socialize the project with stakeholders from regions with the highest consumption of firewood for cooking in the residential sector, in order to identify lessons learned from previous programs, proposals from the communities and information that will enrich the program proposal, as well as to identify and prioritize actions planned for the progressive firewood substitution program.

- Discuss the cultural, gender, and ethnic aspects related to the use of firewood as an energy source for cooking in Colombia, listening to the perceptions of the attendees regarding the customs and/or traditions that should be taken into account in the identification of energy and technological alternatives.
- Generate a dialogue on the energy, technological and institutional alternatives that would allow the implementation of a new program for the substitution and/or more efficient and cleaner use of firewood in the residential sector of the country.

Machine Translated by Google

Workshop order

Presentation

The workshop began with a brief presentation of the project and its objectives to contextualize the attendees on the state of the study and the objective of the workshop, which was not to present results, but on the contrary, to discuss and listen to their perception on the topic of current firewood consumption and the alternatives for its replacement identified so far, as well as the aspects to take into account in a future program of progressive replacement of firewood as an energy source for cooking.

This presentation showcased data from the National Planning Department, including an updated study on the costs of deaths and illnesses associated with environmental degradation caused by indoor air pollution. The study, which compares different energy sources, also examined the pollutants that affect health per unit of energy provided. It showed that firewood contains 26 times the amount of particulate matter as LPG, among other findings. The aim was to demonstrate that the focus of the firewood substitution plan should be to reduce the health impacts of cooking with this fuel in households.

Cooking in the home

Traditionally, cooking was done in homes using a tulpa, a hearth with three stones. The women attending the workshop who cook with firewood in the La Cocha lagoon area of Nariño mentioned that they have always cooked with firewood, although they now have improved wood-burning stoves at home, obtained through a CORPONARIÑO program. They say this stove is good; it doesn't produce much smoke (except when lighting or tending the fire) because it has an external vent, and they find the iron plate useful for keeping food warm without needing to constantly tend the fire.

Regarding firewood collection, they explain that it is obtained primarily from natural tree pruning, from other pruning, and from dead trees. However, sometimes when many people gather, such as during a minga (community work day) or an indigenous council meeting, it may be necessary to cut down some trees. The women mention that they have planted some trees, such as eucalyptus and cypress, to repopulate the area and use them as firewood later. They are committed to caring for the forest and its trees so that future generations can enjoy some of the quality of life they have had. In this regard, they mention that the forest has grown in the last 30 years due to their care, since trees are no longer cut down to sell charcoal, as was done previously. However, they also mentioned a study conducted with master's students from Javeriana University, which concluded that it is beneficial to clear some areas of the forest in this region to encourage greater biodiversity.

Regarding the possibility of replacing firewood with another energy source, the workshop participants said they would consider it, but would need to conduct a trial run before making the switch, along with the households in their nature reserve associations. However, they mentioned that they have LPG in their homes, which they use to heat food and cook some dishes, and that this energy source is very important, but they said that many families cannot.

Machine Translated by Google

Buying it is a cost-effective option, but it also has some drawbacks, such as the fuel capacity, which is sometimes insufficient or doesn't meet the needs of large groups or families. Furthermore, the stoves provided by Montagas (the local LPG distributor) as part of their corporate social responsibility program are too small for cooking for large families. They also mention concerns about replacing firewood with an energy source like gas, which is non-renewable and will eventually run out.

Other workshop participants mentioned that firewood is culturally ingrained in family life, the flavor of food, and so on. Even with the availability and access to other energy sources, this energy source remains in use by households.

Along the same lines, some workshop participants mentioned that in some upper-class households, cooking is done with firewood, even when there are resources to pay for a more expensive energy source like gas. At this point, culinary customs may also play an important role.

Other customs

Besides cooking, in the area near La Cocha Lagoon and probably in other cold-climate areas, firewood is used to heat homes when it's very cold and a fire or fireplace is needed. In tourist areas, in particular, fireplaces are used in the rooms.

Regarding this issue, a professor from the University of Nariño mentioned that the PERS of Nariño found that both low-income and high-income households use firewood for heating.

It is also important to consider the customs of the ethnic groups that have ancestrally inhabited this area, for example, the Quillacinga people, who are "men of fire" and believe that if the fire goes out, life goes out. Besides cooking, firewood has a cultural use for these communities, who gather around the *tulpa (hearth)* as a sacred space for education, discussion, and rituals.

Health

A representative from the Nariño Departmental Health Institute mentioned that this is a relevant issue in the region. In approximately 90% of Nariño's municipalities, COPD (chronic obstructive pulmonary disease) is one of the leading causes of death. However, the population is not always aware of the impact that cooking with firewood has on their health, so it would be important to integrate firewood substitution programs with health programs or projects in the municipalities.

Willingness to pay

The willingness of firewood users to pay for alternative cooking fuels was discussed at various points during the workshop. Participants seemed to agree that higher incomes do not necessarily lead to a change in cooking fuel. People's priorities often lead them to buy other things like appliances or cell phones, but culturally, cooking fuel is not a priority, especially when firewood is very inexpensive or free.

Machine Translated by Google

Cost. Gas is not part of the household budget, therefore, paying a bill or buying a gas cylinder is not well received by users.

Alternatives

Regarding technological and energy alternatives for replacing firewood as an energy source in the residential sector, the use of LPG was mentioned in the region, where it has a wide distribution and there is no natural gas alternative.

Improved wood stoves

Regarding wood-burning stoves, these have been widely distributed in the region primarily by CORPONARIÑO and appear to be well-received by users. However, the academic sector considers these stoves to have an efficiency of less than 20% and that a significant amount of heat escapes through the chimneys. Therefore, it is important to finalize the technical standard for this type of stove, which has been under development for some time.

The Montagas project

As part of its corporate social responsibility program, Montagas SA ESP has implemented a project in which it has distributed approximately 16,800 LPG cooking kits. Although in many cases firewood has not been completely replaced, Andrés de La Vega believes that firewood consumption has at least been reduced because households now heat food with gas and cook with firewood. When asked about the long-term sustainability of this project, he commented that the rotation of kits is currently being maintained.

approximately 43% of the cylinders that have been delivered.

Other relevant comments

- One topic addressed by Professor Darío Fajardo of the University of Nariño was firewood consumption by sectors other than residential, such as brickyards, sugar mills, and lime production. These sectors have very low efficiency and high firewood consumption.

- It is important to include health as a sector in the replacement of firewood, not only take it as a source of data.

Solar energy is not very viable in the region due to climatic conditions; generation is very low.

- One of the attendees had a biodigester on her farm, but a tree fell on it pig farm and production stopped.
- One alternative to consider for subsidizing LPG is the exclusive card strategy. for gas used in Ecuador.
- The issue of LPG emissions was discussed in relation to firewood emissions.

Conclusions

- There is a strong tradition in the area regarding the use of firewood, both due to cultural factors and a lack of interest in paying for an alternative energy source.

Machine Translated by Google

firewood, since even if it increases income, is preferred to acquire other goods.
consumption.

- The health sector has an important role to play in promoting the replacement of firewood with other energy sources in the residential sector, and should be taken into account in the plan.

- Improved stove and LPG programs have been implemented in the region, although these have not been part of a coordinated national or regional strategy.
- The debate on the sustainability of firewood compared to other energy sources is important in the matter of firewood replacement.
- In addition to the residential sector, there are productive sectors that use firewood in large quantities, and their consumption should also be considered.

Photographic Images

PHOTOS REGIONAL WORKSHOP - NARIÑO	
	
Photo: Presentation of the Regional Workshop	
	
Photo: Participant participation	Photo: Workshop attendees

Machine Translated by Google

16.1.3. Antioquia Regional Workshop

Reason: Regional Workshop

Date and time: October 10, 2019, 8:00 am – 12:30 pm.

Location: Medellín, Antioquia.

Assistants

N°	Name	Entity	Email
1	Rosa Elvira Noreña Giraldo	Horn	enorena@cornare.gov.co
2	Jonatan Gutierrez Vanegas	University of Antioquia	jonatan.gutierrez@udea.edu.co
3	Ligia Maria Millán	4 Stefania	EPM	ligia.millan@epm.com.co
Uribe Palacio	5 Yenny Serrano	6	EPM	stefania.uribe@epm.com.co
Claudia C. Gómez	7 Jorge	EPM	yenny.serrano@epm.com.co
Ramirez	8 Santiago Bernal	9	EPM	claudia.gomez.maya@epm.com.co
Carlos Mario Chica Arrieta	Consultant - Corpoema	jorgeramirez90@hotmail.com
	Excelsar	bernalsantiago@hotmail.com
	HACEB	carlos.chica@haceb.com
10	Enrique A. Henas	11 Luisa	Medellín Health Secretary	enrique.henas@medellin.gov.co
Salazar	12 Marylin	Women's Secretariat of Antioquia	lsalazarga@antioquia.gov.co
Gomez M	Women's Secretariat of Antioquia	economic developmentwomen@antioquia uia.gov.co
13	Yuly Barrientos Gómez	Ministry of Energy	yabarrientos@minenergia.gov.co
14	Luis Augusto Yepes	Corpoema	layg15@gmail.com
15	Fabio González	Corpoema	fgonzalezb@unal.edu.co
16	Ingrid Quiroga Mojica	Corpoema	Ingrid.quiroga@corpoema.net

Goals

- Socialize the project with stakeholders from regions with the highest consumption of firewood for cooking in the residential sector, in order to identify lessons learned from previous programs, proposals from the communities and information that will enrich the program proposal, as well as to identify and prioritize actions planned for the progressive firewood substitution program.
- Discuss the cultural, gender, and ethnic aspects related to the use of firewood as an energy source for cooking in Colombia, listening to the perceptions of the attendees regarding the customs and/or traditions that should be taken into account in the identification of energy and technological alternatives.
- Generate a dialogue on the energy, technological and institutional alternatives that would allow the implementation of a new program for the substitution and/or more efficient and cleaner use of firewood in the residential sector of the country.

Machine Translated by Google

Workshop order

Presentation

The workshop began with a brief presentation of the project and its objectives to provide context for the participants regarding the status of the study and the workshop's purpose, which was not to present results, but rather to engage in dialogue and listen to the participants' perceptions of current firewood consumption and the alternatives they could identify for its replacement. They also discussed aspects to consider for the future program of progressively replacing firewood as a cooking fuel.

This presentation showed data on firewood consumption by region in the country, identifying Antioquia as the fifth largest consumer (7%). It also illustrated the effects of deforestation caused by firewood consumption in rural households. Among the findings, it was shown that for every 100 families that consume firewood annually, one hectare of dense forest is being degraded globally. Burning trees (firewood) increases CO2 concentration levels in the atmosphere, contributing to global warming.

On the other hand, data from the National Planning Department was presented, including an updated study on the costs of deaths and illnesses associated with environmental degradation caused by indoor air pollution. The study, which compares different energy sources, also analyzed the pollutants that affect health per unit of energy provided, showing that firewood contains 26 times the amount of particulate matter as LPG, among other findings. This was presented to demonstrate that the focus of the firewood substitution plan would be to reduce the health impacts of cooking with this fuel in households.

Similarly, a table was presented showing the thermal efficiencies of three prototype efficient stoves (named Huellas-Cornare, FN-Fundación Natura, and SCE-Sistema de Cocción Eficiente), which underwent the Water Boiling Test, applying the Kitchen Performance Test (KTP). Among the findings, it was highlighted that, although the implementation of improved stoves nationwide has increased cooking efficiency in households, the percentages are still very low and not significant enough to determine an improvement in the quality of life for these households. This topic was left open for discussion by the workshop participants. The main question revolved around the lack of clarity regarding the stoves' efficiency.

To conclude the presentation, the alternatives identified so far by the consulting group were shown, which included: Improved stoves, LPG, Biomass and Electricity.

Under this scenario, the agenda was developed with the following participations:

Cooking in the home

Regarding the improved stoves, Rosa Elvira Noreña Giraldo, assistant delegate for CORNARE, commented that CORNARE's policy is to go where the need is greatest. In some

Machine Translated by Google

In municipalities, families contribute whatever resources they can. He also noted that the corporation is working on a program to provide 2,000 stoves, bringing the total to 31,000 in its jurisdiction, a program that has been running for 30 years. Of this total, the Ministry of the Environment has helped finance approximately 3,000 stoves.

Similarly, Rosa Elvira explained that during her trips to the fields, she observed that those who cook with firewood complain of headaches, burning eyes, and the lingering smell of smoke on their clothes. Switching to other cooking methods offers advantages and health benefits, as well as time savings in harvesting and picking. However, a limitation is the price of fuel, which sometimes forces families, even those with LPG stoves, to resort to wood-burning because it is free.

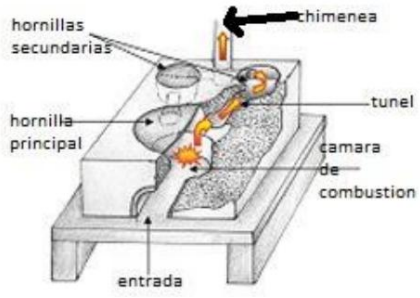
When asked if efficient stoves are commercially available in Antioquia, the answer was yes, explaining that contracts for these types of projects are awarded through a bidding process. Those who decide to build them themselves are advised to use the same suppliers to ensure the stoves are properly constructed. The price of an installed stove is $1.2 million. During a follow-up visit to the project, the constructed stove is evaluated to determine the system's operation and performance and to carry out initial maintenance.

When CORNARE is asked about the possibility of improved stoves in a massive wood replacement program, they state that it is certainly a solution.
While firewood is suitable for homes located far from other sources, a gas connection should be the solution in other cases. Other workshop participants mentioned that firewood is culturally ingrained in family life, influencing the flavor of food, and so on. Even with the availability and accessibility of other energy sources, this fuel continues to be used by many households.

On the other hand, Luisa Salazar, delegate of the Antioquia Governor's Office for Women, points out that there is a gendered division of labor. Women are more involved in domestic care. The Women's Secretariat works on prevention and support for rural mothers to improve their quality of life. During one of her visits, she observed the use of Lorena charcoal stoves designed for two burners. This stove prototype is portable and has the advantage of its low cost, ranging between 400,000 and 500,000 pesos. This speeds up food preparation time, makes maintenance easy, and allows women to better coordinate their work and have more free time, resulting in an improved quality of life.

Machine Translated by Google

Illustration 18 Lorena Stove



Source: Extracted from: http://jopoyveracruz.blogspot.com/2011/08/fogon-mejorado-estufa-ahorradora-de.html

Luisa later argued that in the department of Antioquia there are rural communities located three days' walk from the municipal centers, and in these areas, solutions must be practical. The Lorena stoves, of which there are approximately 500,000, while not a large number, are helpful in remote areas and can be affordable for households located at these distances. Based on the topic of efficient stoves, the question arises: what is being done?
Thinking in terms of sustainability? Because broadly speaking, it seems that LPG would be the solution, but isn't it possible to transition to cleaner fuels that can be used in regions like Urabá, Urrao, Bajo Cauca, and Northern Antioquia?

The possibility of using solar energy is being discussed; it is said that 1 kW installed requires 8m2 The solar panel installation costs around $15 million, without the possibility of using it for cooking. Antioquia is a department with a very high rural population and many remote homes that can only be reached after a 3- to 4-day mule ride.
Similarly, the use of solar panels requires direct contact with the sun's rays, so this technology would not be optimal in cloudy weather.

EPM representatives explained that approximately 120,000 homes in the department have been electrified, but due to cultural traditions, replacing firewood is difficult despite having electricity. Of the total number of homes electrified and with the possibility of replacing firewood, 62% do not do so because of these cultural beliefs. The average electricity bill is around $25,000 per month for consumption plus connection financing, and $10,000 for cooking. Approximately 20,000 homes still need to be electrified. One of the limitations has been the operational and logistical management of bill delivery, as payment by users is very complex and costly. If LPG were the solution, larger cylinders could be considered.

It has been argued that the fuel neutrality of Improved Stoves should go hand in hand with the replanting and establishment of woodlots. However, this model has not been successful, and there are no positive implementation cases. If woodlots are planted but not maintained, the investment is lost, which could lead to inquiries from regulatory bodies and consequently, financial losses. The scheme should be reconsidered, taking into account that farmers do not have the time to manage this activity. If the woodlot idea is to continue, the activity must be separated from the farmer's responsibility, and it must be established that the tasks of creating and managing the woodlot are the responsibility of the farmer.

Machine Translated by Google

The orchards should be managed by someone else (local governments). It is important to contact the DNP (National Planning Department) regarding the issue of woodlots to ascertain their position on the matter.

In biogas, it is necessary to determine in which cases it works; EPM knows that pig farms can be successful in the management of biodigesters given that there is raw material available and they lend themselves to processes in individual and dispersed rural homes, but the operation of biogas implies an activity without remuneration for the collection of biomass and, on the other hand, there is not always someone to do it.

EPM suggests considering subsidies for families living three to four days away from urban centers, subsidies that should differ from those offered to, for example, a user in socioeconomic stratum 1 in Medellín. Regarding this issue, Jorge Ramírez explains that Law 1099 of 2006 establishes a differentiation of rural subsidies, which could provide a way to effectively differentiate them in the case of isolated solutions. Special consideration must be given to rural areas where the poor are even poorer than those in the city's lowest socioeconomic stratum. The fact is that, despite Antioquia having electrified a significant portion of its population, the situation remains unresolved.

96%, a good number of homes still cook with firewood.

Therefore, if a massification plan is to be implemented, a characterization by municipalities must be taken into account so that a portfolio of alternatives can be provided that is in accordance with the needs and possibilities of each region.

Meanwhile, Enrique Henao, the delegate from the Ministry of Health, stated that the strategy is flawed. He explained that while improving stove efficiency is important, the real question is how severe the health problem truly is. Health insurance providers (EPS) must intervene because they are obligated to invest in health promotion and prevention, which they consistently fail to do.
We need to work with other entities that haven't been involved in the process to answer questions like: What is the cost to the health system of not having electricity and what is the cost if it switches to a gas system?

Ultimately, we won't move forward until we address poverty. It's essential to coordinate these plans with the Ministry of Housing as well, in order to design sustainable rural housing solutions that take into account new cooking needs. While it's common to see housing improvements, nothing is done to address cooking.

Regarding cooking efficiency, Carlos Mario Chica, representative of Haceb, explains that induction cooktops can reach 95% efficiency, while LPG cooktops reach 61%. This information should serve as a basis for achieving the strategic goal of replacing firewood consumption with LPG in 100,000 homes. Therefore, it is also recommended to contact the Energy Transformation Mission to understand its perspective on biomass and to grasp the National Planning Department's (DNP) vision. The importance of

talk to Hector Herrera from UPME.

Later, EPM was asked about the use of LPG in remote areas, to which it responded that EPM has used compressed natural gas (CNG) to compete with Gases de Antioquia for market share. If LPG were the solution, cylinders would have to be delivered to remote rural communities, and it would likely be necessary to form alliances with suppliers, using a combination of improved stoves and larger cylinders.

Machine Translated by Google

Optimize logistics, considering that the transportation costs per mule, which is the only way to reach those remote regions, are $70,000 per mule day.

The LPG program in India is discussed, its main characteristics, the transfer of subsidies from high-income consumers to low-income consumers, and the introduction of banking services for women, which makes it possible to pay for the cylinder and immediately receive the subsidized component of the price in their account.

Regarding the institutionalization of these processes, sometimes it's necessary to move beyond the sector itself and propose new institutional frameworks. The institutionalization and definition of the program must be carried out at the highest level of government. These types of plans, which define public policies, should be presented to the National Council for Economic and Social Policy (CONPES), outlining the scope, objectives, cultural and educational aspects, and the necessary coordination among all state agencies to establish the characteristics of the implementing agency.

Regarding the strategy to be implemented by the Agency, each energy source has its place; for example, LPG with broader coverage and improved cookstoves in areas where other energy resources are unavailable. This Agency will also need to identify national and international sources of financial support, such as the Green Climate Fund and finanzasdelclima.dnp.gov.co. Currently, international cooperation resources are available to finance these types of programs, and the Agency in charge of the program must make appropriate use of these resources.

Finally, Jonatan Gutiérrez, representative of the University of Antioquia, presented the progress on the construction of a steam-powered gasification stove. This stove is designed to address the steam energy escaping from the truck bed, allowing the heat produced to be harnessed. The stove has a burner, uses pellets, and achieves 40% efficiency. It reduces fuel consumption and emissions and would have a market cost of $500,000. It is still under development.

Conclusions

Some conclusions from the workshop are:

- The health sector has an important role in promoting the replacement of firewood with other energy sources in the residential sector, and should be taken into account in the plan.

- The region has implemented programs for improved stoves and LPG, although the former have not been part of a coordinated national or regional strategy.
- The debate on the sustainability of firewood compared to other energy sources is important in the context of firewood substitution and the use of non-conventional sources.

- We must find a way to coordinate the development of projects like this with the government entities involved, such as the Ministry of Environment, the Ministry of Mines and Energy, the Ministry of Housing, and the Ministry of Health.

Machine Translated by Google

- The formulation of a CONPES document on firewood is being considered, given the high percentage of families that still use this energy source as the main resource for cooking.
- The issue of climate change must be emphasized in order to seek international funding, taking into account energy efficiency indicators and carbon footprint reduction indicators.

Photographic images:

PHOTOS REGIONAL WORKSHOP - MEDELLÍN	
	
Photos: Development of the Medellín Regional Workshop	
	
Photo: Attendees of the Medellín Regional Workshop	Photo: Haceb stove laboratory

Machine Translated by Google

16.1.4. Boyacá Regional Workshop

Reason: Regional Workshop

Date and time: October 17, 2019, 8:00 am – 12:30 pm.

Location: Sogamoso, Boyacá.

Assistants

N°	Name	Entity/position	Email/phone
1	Laura Constanza Guevara	CorpoBoyacá	lauguevara.2546@gmail.com
2	Carlos Alberto López Monsalve	Boyacá Government	agrocalm2662@gmail.com
3	Fabio José González	Sogamoso Development Secretariat	angelgon188@gmail.com
4	Gloria Cristancho	Housewife 3105505014	
5	Cesilia Camargo	Housewife 3232327727	
6	Rosalba Camargo	Housewife 3232327727	
7	Luz Ángela Martínez	Student/housewife	luzangel548@hotmail.com
8	Rosa Helena Tinjacá	Housewife 3115475905	
9	Teresa Cristancho	Housewife 3208403263	
10	Hilda María Mascón	Housewife 3208849606	
11	Luz Marina López	Housewife 3214521680	
12	Ingrid Quiroga Mojica	Corpoema	ingrid.quiroga@corpoema.net
13	Luz Ángela Pérez	Corpoema	luz.perez@corpoema.net
14	David Cortés	Corpoema	david.cortes@corpoema.net
15	Fabio González	Corpoema	fgonzalezb@unal.edu.co
16	Gerson Rodriguez	Sogamoso Mayor's Office	gersonvet.alcsogamoso@gmail.com
17	Luis Augusto Yepes	Corpoema	layg15@gmail.com
18	María Cristina Merchán	Secretariat of Agriculture Boyacá Governorate	3134572022

Goals

- Socialize the project with stakeholders from regions with the highest consumption of firewood for cooking in the residential sector, in order to identify lessons learned from previous programs, proposals from the communities and information that will enrich the program proposal, as well as to identify and prioritize actions planned for the progressive firewood substitution program.
- Discuss the cultural, gender, and ethnic aspects related to the use of firewood as an energy source for cooking in Colombia, listening to the perceptions of the attendees regarding the customs and/or traditions that should be taken into account in the identification of energy and technological alternatives.

Machine Translated by Google

- Generate a dialogue on the energy, technological and institutional alternatives that would allow the implementation of a new program for the substitution and/or more efficient and cleaner use of firewood in the residential sector of the country.

Workshop order

Presentation

The workshop began with a brief presentation of the project and its objectives to provide context for the participants regarding the status of the study and the workshop's purpose, which was not to present results, but rather to engage in dialogue and listen to the participants' perceptions of current firewood consumption and the alternatives they could identify for replacing it. They also discussed aspects to consider for the future program of progressively replacing firewood as a cooking fuel.

This presentation showed data on firewood consumption by region in the country, where it was identified that Boyacá ranks sixth in firewood consumption by department.
Similarly, the effects of deforestation caused by firewood consumption in rural households were shown. Among the findings, it was evident that for every 100 families that consume firewood annually, one hectare of dense forest is being degraded worldwide.
Burning trees (firewood) increases CO2 concentration levels in the environment, contributing to global warming.

On the other hand, data from the National Planning Department was presented, including an updated study on the costs of deaths and illnesses associated with environmental degradation caused by indoor air pollution. The study, which compares different energy sources, also analyzed the pollutants that affect health per unit of energy provided, showing that firewood contains 26 times the amount of particulate matter as LPG, among other findings. This was presented to demonstrate that the focus of the firewood substitution plan would be to reduce the health impacts of cooking with this fuel in households.

Similarly, a table was presented showing the thermal efficiencies of three prototype efficient stoves (named Huellas-Cornare, FN-Fundación Natura, and SCE-Sistema de Cocción Eficiente), which underwent the Water Boiling Test, applying the Kitchen Performance Test (KTP). Among the findings, it was highlighted that, although the implementation of improved stoves nationwide has increased cooking efficiency in households, the percentages are still very low and not significant enough to determine an improvement in the quality of life for these households. This topic was left open for discussion among the workshop participants. The main question concerned the efficiency of the stoves, as it was not clearly defined.

To conclude the presentation, the alternatives identified so far by the consulting group were shown, which included: Improved stoves, LPG, Biomass and Electricity.

Under this scenario, the agenda was developed with the following participations:

Machine Translated by Google

Cooking in the home

Regarding the topic of improved wood-burning stoves in the department of Boyacá, Laura Constanza Guevara, an official with CORPOBOYACÁ, commented that no efficient wood-burning stove projects have been developed in the department under the Corporation's jurisdiction. The main reason is that, in the region near Sogamoso, households use charcoal, not firewood, for cooking. Therefore, they have adapted their stoves so that the smoke produced by the charcoal is vented outside the houses through a flue called a "buitrón." As explained by Hilda María, a homemaker attending the workshop, firewood and charcoal are used to prepare foods like mute (a traditional stew), reducing cooking time. They used to use petroleum jelly, which was stored in a jar or bottle with a hole in the lid. A rag was placed inside, and when lit , *it would catch fire, and the next day, your nose would be black from the smoke. That's how we were raised, and that's why I'm so sick today from inhaling so much smoke.* Currently, she uses a 40lb LPG cylinder (costing 60,000 pesos) that lasts her approximately eight months. She prepares her lunch on a charcoal stove and cooks by herself, meaning she alternates between different fuel sources depending on the meal.

Doña Hilda María explains that food cooked on a charcoal stove retains its heat, unlike food cooked on an LPG stove, which needs to be reheated several times. The charcoal stove stays lit all day; it's stoked regularly. A sack of charcoal costs 20,000 pesos and lasts approximately 15 days. Firewood is only used to start the fire.

Doña Rosa Elena says that at first everything was cooked with firewood, then it switched to charcoal and that is what is consumed now; the stove is kept warm and it is stoked by inserting a charcoal.
She believes it's very rare to find someone in Sogamoso who uses firewood as fuel. The charcoal stove is designed to use both firewood and charcoal.

A coal stove can cost around 400,000 pesos in materials, not including the hardware and assembly, with dimensions of 1m by 0.50m.

Culturally, the flavor of cooking with wood or charcoal is considered to be very different from that of gas.

On the other hand, Gloria Cristancho commented that in the rural communities near Duitama, firewood is still used for cooking. Firewood consumption is linked to the poverty rate of some farming families who lack their own homes and work intensively to earn money to buy food and pay rent. This is the case for a family consisting of Cecilia Camargo, her daughter Rosalba Camargo, and her grandson. Their stove is in poor condition. The emission of particles and gases is at critical levels, and their health has been threatened by the constant inhalation of smoke in their home. Cecilia and her grandson have been the most affected by the indoor air pollution, and both have been hospitalized due to lung problems and severe flu-like illnesses associated with using firewood for cooking. They have difficulty using a gas stove due to a lack of knowledge.

Therefore, they collect wood from a forest near their home via the trail

Machine Translated by Google

ecological Zarza in Duitama Boyacá, a similar case in Sativasur-Ticuaquita where mostly elderly people live who currently cook with firewood.

Chimney sweeping maintenance is done every 8 days, maximum 15.

On the other hand, Mr. Gerson Rodríguez, representative of the Sogamoso mayor's office, explained that currently no initiatives have been implemented regarding deforestation caused by firewood consumption, but this phenomenon does exist. He stated that it is important to replant trees as quickly as they are cut down. He considers water vital to any environmental chain. The case of Aquitania, concerning the problems with trout farming in the lagoon, was also presented. Deforestation occurs more frequently in areas associated with productive activity. People in the region who cook with firewood do not cut down trees; rather, they engage in a process of harvesting wood, which, although it does contribute to forest degradation, is not the primary source of deforestation in the department. He also called on the regional environmental authority to ensure that appropriate sanctions are applied to the individuals and organizations that are damaging the environment. The number of single mothers in the region has increased, as have young people who lack the resources to buy charcoal. Mr. Rodríguez concluded by stating that these are difficult issues to address and that efforts must be combined to provide effective solutions. He believes that in the case of LPG, the gas is a "pollutant," but only for the users' wallets. Coal is much cheaper, and the design of the stoves doesn't generate smoke or soot inside homes.

There have been cases where people from other regions rent farms and burn native forests and frailejones (a type of plant) to plant potatoes. The mayor's office has taken steps to control these activities and has issued fines to those found guilty.

On the other hand, the housewives argued that cooking with charcoal generates a more "greasy" smoke than cooking with firewood, which means that the time spent cleaning their kitchens and especially the pots is longer.

Engineer María Merchán, Secretary of Culture, explained that about three or four years ago they explored the option of implementing improved stoves, but the effort was unsuccessful because families' strong cultural attachment to their existing stoves makes it difficult to introduce a new technology, as households are satisfied with their current coal-burning stoves. She also stated that the issue had not been addressed from a health perspective. The development and creation of a project are crucial. This project should be aligned with a comprehensive training program and ensure the consent of households, as well as fostering institutional collaboration. Ideally, the project should be presented to the incoming administration for inclusion in their government plan.

To conclude the workshop, representatives from the department's government, the mayor's office, and CORPOBOYACÁ pledged to join forces to support families who still cook exclusively with firewood and lack the resources to improve their cooking technology. While the environmental impact of cooking with firewood is debatable, there is agreement on the repercussions for people's health, especially for women and children.

Machine Translated by Google

Conclusions

- Improved stove programs have not been implemented by CorpoBoyacá or by previous government administrations.
- Biomass consumption revolves around the use of coal as the main energy source in their homes.

They have stoves suitable for cooking with charcoal, but in the past they prepared their food with firewood. Due to health reasons (pulmonary problems, burning eyes, facial paralysis) they stopped using firewood.

- Some households use both LPG (for cooking breakfast and dinner) and firewood. (for cooking lunch).
- Alliances should be sought between local governments and national institutions in order to provide solutions that guarantee the sustainability of possible mitigation alternatives.

Machine Translated by Google

Photographic Images

PHOTOS REGIONAL WORKSHOP - SOGAMOSO	
	
Photo: Regional Workshop Presentation	
	
Photo: Attendees of the Regional Workshop	Photo: House with improved stove
	
Left and right photos: House with a wood-burning stove in poor condition and walls contaminated with soot	

Machine Translated by Google

Annex 2 – Expert Workshop

Bases of the Progressive Replacement Plan for Firewood

Workshop Memoirs for Experts

Reason: Experts Workshop

Date and time: October 24, 2019, 8:00 am – 1:00 pm.

Location: Bogotá DC

List of Attendees:

N°	Name	Entity/position	Email/phone
1	Carlos A. Berbén	Alternatives LDTA	carbermon@gmail.com
2	Felipe Gomez	Agremgas	fgomez@agremgas.com
3	Martin Mora	Gombel Gas	martin.mora@gasamigo.com.co
4	Sarah Lucia Mira Roa	Haceb	sara.mira@haceb.com
5	Adriana Arévalo Mojica Andesco		adriana.arevalo@andesco.org.co
6	Carlos Torres Duque	Pulmonology Foundation	ctorres@neumologica.org
7	Alba Duarte	Gasnova	albaduarte55@gmail.com
8	Esteban Gómez Correa UPME		esteban.gomez@upme.gov.co
9	Carlos Eduardo Niño	UPME	carlos.nino@upme.gov.co
10	José Alfredo Ladino	Indi	diag@indisel.com.co
11	María Isabel Varela	ITSELF	mariavarela@ipse.gov.co
12	Germán León	Gasnova	gleon@gasnova.co
13	Heriberto Dorado	Biotech	heriberto.dorado@biotec.net
14	Dora Castaño	UPME	dora.castano@upme.gov.co
July	15 Flórez	Particular	julioflorez-25@hotail.com
16	Luz María Ospina	CREG	maria.ospina@creg.gov.co
17	Alejandro Martínez	Gasnova	amartinez@gasnova.co
18	Hector Herrera	UPME	hector.herrera@upme.gov.co
19	Marcela Bonilla	UPME	marcela.bonilla@upme.gov.co
20	David Cortes	Corpoema	david.cortes@corpoema.net
21	Diego Miguel Piñeros MME		dmpineros@minenergia.gov.co
22	Javier Aristizabal	FOOD	jaristizabal@minambiente.gov.co

Machine Translated by Google

23 Felipe Villegas	Corpoema	felipe@nativasconsultores.com
24 Juan Uribe	Senate of the Republic	juan.uribe@senado.gov.co
25 Carlos Castellanos	CREG	carlos.castellanos@creg.gov.co
26 Sara Vélez C.	MOTHER	svelez@minenergia.gov.co
27 Oriana Serrano	Natura Foundation	oserrano@natura.org.co
28 Nidya Chaparro	FOOD	nchaparro@minambiente.gov.co
29 Paula Andrea Roa	National Congress	paula.roa@camara.gov.co
30 Samuel Osorio	Secretary of Health	sdosorio@saludcapital.gov.co
31 María Báez	CAR. Cundinamarca	mbaezc@car.gov.co
32 Carlos David Beltrán	Particular	cdbeltran@yahoo.com
33 Johanna Vanegas	CAR	johanavanegas07@gmail.com
34 Diana Becerra	ANDI	plectrodomesticos@andi.com
35 Guillermo Cajamarca	Celsia Energy	gcajamarca@celsia.com
36 Hector Herrera	UPME	hector.herrera@upme.gov.co
37 Carlos Mauricio Silva	Ministry of Environment	carsilva@minambiente.gov.co
38 Darío Mayorga	Corpoema	mayorgadario@gmail.com
39 Laura Hortua	Corpoema	laura.hortua@corpoema.net
40 Luz Ángela Pérez	Corpoema	luz.perez@corpoema.net
41 Ingrid Quiroga	Corpoema	ingrid.quiroga@corpoema.net
42 Fabio Gonzalez	Corpoema	fgonzalezb@unal.edu.co
43 José Rincón	Corpoema	joserinconmartinez@gmail.com
44 Luis Augusto Yepes G	Corpoema	layg2015@gmail.com

Goals

- Socialize the project with the experts of each proposed alternative, in order to identify lessons learned from previous programs, proposals from the communities and information that will enrich the program proposal, as well as identify and prioritize actions planned for the progressive replacement of firewood.

- Discuss the cultural, gender, and ethnic aspects related to the use of firewood as an energy source for cooking in Colombia, listening to the perceptions of the attendees regarding the customs and/or traditions that should be taken into account in the identification of energy and technological alternatives.

Machine Translated by Google

- To generate a dialogue on energy, technological and institutional alternatives that allow the implementation of a new program for the substitution and/or more efficient and cleaner use of firewood in the residential sector of the country.

Workshop order

Presentation

The workshop began with a brief presentation of the project and its objectives to provide attendees with context on the progress of the study and the workshop's purpose. In addition to presenting preliminary results, the workshop aimed to foster dialogue and gather participants' perspectives on current firewood consumption and potential alternatives for its replacement. It also sought to identify other aspects to consider for the future program of progressively replacing firewood as a cooking fuel.

This presentation showed data on firewood consumption by region in the country. It also showed the effects (I removed this part due to the ongoing debate about whether or not firewood consumption causes deforestation) caused by firewood consumption in rural households. Among the findings, it was shown that for every 100 families that consume firewood annually, the equivalent of one hectare of dense forest is being degraded globally. Burning trees (firewood) increases CO2 concentration levels in the atmosphere, contributing to global warming.

On the other hand, data from the National Planning Department was presented, including an updated study on the costs of deaths and illnesses associated with environmental degradation caused by indoor air pollution. The study, which compares different energy sources and shows that firewood contains 26 times the amount of particulate matter as LPG, among other things, reveals that the focus of the firewood substitution plan will be to reduce the health impacts of cooking with firewood in households.

Similarly, a table was presented showing the thermal efficiencies of three prototype efficient stoves (named Huellas-Cornare, FN-Fundación Natura, and SCE-Sistema de Cocción Eficiente), on which the Water Boiling Test was performed, applying the Cooker Performance Test (KTP). The main concern regarding the Improved Stoves revolves around their actual efficiency and the relatively low number of stoves in the country, which makes it difficult to structure a nationwide firewood replacement program.

To conclude the presentation, the alternatives identified so far by the consulting group were shown: Improved stoves, LPG, Biomass and Electricity.

Panel development

Under this scenario, the agenda was developed, resulting in the following panel participations:

Improved Stove Panels

Panelists:

Machine Translated by Google

- Javier Aristizabal - Directorate of Climate Change and Risk Management of the Ministry Environment and Sustainable Development.

- Fabio Gonzalez - Physicist and consultant on energy efficiency issues in the country.

Development:

The Ministry of Environment's first National Efficient Stove Plan aims to build 660,000 stoves nationwide over 15 years, with 200,000 planned for this four-year period. Guidelines have been developed in conjunction with the National Planning Department (DNP) to promote similar projects, providing criteria for local governments to formulate efficient stove projects. Regional autonomous corporations have also been pioneers in the development of improved stoves. One limitation observed is the lack of diversity in the country's prototypes; instead, the promotion of the "Huellas" type stove has predominated. Regarding the technology of improved stoves, regulations are expected to be issued soon, which will be fundamental for developing a new National Improved Stove Plan in 2020.

During the panel discussion, it was noted that the efficiency of improved stoves reaches up to 20%, but varies depending on the thermal zone where they are installed. It was also identified that stove prototypes vary according to the geographical and thermal characteristics of the regions and must be adapted to the needs of households.

Families that still use firewood do so because they lack access to other energy sources for cooking or because alternative fuels are too expensive. Firewood is readily available in their communities at no direct cost, aside from the time spent collecting it, and is preferred for its efficiency and short cooking time.

Regarding woodlots, although the principle behind their development is clear—replenishing consumed firewood through reforestation—implementing the concept is complex. Farmers themselves have daily concerns about meeting their basic needs, so long-term reforestation is not a viable option for them.
priority. Furthermore, the area that has to be dedicated to the woodpile may conflict with the needs for planting other products, especially in areas of smallholdings.
Not to mention that the resources used for these purposes, due to a lack of proper management, might not achieve the proposed objectives, which translates into asset deterioration if they turn out to be public funds.

It is indicated that the cost of an improved stove in the country is on average 1,200,000 pesos, although it is expected that once the technical standard is in place, the cost will increase due to the need to adapt the stove to the requirements of the new regulations.

It is argued that cooking with firewood does not cause deforestation in the sense of impacting native forests, but rather that people go to the forest to collect fallen tree debris such as branches and dry trunks. In other words, they use the necromass produced by the forest to prepare their food, which nonetheless contributes to ecosystem degradation by using biomass that normally decomposes and whose nutrients help nourish the forests.

Machine Translated by Google

Today, families are more aware of the health effects of inhaling wood smoke. Many families are willing to change their cooking methods and switch to LPG.

It is observed that, when making the transition, families prefer to cook their breakfast and dinner on the gas stove and lunch on the wood stove, but due to a limitation of household income, it is inevitable that they return to cooking with wood for two of the three meals in order to extend the duration of the LPG cylinder.

There must be a transition to an energy source that guarantees a solution to the health problems associated with cooking with firewood.

The efforts of the Ministries of Environment, Mines and Energy, Housing and Health must be integrated to unify objectives and ensure the effectiveness of the measures that will come out of this consultancy.

Panel Gas Natural y GLP

Panelists:

- Sara Veléz - Advisor to the Ministry of Mines and Energy
- Alejandro Martinez - President of the company GASNOVA Felipe Gómez – President of AGREMGAS

Development:

Efforts should be combined with the Ministry of Education to raise awareness of the risks associated with cooking with firewood. Successful programs exist worldwide where not only is there a shift in mindset in rural areas regarding the health repercussions of inhaling wood-burning fumes, but children are also driving the pressure to make the substitution a reality.

In Córdoba, Barbada, the sea brings in large quantities of wood, but residents cook with LPG because they know that firewood is bad for their health. An LPG cylinder costs them $75,000 and lasts them a month. Electric stoves are also used. There is a growing awareness that even with free access to firewood, its use is harmful to health, and that it is preferable to use other resources like LPG for cooking.

In the Congress of the Republic, progress is being made on a regulatory framework that will allow progress towards an energy transition to address this problem.

The Ministry of Housing, through its rural housing program, must include the issue of housing and safe cooking spaces. This topic should be brought to the media to raise awareness about the harmful effects of cooking with firewood.

The fieldwork conducted by the Working Team shows that people cook with firewood because the food doesn't taste the same as when using LPG. Luz Ángela Pérez, a social worker, explains that when single mothers were asked, "If there were a firewood substitution program, would you stop using it?" they answered yes. They understand that there is a health problem, and also

Machine Translated by Google

Gathering firewood requires more work and physical effort, and it is they who must do it. During droughts, they sometimes have to travel up to three hours to collect firewood.

Julio Flórez, a businessman, argues that LPG is an excellent alternative fuel, provided that those who can use it have the money to buy it. In downtown Puerto Carreño, a 100-pound LPG cylinder costs 180,000 pesos.

In indigenous communities, the issue of available cooking options is very complex. They typically cut down the trees and store them for six months to dry. This is where the issue of deforestation arises.

In the Orinoquía region, cashew waste is used to make briquettes for cooking. Biomass would be one of the best alternatives, but the costs are still very high.

The traceability of the cylinders must be guaranteed to better monitor consumption. In rural areas, a farmer travels to the town center every 30 to 40 days to make purchases. For this reason, the 30-pound cylinder best meets their needs.

However, other supply options should be explored. Could there be a prepaid LPG option? Or could smaller cylinders be manufactured?

The country has a prepaid energy system, which could be an alternative to facilitate LPG purchases. Furthermore, although larger cylinders are more complex to transport and would entail higher costs, this system could be used to serve very isolated areas, billing consumption on a monthly basis. Additionally, smaller cylinders don't last a month, requiring farmers to travel more frequently to purchase them. However, the use of smaller cylinders could be promoted in areas near population centers, offering lower purchase costs for users.

According to industry associations, a political decision is needed to make LPG the preferred energy source for replacing firewood, with a target of transitioning 100,000 families to LPG within four years. However, this is a very low target considering the 1.6 million families who currently cook with firewood. LPG subsidies, totaling 100 to 400 billion pesos in social investment, should be requested to guarantee these families access to this energy source.

It's important for us as an industry to emphasize that we are a top-tier alternative. The population has found a solution to their problem in LPG, so we must stop calling what is a social investment—aimed at satisfying basic unmet needs—a subsidy. Healthcare costs are high, and using LPG would lessen the burden on the healthcare system. An inter-institutional effort is required.

Biogas Panel

Panelists:

Machine Translated by Google

- José Rincón - Chemist, academic and consultant on biogas and efficiency issues energy in the country.
- Héctor Herrera Flórez - from the Hydrocarbons Sub-Directorate of the Unit of Mining and Energy Planning.

Development:

Why Biogas? Biogas is the same as natural gas with a few more impurities.
It comes from biomass, where 60% is natural gas and 30-40% is carbon dioxide , with some percentages of hydrogen sulfide and other impurities. Biogas production began in China in 1920. The raw materials used include animal manure and waste from corn, sugarcane, and other sources. Various processes have been developed using polyethylene bags and other more sophisticated systems, which can cost up to $5,000.

Biogas has two benefits: it's an energy source and the waste can be used as a biofertilizer. It has a lifespan of 5 years, and no waste is discharged into waterways.
Its value for the purposes of the analysis is estimated at $5,200,000 pesos, including the stove and the gas system.

The aim is to establish a rural biogas production program as a sustainable alternative for families with easy access to this energy source. The program should be divided into stages: a learning stage, where personnel are trained to build and maintain biodigesters; a transfer stage, involving the construction of pilot plants for different products to determine which plants are suitable and in which regions of the country (selecting the technology and considering the effectiveness of microorganisms at 30-35 degrees Celsius); and finally, a sanitation and implementation stage, where a standardized system is established and quality control of the technologies can begin.

Among the advantages of biogas are: the use of waste, the creation of biofertilizers, the creation of a circular economy because resources are being used, and the sale of avoided emissions.

We must take advantage of the international support that underpins these types of technologies.
Programs have been implemented in Peru, with support from the German embassy, the IDB, the FAO, among others.

A working group should be created, comprised of government representatives and research and development centers. The beneficiaries would be the rural sector and entrepreneurs.

Finally, the creation of a biogas research center is vital for the development of this energy source in the country. On the financial side, it is necessary to create and train human resources and ensure the consolidation of a commercial market.

From the Hydrocarbons Subdirectorate of the UPME:

The issue of biogas has been under discussion for about 3 years, since 2016 following the issuance of the CREG Resolution to allow the commercialization of Biogas.

Machine Translated by Google

A request was submitted to CONPES to establish a comprehensive waste management system. A consultancy was conducted with the National University, which included an estimate of residual biomass. Additionally, an energy valuation tool for biogas is available in the country and has been used in projects in cities like Neiva.

The National Development Plan (PND) includes the use of bioenergy. There have been initial approaches to some municipalities through the Regional Autonomous Corporation (CAR), but these have not been successful. Therefore, the proposal is to implement community biogas systems. The Inter-American Institute for Cooperation on Agriculture (IICA) and the National Training Service (SENA) would act as facilitators for the implementation of biogas systems in homes.

Among the emissions reduction commitments, a Nationally Appropriate Mitigation Action (NAMA) for biogas is being developed, involving the Ministry of Mines and the Ministry of Environment, with the collaboration of the Ministries of Agriculture and Housing. The aim is to promote small-scale, community-based biogas production. It is a clean energy source (it does not produce CO2), is easy to work with, and can generate additional energy, such as electricity.

What is the value in terms of time and labor for users? Rural biogas should incentivize work by users, but its economic valuation must include the labor costs required to operate the system. The benefits of implementing biogas are energy independence and savings from not paying for other energy sources.

Regarding the material required to produce energy, the waste from a cow is estimated at 10 kilos per day; with 20 kilos, one cubic meter of gas is produced for cooking for two hours.
There must be at least two cows and some pigs in the house to produce biogas for cooking.

The issue is education and the transformation of thinking to guarantee the success of this fuel.

In Puerto Leguizamo, community biogas projects have been established. There are initiatives,
But government support is lacking.

New concept – Renewable biogas related to biomethane. A carbon market can be introduced, which will allow projects to become viable.

There are different geographical conditions among families who use firewood, and as part of the proposed replacement policy, a comprehensive census must be conducted to determine the best technology. Biogas has the significant advantage of being produced from local biomass, but not just any biomass. This fuel originates from a biological process that must be sustainable, and this can only be achieved by involving regional corporations in programs that raise awareness and train users.

A sense of ownership of the technology must be fostered, as this facilitates subsequent developments. Not everything organic found on the farm should be put into the biodigester because the reactions change according to the proportion of biomass, as is the case with cow manure, which produces a scum-like substance, or chicken manure, which behaves differently. This point must be taken into account and considered in the following stages. Professor Rincón

Machine Translated by Google

He clarifies that this is the objective of the Biogas Research Center which, together with continuous training, will allow the success of the biodigesters.

There must be sustainability that allows for closure. There are many research centers seeking the implementation of the NAMA and hoping that, through the research center, results can be provided regarding the appropriate biomass for each region.

The International Biogas Research Center produces biogas from a mixture of 30% pig manure and 70% cattle manure. This process yields enough biogas to power light bulbs. These achievements are the result of extensive research, which underscores the need to develop a comprehensive biogas program in the country.

Colombia is a significant producer of palm oil, which could be a community-based solution.
However, the limitation lies in the fact that these crops are not interconnected. Distance is a limiting factor for the substitution option in the consulting. But since it is a manageable gas, it can be produced and stored for distribution.

Panel on Health, Culture and Environment

Panelists:

- Carlos Torres - Director of the Research Department of the Specialization Program in Pulmonology and Director of the Colombian Pulmonology Foundation.

- Luz Ángela Pérez - Social worker with experience in community management and territory.

- Samuel Osorio - Medical delegate of the District Health Secretariat.

Development:

Cooking with firewood contributes to air pollution, which can be divided into indoor and outdoor air pollution. To put this in context, air pollution is the fifth leading risk factor for death worldwide, just behind smoking. In 2016, 7 million people died globally, almost 50% of them from air pollution, primarily from cooking with firewood.

Last year, the Ministry of Health put this problem into context, also indicating that in Bogotá there is no high impact because cooking with firewood is not done in the rural sector.
But there is 0.2% of the population, that is, 16,600 people, who may be using this fuel.

A study is underway to determine the percentage of people who cook with firewood and its impact on their health. However, some individuals with a history of cooking with firewood who have migrated to Bogotá have created a health liability that is now becoming apparent and placing a strain on the healthcare system, significantly impacting hospital emergency rooms.

Machine Translated by Google

Six thousand surveys are conducted annually, providing a diagnosis of the significant risk of respiratory problems among people who have cooked with firewood. Those who cooked with firewood have a 70% higher risk than those who did not (breathing difficulties, asthma, among others). A committee is being formed to prioritize these individuals and avoid overburdening the healthcare system.

While the total number of respiratory illnesses and their cost to the system have been quantified, there are currently no economic indicators of the cost to the country of addressing passive respiratory illnesses related to indoor air pollution from wood consumption.

The risk of inhaling smoke from a wood-burning stove is greater than the risk of secondhand smoke, that is, inhaling the smoke from someone smoking next to you. (Let's discuss this conclusion.)

Dr. Carlos Torres

Pulmonologists are going to have a lot of work due to air pollution. Air is composed primarily of oxygen and nitrogen, and anything other than oxygen and nitrogen is potentially harmful. Therefore, when discussing energy sources, the best would be the one that releases the fewest pollutants or emissions into the environment.

The health impacts must be taken into account and are not easy to quantify since they manifest in the long term and it is not easy to associate them with that source of pollution, in this case firewood.

The topic of exposure to wood smoke has been a subject of research by the Colombian School of Pulmonology. Ten years ago, the first worldwide study was conducted with Prepocol, which showed that exposure to wood smoke is a risk factor for COPD (Chronic Obstructive Pulmonary Disease).

Many people still cook with firewood, so the effects will be seen for years and decades to come. One in ten people over 40 meet the criteria for having COPD (8.9%). A third of them have exposure to firewood as their only risk factor. Children exposed to firewood have greater lung dysfunction than those who have not, due to early exposure to the smoke from this fuel. Firewood is also a risk factor for asthma, according to international surveys.

The disease caused by wood smoke is different from that caused by cigarette smoke. Since there is no distinction between COPD caused by wood smoke and COPD caused by cigarette smoke, the treatment currently provided is the same. They are different diseases because the particle deposition is different; the bronchi are blackened in the case of wood smoke, which is not observed in COPD caused by cigarette smoke.

One-third of people who use oxygen today are associated with exposure to wood smoke. This specific condition needs to be characterized, and the contributing factors and risk factors associated with wood smoke exposure need to be determined.

Machine Translated by Google

Any preventative activity can generate savings in the health system and this does not only apply to firewood.

Currently, vaping is being promoted because it is not addictive and does not cause health problems, but many cases of lung disease are now being linked to this activity. If preventative measures were implemented, consumption rates would most likely fall, thus easing the burden on the healthcare system and reducing associated costs.

The healthcare system and health insurance providers (EPS) reflect a problem that extends to other sectors. Very few people utilize informational strategies to change behaviors. Strategies must be developed not only within the healthcare sector but also by involving other factors such as culture, energy, and all branches of government to shift this mindset.

Through surveys conducted by the Ministry of Health with people over 60, children under 5, and children aged 5 to 14, risk factors are identified. While the questions doctors ask during consultations are not systematically recorded, healthcare professionals are increasingly familiar with the risk factors associated with cooking with firewood, and it is becoming more common for them to ask if patients have ever been exposed to firewood.

Based on cultural behaviors, Luz Ángela refers to the fact that people, especially housewives, are willing to change their cooking fuel because they have identified and internalized that cooking with firewood is having repercussions on their health and that of their children (not only pulmonary but also related to burning eyes, cataracts, and headaches). Similarly, Dr. Carlos Torres explained that it has been identified as a risk factor for problems associated with high blood pressure, tuberculosis, lower cognitive performance, and lung cancer.

In Boyacá, when their economy improves, the transition from firewood to charcoal occurs and later switching from coal to LPG. In one particular case, a housewife forgot that she had health problems caused years earlier by the smoke from her wood stove and went back to using wood. When she became ill again, she switched back to LPG.

In Bahía Solano, they use firewood when they run out of gas because resources are limited. Firewood should be an option, not an obligation.

Electrical Panel

Panelists:

- Guillermo Andres Cajamarca - Commercial Business Unit Manager at Celsia Energy.
- Fabio Gonzalez - Physicist and consultant on energy efficiency issues in the country.

Development:

Efforts are underway to find a balance that allows for the distribution of energy as a fuel source for cooking. Technologies are moving towards the use of electricity.

Machine Translated by Google

The advantages of electric heating include: improved health, as these stoves produce no emissions; reduced risk; and more affordable costs. Energy efficiency is coupled with technological improvements.

In the country until the 1980s almost all households used electricity for cooking, and that's where the design of the cookware set, stoves and other appliances came from.

The electric (induction) stove is considered the most modern cooking technology and is used in the United States, Canada, and Europe. The design of an induction stove allows for efficiencies exceeding 90%, and with the new long-term energy contract auction, there will be enough energy available to encourage households in socioeconomic levels 5 and 6 to switch to electric power, which will help with cross-subsidies for interconnected areas.

In unconnected areas, there's the specific case of Mitú, where all fuel arrives by air, including LPG cylinders, which arrive by air at very high prices. People cooked with firewood because it was free. A small hydroelectric plant (PCH) was installed, with a 2MW fuel-powered generation capacity. For people in socioeconomic stratum 1, energy is relatively cheap due to subsidies. They've had 24-hour electricity for six years, and 90% of the population has access to it.

Many residents cook with electricity because they don't end up paying for it. Generating 1 kWh costs 1,300 pesos in Mitú, and the government ends up paying for it. Now that subsistence consumption is being debated, cooking is included as part of subsistence consumption in Mitú. One option is solar power; however, the costs are high for a single burner. Currently, 700-watt systems are being installed for 15 million pesos, which wouldn't be enough for one burner. In Mitú, installing 1 kW for cooking in a rural area could cost up to 20-25 million pesos, plus the burner and other expenses. At that level, electric cooking is beyond the country's economic reach given the associated costs.

Subsistence consumption means that if a cheaper substitute is available, electricity is not used, and in Colombia's case, that would be LPG. Subsistence electricity consumption is 100 kWh per month, and cooking alone consumes between 60 and 80 kWh per month. Therefore, it is not feasible to implement electricity as a cooking energy source for socioeconomic levels 1, 2, and 3, but it could be for levels 5 and 6.

The solar stove as a solution, proposed by Celsia, would be an electric stove and water heater with an induction cooktop option.

A kit is currently being developed, with an estimated cost of 7 million pesos, that will have a capacity of 120 kWh or 140 kWh per month, covering cooking and heating needs. It poses no health risks and requires no batteries, as surplus energy is fed back into the grid.

With the surplus energy scheme, the energy balance allows for grid backup, where you pay for what you consume. One financing option could be $2 million in government subsidies and $5 million in loans for households, with full repayment due six years after installation. This would allow the system to balance.

This initiative is expected to work as long as there is a supporting public policy. Similarly, if there is no market, the price is relatively high. Where a market exists, prices decrease significantly.

Machine Translated by Google

In the auction a few days ago, the price closed at $90 pesos per kWh, meaning that installations could cost $400 per kWh and would be possible for cooking. It's a market issue.

What does the kit include? The kit includes assembly, design, engineering, construction, commissioning, the stove, and the water heater. Internal wiring is not included.

Closing

Thanks to each of the panelists who participated in the development of the Expert Workshop and later the trip to share lunch.

Photographic Images

PHOTOS EXPERT WORKSHOP



Photo: Development of the Expert Workshop

Machine Translated by Google

Annex 3 - Databases

The following files are attached:

- **Annex 1- 20191209 Firewood Consumption Scenarios** (Excel)
  It contains estimates of scenario 1 according to the BECO and the 2018 Quality of Life Survey, and of scenario 2 where 30% of those who cook with firewood diversify with another energy source.

- **Annex 2- 20191209 Tables and calculations Deliverable 2 Contract C031** (Excel)
  It contains the graphs and tables of deliverable 1 and the estimation of cooking costs in rural and urban households that consume firewood as the main energy source for the different substitution alternatives, differentiating by technology used.

- **Annex 3- 20191209 Laboratory report on wood consumption - 2019** (PDF)
  It contains the results obtained from measurements in the wood laboratories carried out at the national level, regarding the estimated daily wood consumption per household.

- **Annex 4- 20191209 Qualitative and quantitative survey of firewood** (Excel)
  It contains the information provided by the families subject to Annex 3- 20191209 Report laboratories firewood consumption- 2019 with their respective graphs and tables.

- **Annex 5- 20191209 Emissions Calculation** (Excel)
  It contains estimates regarding the reduction of emissions associated with the implementation of the program to replace firewood as an energy source in the residential sector of Colombia. $_2$

- **Annex 6- 20191209 Strategy by department and cost of the plan** (Excel)
  It contains the technology replacement strategy with its respective implementation phases (short, medium, and long term), differentiating the alternatives by phase and considering the feasibility of implementing each technology within the established timeframe. Likewise, the total costs of the strategy are presented for each phase and each replacement technology, broken down by region and department.